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# Job recommender systems: a systematic literature review, applications, open issues, and challenges

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## Abstract

**Background:** The HR department has undergone substantial transformations, with a focus on improving recruitment, employee satisfaction, and training through technological integration, particularly with Recommender Systems (RSs) and Job Recommender Systems (JRSs). In addition, the adoption of remote working across multiple industries has significantly increased the need for smart technologies in Human Resources (HR) processes over time. The COVID-19 pandemic accelerated this shift, placing even greater emphasis on the utilization of these technologies.

**Objective:** Over the past decade, research on JRSs has increased significantly; however, no comprehensive Systematic Literature Review (SLR) has been conducted in this field. To fill this gap, this study conducts an SLR covering relevant RS and JRS studies published in the last 13 years. It aims to critically analyze commonly used methodologies, identify key research trends and gaps, provide a comprehensive overview of existing literature, define application areas in HR processes, classify performance evaluation metrics used, highlight underexplored themes, discuss limitations of current research, and offer recommendations for researchers and HR practitioners for future studies in the JRS field.

**Method:** In this SLR, a total of 19 survey articles (including both RSs and JRSs) and 57 research articles on JRS solutions, published between 2010 and 2023, were reviewed and categorized based on publication years, publishers, application themes, applied methodologies, RS filtering techniques, utilized tools, and future studies. Moreover, a comprehensive taxonomy was developed to classify the filtering techniques and methodologies used in both RSs and JRSs. The study also categorizes and classifies performance assessment metrics used in both RS and JRS solutions into four categories: Predictive Accuracy, Classification-Based Accuracy, Ranking-Based Accuracy, and Non-Accuracy-Based Metrics.

**Results:** Key findings reveal that while JRS solutions utilizing Collaborative, Content-Based, Hybrid-Based, and Knowledge-Based Filtering are well-represented in the literature, those applying Demographic-Based, Reciprocal-Based, Popularity-Based, and Non-Personalized Filtering remain underexplored. The methodologies highly employed in JRS research include similarity-based techniques (e.g., TOPSIS, Cosine Similarity), machine learning models (e.g., XGBoost, LSTM), and knowledge-based approaches (e.g., Ontology, Semantic Web). Evaluation metrics vary significantly based on objectives

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and filtering techniques, with Classification-Based Accuracy Metrics (e.g., precision, recall, F1 score) being the most commonly used. An analysis of historical trends in JRS publications revealed fluctuations in research interest, with a decline in 2011 and notable peaks in 2017 and 2023. The study also identified a lack of JRS solutions addressing broader HR processes beyond candidate-job matching.

**Conclusion:** As a result, JRSs are still under development and need further improvements, particularly in reducing bias and increasing efficiency in HR processes through advanced AI technologies. Over the past 13 years, JRS studies have explored a variety of themes, including recommending suitable team members, identifying training modules based on skill gaps, developing career path planning systems, assessing employee satisfaction, standardizing professional knowledge, analyzing job postings to determine the busiest job fields, and estimating job tenure and turnover intention. However, research on other crucial HR processes, such as workforce planning, personalized JRSs such as recruitment of special groups, employee engagement strategies, and talent retention frameworks, remains limited, highlighting opportunities to explore in the future.

## Highlights

- This study presents a systematic literature review on Job Recommender Systems (JRSs).
- The study covers a discussion about challenges, future developments, and open issues on JRSs via 6 research questions.
- Also, other review studies (8 RSs field and 11 JRSs field) were examined and compared.
- 527 articles were retrieved through our search query from both WoS and Google Scholar between January 1, 2010—April 30, 2023.
- Of these, 57 were deemed qualitatively appropriate and were included in the systematic review.
- A taxonomy of methodologies applied in both RSs and JRSs are developed.
- RSs taxonomy developed was associated with 6 categories: (1) Collaborative, (2) Content, (3) Hybrid, (4) Knowledgebase, (5) Context aware, and (6) Others.
- JRSs taxonomy developed was associated with 6 categories: (1) Collaborative, (2) Content, (3) Hybrid, (4) Knowledgebase, (5) Context aware, and (6) Reciprocal.
- Types of accuracy metrics used in both RSs and JRSs were categorized and compared.
- This study draws attention to the requirements in the development of effective JRSs within the scope of current RS technologies.
- Findings is a guidance for future R&D studies in the field of JRS for developers, practitioners and academics.

**Keywords:** Systematic literature review, Recommender systems, Job recommender systems, Human resources management

## Introduction

The activities and strategies of Human Resource Management (HRM) can be classified into two categories, as suggested by Vrontis et al. [148]: HR activities (such as hiring, training, and assessing employee performance) and HR strategies (e.g., assessing organization current workforce, creating employee development plans, creating succession plans, performing a gap analysis, etc.). The main processes of HRM are planning, recruitment, orientation, training and development, performance management and

motivation. In this research, we have focused on the recruitment and hiring processes of HRM. During the hiring process, the HR authorities aim to make the best possible match between a job seeker and the right job posting. However, this process may take as long as days, weeks, or even months. In a world where competition is rapidly increasing, HR authorities have needed advanced Job Recommender Systems (JRSs) to ease the processes such as the collection and processing of all data related to the recruitment process, staff training, staff orientation, and staff performance evaluation. To make the recruitment process more efficient, JRS solutions are able to analyze the unstructured/semi-structured resumes uploaded by job seekers through the HRM systems. This analysis may involve emerging Artificial Intelligence (AI) technologies like Machine Learning (ML), Natural Language Processing (NLP), Semantic Web, and others, to match the resumes with existing job postings. In other words, efficient JRS solutions have the capability to identify the most suitable candidate for a specific job opening, facilitate immediate job interviews, reassign qualified personnel to suitable positions, monitor the competencies of the workforce, propose relevant training programs, conduct personnel performance assessments, and others [55, 148]. Particularly in the recruitment process, accurately extracting relevant information from a job seeker's resume, including profile details, past job titles/experiences, educational background, and competencies, is crucial. AI technologies have reached a level of maturity where they can effectively perform this requirement.

Accordingly, it is possible to make "job-candidate matching" with high accuracies using effective AI solutions in matching a job seeker's qualifications with existing job openings [55, 148]. Indeed, JRSs can be integrated into HRM systems, and they can also function as modules within AI-supported chatbots operating in the web/mobile environment. This integration can empower chatbots to not only recommend suitable job positions based on a user's profile but also to conduct online job interviews with job seekers [8, 94, 107, 145]. The utilization of JRSs within chatbots can enhance the overall efficiency of HRM recruitment processes, offering a seamless and interactive experience for both employers and job seekers in the digital space. For instance, MYA, a chatbot developed by start-up MYA Systems, uses speech-based AI algorithms to facilitate the recruiting process [94]. MYA drives NLP and comprehension technologies. By initiating interviews with candidates, it can measure whether the candidates are suitable for the job posting and guide the candidates through all processes. MYA agent arranges an interview with an HR staff member during the final recruitment process and supplies information to the HR staff. The HR staff makes the final decision. In less than 2 years, MYA has acquired 120 corporate customers, 40 of which are on the Fortune 500 list, and announced that 3 of the top 5 recruitment agencies use MYA [152]. It is claimed that firms that benefit from MYA save more than 70% of their time in their recruitment processes. Another similar solution, RAI, is developed by Hiring Solved and is used as an alternative to MYA [123]. Additionally, other chatbots such as JobPal, MyAlly and Debra have also been developed to support this process [8, 145].

Another application area of JRSs is the AI-based video recruitment process [78]. The voice and facial recognition software developed by HireVue incorporates a special algorithm that analyzes candidates'body language, tone of voice, and facial expressions to determine which candidates are more suitable for a job posting. With the AI-based

capability of HireVue, researchers aimed to reduce bias by eliminating unreliable and inconsistent variables typically found in resumes and phone calls. Thanks to effective AI-based applications, job seekers can easily get the necessary job details to make the right decisions, and companies looking for personnel can easily access sufficient information about job seekers. In the field of HRM, it can be said that JRS solutions are still in the maturation process due to the complexity of recruitment processes, the emergence of new professions and competencies over time, the multitude of parameters in the resume-job posting matching process, and the wide application area of JRSs [55, 148]. As in the case of MYA, JRSs have entered the industry and are used by HRM authorities to manage their recruitment processes, however, JRSs have a long way to go before they can be personalized, further developed, and integrated into challenging HRM processes over time [79, 148]. Because, in JRS solutions, the primary goal is to keep efficiency, robustness, speed, and accuracy high, as well as ensure its sustainability.

To identify, classify, synthesize and compare the current state-of-the-art developments in both RSs and JRSs [14, 25, 42, 43, 55, 64, 67, 137, 143, 148], resulting in a comprehensive Systematic Literature Review (SLR), drawing on our previous review experiences [28], we identified the following research questions:

RQ1: What are the approaches and methodologies applied in the JRSs?
RQ2: What are the main ideas focused on in JRSs?
RQ3: What are the quality and performance measurements applied in the evaluation of JRSs?
RQ4: What are the popular evaluation and modeling tools used in developing JRSs?
RQ5: What are the current challenges and future improvements in JRS implementations?
RQ6: What are the available standards applied in the JRSs?

The objectives of this study are to review and categorize existing literature on JRSs and RSs published between 2010 and 2023, develop a comprehensive taxonomy for classifying JRS methodologies based on recommendation filtering techniques, analyze and categorize evaluation accuracy metrics used in JRSs and RSs, including Predictive Accuracy Metrics, Classification-Based Accuracy Metrics, Ranking-Based Accuracy Metrics, and Non-Accuracy Based Metrics, review and classify research and development articles on JRSs, focusing on publication trends, application areas, filtering techniques, tools, methodologies, and future research directions, and identify gaps in current JRS research and propose areas for future development to enhance HR processes using advanced AI technologies.

Therefore, in this study, we conducted a SLR study on the advantages, disadvantages, tools used, applied methods, and assessment performance metrics in JRSs, adhering to the research questions given above. The aim is to reveal the important gaps and challenges identified in JRS solutions. Thus, it will be possible to identify important parameters and factors where future research can improve the capabilities of JRS approaches. Additionally, in this work, we put forward a comprehensive taxonomy to classify the methods and methodologies used in both RSs and JRSs. In this study, we adhered to the

sequential steps outlined in Fig. 1 and provided a detailed explanation of each step in the corresponding sections of the manuscript, aligned with the stages depicted in Fig. 1.

*Step 1- Analysis of HR Functions and Processes* The study began with a detailed examination of HR functions and processes, including recruitment, training, and job performance. The recruitment process was explained and illustrated through a comprehensive diagram, particularly to facilitate understanding.

*Step 2- Identification and Comparison of Performance Metrics* The study identified and compared performance metrics utilized in the evaluation of RSs. This included analyzing metrics such as Predictive Accuracy Metrics, Classification-Based Accuracy Metrics, Ranking-Based Accuracy Metrics, and Non-Accuracy Based Metrics. In addition, challenges in setting up RSs is discussed.

*Step 3- Review of Literature Survey Articles and Taxonomy Establishment* Nineteen literature survey articles were reviewed, with 8 focusing on RSs and 11 on JRSs. The review identified various limitations, such as outdated reviews and insufficient details

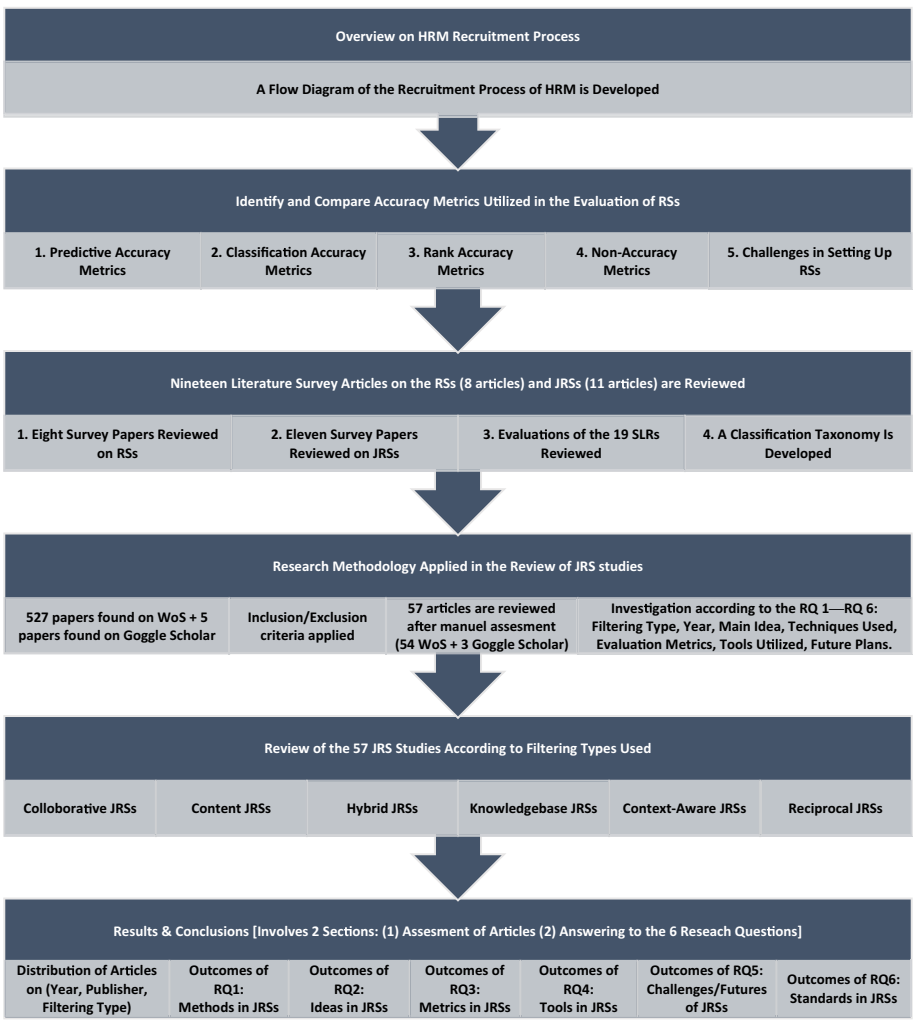


**Fig. 1** The contributions and obtained results of this SLR study conducted are shown via a flow diagram

on JRS solutions. Subsequently, a broad taxonomy was established to categorize the filtering techniques and methodologies used in both RSs and JRSs. This classification aimed to reveal the impact of existing RS methodologies on JRS methodologies.

*Step 4- Research Methodology Applied in the Review of JRS studies* A comprehensive search query was formulated for articles published between January 1, 2010, and April 30, 2023, using specific keywords related to JRSs. Of 527 retrieved articles, 57 relevant research articles were examined. These articles were classified and assessed based on publication year, focus, techniques, data, tools, performance metrics, and future development plans.

*Step 5- Review of the 57 JRS Studies According to Filtering Types Used* involves a thorough review of the 57 JRS studies, specifically analyzing the filtering types used. This review categorizes the studies based on the various filtering techniques applied, such as content-based, collaborative, hybrid, context-aware, and popularity-based filtering. By examining how these filtering methods are utilized across different JRS solutions, the step aims to assess their effectiveness and application in various contexts, providing insights into the strengths and limitations of each approach in enhancing JRSs.

*Step 6- Results & Conclusions* involves the results and conclusions of the study, structured into two main sections. The first section,"Assessment of Articles,"examines the distribution of the 57 reviewed JRS studies by publication year, publisher, and filtering type. The second section addresses the six research questions: RQ1 focuses on the methods employed in JRSs; RQ2 highlights the key ideas explored in JRS studies; RQ3 analyzes the evaluation metrics utilized in JRSs; RQ4 reviews the tools and technologies used in JRS development; RQ5 identifies the challenges faced by JRS solutions and proposes future research directions; and RQ6 examines the standards applied in JRSs.

The rest of this study is proceeded as follows. Sect."Background"presents the background of business processes in HRM, overview of RSs, and various accuracy metrics utilized in RS solutions. Sect."Other Similar Literature Review Studies"discusses and compares similar SLR studies on both RSs and JRSs, and obtained a comprehensive taxonomy based on filtering techniques and methodologies used. Sect."Research Methodology"discusses the research methodology selected, article selection process, and research motivation applied in this SLR study. Section "Review of the Selected Studies on JRSs" categorizes, summarizes, and classifies the selected JRS articles, while Sect."Results and comparisons"presents findings, comparisons, and describes the study limitations. Finally, this article is concluded in Sect."Conclusion", where the achievements are shared.

## Background

This section provides an overview of the functionality and main processes of HR departments, presents an overview of RSs, and analyzes the evaluation accuracy metrics used in the RS solutions. Sects."Human Resources Management and Recruitment Process","Architecture of Recommender Systems","Metrics Used in the Evaluation of Recommender Systems", and"Challenges in Setting Up Recommender Systems"delineate the recruitment process steps in HRM, provide an overview of RSs and its modules,

discuss the accuracy/non-accuracy metrics used in RS solutions, and outline the challenges faced by RSs, respectively.

#### Human resources management and recruitment process

Human Resources (HR) is responsible for administering all management decisions and practices that impact on the relationship between an organization and its employees. The internet has created numerous opportunities for organizations, allowing them to observe the global stages of digitalization and influencing their HRM and its processes. In addition to accelerating digitalization, special conditions such as COVID-19 may cause various changes or innovations in the processes carried out by the HRM side (e.g., remote performance evaluation of online personnel, online recruitment interview, etc.). Furthermore, social network sites have helped job seekers create their own profiles and search for corporate information, both formally and informally on online [92, 100].

HRM can be categorized into two main areas [148]: (1) HR Activities, which include functions such as hiring, training, and managing employee performance, and (2) HR Strategies, which involve broader aspects like corporate decision-making, addressing job displacement issues, developing new HR solutions using AI technologies, and providing learning opportunities for employees. Main well-known processes of HRM are (1) planning, (2) recruitment, (3) orientation, (4) training and development, (5) performance management and motivation. In this research, we mainly focused on "recruiting/hiring" activities of HR. Researchers [77] defined "recruitment" as the HR process of finding and assigning the best candidates to roles where they can work most efficiently.

Figure 2 illustrates an example of the HRM recruitment process. It starts with identifying a staffing need within a department, followed by defining the job position and establishing the criteria for selecting the ideal candidate. Subsequently, the department manager fills in an employee request form for the open job position, which is then forwarded to the HR staff responsible for suitability verification. The form submitted to HR is evaluated according to the characteristics, titles, scope, and complexity of the job for the classification step and its suitability is checked by responsible HR staff. If the open job position is deemed suitable, an internal job posting is initiated, and efforts are made to find a suitable candidate within the company. If one or more personnel are found suitable for the open job position, separate interviews are scheduled for these individuals. Each interview requires a pre-interview phase, involving the preparation of interview questions. After identifying a successful candidate(s) in the interviews, the HRM selects the most qualified or suitable candidate and assigns them to the job position. If a suitable candidate is not found for the open position, HR searches for potential resumes in the HR pool. If qualified candidates are identified, they are invited for an interview, following the same process. If no suitable candidates are found, HR starts a new search using a different strategy.

#### Architecture of recommender systems

Recommender Systems (RSs) are web applications or software that give personalized recommendations to users. The development of technology and the increase in web data consumption on the internet led to the emergence of RSs. According to Negroponte [105], RSs were first proposed in the 1970 s and were further developed and prototyped

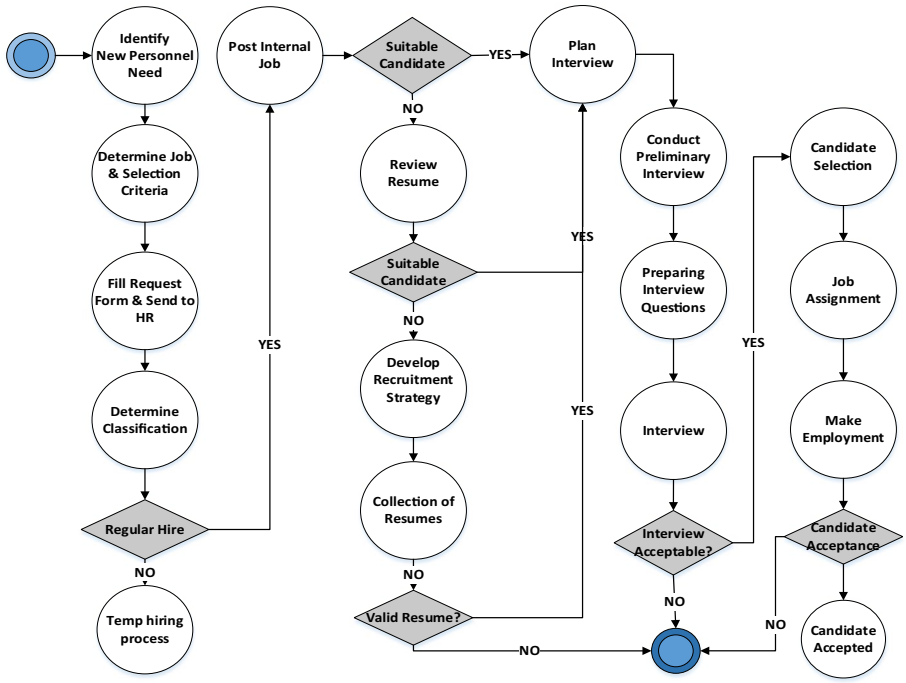


**Fig. 2** An example of a recruitment process conducted by HRM

by Goldberg et al. [58] in the early 1990 s. It began to be implemented and commercialized for the first time in the mid-1990 s [95].

Since RSs consider the profile data, demographic/cultural characteristics, past/current preferences, etc. of the users, they can enable the user to reach the desired service/product faster. RSs have become very important software solutions today as they enable customization for both consumers and companies and save time in process management. Similar to a search engine, RSs can provide the user with the best recommendations without requiring the user to be specific about what they want [20].

Although users'profiles do not generally change, it is difficult to evaluate user preferences/tastes that may change over time. This difficulty forced RSs to evolve over time and their use in different industries became popular [1, 16, 21, 28, 51, 71, 88, 89, 94, 95, 115, 156]. Hence, the evolution of RSs over time has begun with the integration of advanced AI solutions into the core processes of RSs [118]. The following stages, which are commonly applied in RS solutions, are determined below.

(1) *Data Collection* Every user transaction generates data for RSs. For instance, the user's search queries on a website's search engine, products clicked on or ignored, duration of product views, product reviews, items added to the shopping cart, returned products, music played or paused, movies watched or interrupted, and even external factors like weather—all these details are crucial data points for RSs.
(2) *Storage* The gathered data is subsequently stored to be presented to the user as a set of recommendations.

(3) *Analysis* Using various approaches and algorithms of AI or ML, a set of suitable services/products that the users can consume are deduced [118]. For example, a ML-based RS can obtain effective data models on a dataset by analyzing information and predicting users'preferences. The system then produces the most relevant and useful results based on the correlation between demands and interests [71, 118].

(4) *Filtering* As a result of the analysis, the recommendations to be presented to a user are filtered and sorted.

RSs have increasingly contributed to HRM processes, particularly in recruitment, enabling more efficient and timely candidate selection. JRS solutions help reduce HRM effort and costs while improving job seeker accuracy. Given the vast data processed in HRM, the development of advanced JRSs will significantly benefit both the HRM field and the literature.

The common JRS architecture is shown in Fig. 3, where job position recommendations are generated by analyzing user data such as past experiences, titles, profession, and profile information.

A JRS architecture has the capability to provide essential services discussed below to diverse stakeholders, encompassing HR personnel, HR service providers, job seekers, employers, and system administrators. A JRS enables individuals looking for a job opportunity to create their own resumes through built-in resume creation tools or directly upload pre-existing resume documents (e.g., in Word/PDF format) to the system. The JRS then aims to recommend suitable open job positions by processing and analyzing the job seekers'data (e.g. using profile details, education background, past work experiences, certifications/qualifications, etc.). The recommendations obtained are made available to the users either through career-planner websites, HRM portals, or mobile-based employee recruiting applications.

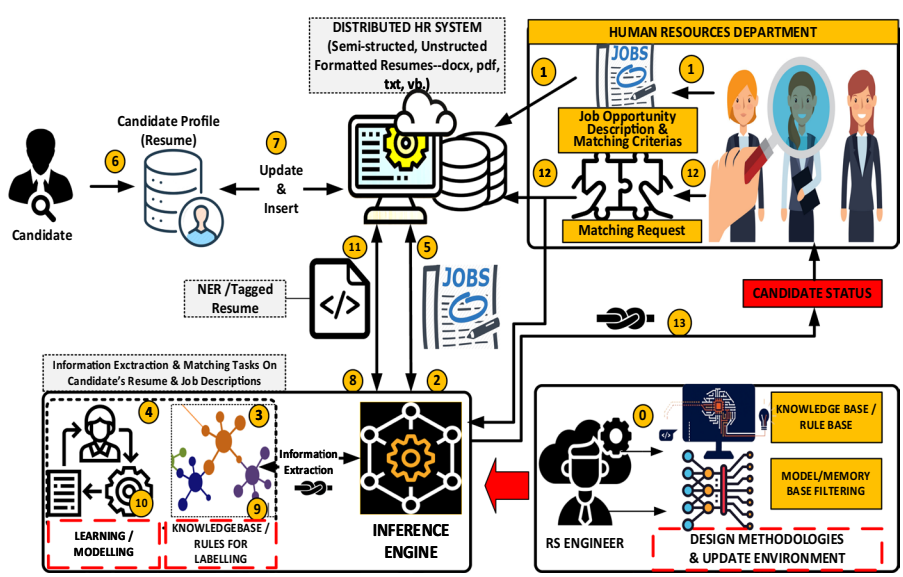


**Fig. 3** General architectural model of a Job Recommender System

- *RS Engineering*. The development of JRS components requires RS engineering (Fig. 3, Step 0). For example, the system's knowledge base has a conceptual framework structure that includes concepts related to resumes and job opportunities, as well as inter-conceptual feature definitions and inference rules. HR personnel can directly or indirectly interact with JRS components through the HRM system or service provider and enable filtering services to find resumes suitable for open job opportunities (Step 1).
- *Data Collection and Storage*. Open job positions are identified by HR staff through HR solutions (Step 1). Subsequently, semi-structured job positions can be transformed into a more structured form using various tagging and parsing methods through learning-based/rule-based approaches (Steps 2–5) and stored in the system's database. Additionally, potential job seekers have the option to upload and store their resumes in an unstructured format into the system's database via the system services (Steps 6–7).
- *Parsing*. The unstructured resumes uploaded by job seekers can be transformed into a structured format (semi-structured, fully structured) through various methods (e.g., Named Entity Recognition—NER). To accomplish this, the terms in the content of each resume are analyzed through the system procedures, which may utilize the system knowledgebase, learning-based model(s), or other approaches (e.g., synthetic matching, regular expressions, statistical approximations, etc.), if any, to be conceptually analyzed and labelled by NER (Steps 8—10).
- *Rule/Model based Inferencing and Matching Module*. Inference engines can be employed for the task of matching between a resume and a set of structured job opening descriptions in the system database, or vice versa. To achieve this, a system service is employed to activate the inferencing module, which matches an uploaded resume(s) to the structured job opportunities (Step 12). To achieve this, the system activates its own matching rules/learning models based on predefined criteria during the matching process. If suitable matches are identified, a matching score is then determined for each resume (Step 12). Suitable open job positions are filtered and formed as a list for the resume considered and presented to HR personnel (Step 13). The system can also operate in reverse. HR personnel can initiate the matching process by requesting a match for an open job position. The reasoning and filtering processes are then executed to match existing resumes stored in the system's database with the specified job criteria.

To reduce time and costs by analyzing candidate resumes within HRM applications and match the right candidates to job positions, improving the quality and accessibility of JRS solutions is essential. AI-driven, customized guidelines have been integrated into JRS solutions to assist HR staff and job seekers, ensuring improved dynamics, accuracy, and user satisfaction. For example, LinkedIn enhanced its RS in 2020 by using deep transfer learning to develop domain-specific models [84]. The basic evaluation criteria and quality metrics used in both RSs and JRSs are identified and categorized in the following section.

### Metrics used in the evaluation of recommender systems

There are various measurement methods and metrics for evaluating success and efficiency of RSs. These metrics can be examined in four main categories: (1) Predictive Accuracy Metrics, (2) Classification Accuracy Metrics, (3) Rank Accuracy Metrics, and (4) Non-Accuracy Metrics [35, 62, 63, 131, 147]. This metrics classification aims to understand the performance evaluation for both RSs and JRSs and to assist researchers in the performance analysis stage. The metrics are explained in detail below, illustrated with examples.

#### *Predictive accuracy metrics*

Predictive accuracy metrics, also known as rating prediction metrics, assess how closely the estimated ratings provided by a RS align with the actual user ratings. These measurements are commonly used to assess rating systems that involve non-binary ratings [147]. It is a suitable method for situations where it is necessary to accurately estimate ratings for all items. Commonly used rating accuracy metrics RSs are Mean Absolute Error, Mean Squared Error, Root Mean Squared Error, Normalized Mean Absolute Error, R Squared, and Explained Variance, which are commonly used in RSs. In addition, it is possible to say that there are several similarity techniques used for prediction during matching tasks.

(1) *Mean Squared Error (MSE)* MSE is one of the most popular metrics (Vijaysinh, 2021). It is a simple metric that calculates the difference (error) between the actual rate and the predicted rate, squares it and then provides the mean of all the errors (e.g., estimated error of a job seeker's competencies with expected competency requirements in a job posting).

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(R_i - P_i)^2 \tag{1}$$

(2) *Root Mean Squared Error (RMSE)* RMSE is based on the standard deviation between the actual rating and the predicted rating. RMSE is the root of MSE and is beneficial since it helps to bring down the scale of the errors closer to the actual values, making it more interpretable. It squares the errors before collecting them, thus giving greater importance to the large errors. Normalized RMSE is applicable for evaluation calculations [35, 63, 147].

$$\begin{aligned} RMSE &= \sqrt{\sum_{i=1}^{k}\frac{(R_i - P_i)^2}{n}} \\ Normalized(RMSE) &= \frac{RMSE}{R_{\max} - P_{min}} \end{aligned} \tag{2}$$

where, $R_i$_is the actual data, $P_i$_is the predicted data, n is the number of predictions.

(3) *Mean Absolute Error (MAE)* MAE is similar to RMSE, but it uses absolute value instead of squaring and taking the root of the mean. Shortly, it is the meaning of the absolute error values (actuals – predictions). MAE measures the average absolute deviation between the predicted rating and the user's actual rating. In other words, MAE is calculated by taking the absolute difference between the predicted values

and the actual values, then averaging it across the dataset. RMSE imposes larger penalties for major errors compared to MAE. There are differentiated and normalized versions of MAE and RMSE [35, 62].

$$MAE = \frac{\sum_{i=1}^{n} |R_i - P_i|}{n} = \frac{\sum_{i=1}^{n} |e_i|}{n} \tag{3}$$

(4) *Mean Absolute User Error (MAUE)* MAUE individually calculates the MAE for each user and then takes the average of these individual MAEs. This method solves any possible bias introduced in the calculation of MAE. Therefore, all the users will have the same weight during the computation [35, 62].

(5) *R Squared ($R^2$)* $R^2$ is a statistical measure of how close the data are to the fitted regression line for a model [54]. Therefore, a higher $R^2$ value signifies that the model is a good fit, whereas a lower $R^2$ value suggests that the model is not a good fit. Thus, $R^2$ serves as a measure of error. A higher $R^2$ value is achieved when the ratio of the sum of the residuals to the sum of the means is smaller: *$R^2$=SSR/SST, where SSR={Sum of the Residuals} and SSR={Sum of the Means}.* A lower $R^2$ value indicates that the model generates more inaccurate predictions, signifying a higher degree of error. Simply put, when the $R^2$ value is too low, the system's estimates become less valuable, leading to uncertain and less accurate predictions.

(6) **(6)** *Explained Variance (EV)* It measures the proportion of variance in the data that the model explains. It is calculated by subtracting 1 from the ratio of the variance of the forecast errors to the variance of the actual values *{EV=1- Var (Prediction Errors)/Var (Actual Values)}*. Scores approaching 1.0 are positive and this means better accuracy in estimating the squared standard deviations of errors [97].

(7) *Similarity* Similarity measurement methods are commonly used in user-to-user or item-to-item collaborative-based filtering RSs. Generally, these methods involve utilizing ratings of items that are commonly rated when calculating the similarity between a pair of users or items. RSs use various similarity techniques such as Cosine Similarity, Pearson Correlation Coefficient, Jaccard Similarity, Tanimato, etc. to find similar users/items for an active user/an item [13, 101].

– *Cosine Similarity* It is vector similarity metric [127–129] that measures the cosine angle formed by the two ratings vectors. When computing cosine similarity, only positive ratings are considered, and negative ratings are disregarded. The output value ranges from 0–1. A value close to 1 indicates that it exists a strong correlation between the two variables, while a value close to 0 indicates no correlation.
– *Pearson Correlation Coefficient (PCC)* PCC measures the extent to which two users are correlated, representing the linear correlation between them. This algorithm as a memory-based collaborative filtering method overcomes the problem of scalability by calculating the similarity between pairs of items rated by the users, or between pairs of users who rate the same item. PCC is a method explicitly created for estimation, where it calculates a predicted score for each item of interest. Although its primary purpose is estimation, this method is also employed for the recommendation task in literature [62]. In this context, it involves predicting a score for each potential

item and subsequently arranging the items based on decreasing predicted scores [62, 89].

- *Jaccard Similarity* Jaccard distance is a method of measuring the similarity between two sets by counting the number of items. It can be defined as a measure of similarity between returned recommendations by a RS. The similarity between sets A and B is determined as the ratio of the size of their intersection to the size of their union [13].
- *Tanimoto Similarity* The Tanimoto measure can be considered an enhanced version of the Jaccard measure. It considers the number of common items rated by the user and ignores the actual rating value provided by the users. The measure calculates the ratio between several items rated by both the users and a total number of items rated by either of the users [132].

*Classification accuracy metrics*

(1) *Accuracy* Accuracy metric indicates how close an amount is to its true value. It measures the accuracy of an ML model and expresses the percentage of correctly classified samples [35, 62, 63].

TN: Number of negative cases correctly classified
TP: Number of positive cases correctly classified
FN: Number of positive cases incorrectly classified as negative
FP: Number of negative cases correctly classified as positive

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

(2) *Precision* It represents the proportion of correctly predicted positive items among all predicted positive items [35, 63, 134].

$$Precision = \frac{TP}{TP + FP} \tag{5}$$

(3) *Recall* Recall is calculated by dividing the number of correctly predicted positive samples by the total number of true positive samples [62, 63, 76].

$$Recall = \frac{TP}{TP + FN} \tag{6}$$

(4) *Inverse Precision* This metric represents the probability that a non-recommended item is genuinely irrelevant. It is calculated as the ratio of truly irrelevant and not recommended items to the total number of all non-recommended items [19, 35].
(5) *Inverse Recall* This metric gives the probability that an irrelevant item is actually not recommended. It is calculated as the ratio of truly irrelevant and not recommended items to the total number of irrelevant items [19, 131].
(6) *Receiver Operating Characteristics (ROC)* ROC refers to the ability of an ML model to distinguish between positive and negative classes. Precision and Recall may not yield the desired results in certain. The ROC curve is particularly useful

for this evaluation [134]. It may be more accurate to use the Precision and Recall methods with the ROC curve, but both can be used separately. For example, if the most appropriate threshold value is determined to recommend a certain amount of news to a user, it may be more appropriate to create a graph and use the ROC curve [7, 71].

(7) *F1-Measure* If Precision-Recall is used with ROC curve, it would be more appropriate to use F1-Measure. Accuracy and F1-Measure can also be implemented as a separate metric [35, 76].

$$F1 = 2.\frac{Precision.Recall}{Precision + Recall} \tag{7}$$

(8) *Logistic Loss (Log-Loss)* Log-Loss is a well-known evaluation measure among binary classifiers. It is known as the negative log probability of the true labels given the estimates of a classifier. Log-Loss is occasionally used as the optimization goal in the context of Logistic regression and Neural Networks. It considers the uncertainty of your estimate based on how much it differs from the actual data. This gives us a more nuanced view of the performance of your model [111, 157].

(9) *Area Under Curve (AUC)* AUC is integral area under the receiver operating characteristic curve [35, 87, 157].

(10) *Specificity (SPE)* This metric, known as the true negative rate, displays true negative samples that were accurately predicted [62, 134, 155, 157]. It is calculated as the ratio of true negatives to the sum of true negatives and false positives.

$$Specificity = \frac{TN}{TN + FP} \tag{8}$$

*Rank accuracy metrics* It is a ranking accuracy or ranking prediction metric in statistic. In RS, an ordered list of recommendation is created for each user in the system, and traditional classification criteria such as precision in RSs are used to find out how many of these recommended items are relevant. However, precision may fail to capture the correct order or sequence of items in the list. Users have a limited attention span, and it is very important to recommend the right items at the top. Measuring the correlation of recommendation rank allows for predicting the correct order of recommended items based on user preferences. As a result, ranking metrics can be useful when a HRM receives a long list of relevant and recommended resumes [5, 99, 147, 150].

(1) *Precision@k or P(k)* Precision@k is a popular metric, which means "the number of relevant items recommended among the top k items". For instance, when you search for "plastic surgery" in a search engine and 7 out of 10 links on the first page are related to "plastic surgery", P@10 for that query equals 0.7 (Shervin, 2020).

$$Precision\ @k = \frac{(\#\ of\ recommended\ items\ @k\ that\ are\ relevant)}{(\#\ of\ recommend\ items\ @k)} \tag{9}$$

(2) *Average Precision (AP)* It is a measure of how well the system ranks relevant items in its recommendations and considers the precision at each relevant position in the ranked list. In the formula, k represents the rank in the position of retrieved items, n is the number of selected items, *P(k)* denotes the precision value at position *k* in the list. Additionally, *rel(k)* is an indicator function equal to 1 if the item at rank k is relevant and 0 otherwise, while $N_r$ stands for the total number of relevant items [133, 135].

$$AP = \frac{\sum_{k=1}^{n} P(k).rel(k)}{N_r} \tag{10}$$

(3) *Mean Average Precision (MAP)* MAP is the average precision calculated for each user and then normalized over all users. When there are K suggested items, P(k) is averaged over all data points from 1 to K [35, 133, 135].

$$MAP = \frac{\sum_{i=1}^{I} AP_i}{I} \tag{11}$$

$where, I is the total number of items$

(4) *Hit Ratio (HitR)* In an RS, the HitR is simply the fraction of users for which the correct item(s) is included in the recommendation list of length L. The larger L in the formula means the higher the hit rate. Because the correct item(s) has a higher chance of being included in the suggestion list. Therefore, choosing an appropriate value for L is an important step [163].

$$HR = \frac{\left|\cup_{hit}^{L}\right|}{|\cup all|} \tag{12}$$

where, $|\bigcup_{hit}^{L}|$ is the number of users for which the correct answer is included in the top L recommendation list, $|\bigcup all|$ is the total number of users in the test dataset.

(5) *Mean Reciprocal Rank (MRR)* MRR is a statistics-based metric and referred to as Average Reciprocal Hit Ratio (ARHR). It provides a relative score that gives the average/mean of the inverse of the ranks at which the first relevant item was retrieved for a set of queries. It doesn't apply if there are multiple correct items in the resulting list. For instance, if an RS retrieves 6 items as returned results and turns out there is a relevant item in the third-highest spot, MRR only takes this into account [36, 91].

$$\begin{aligned} MRR &= \frac{1}{\cup all} \sum_{u=1}^{|\cup all|} RR(u) \\ RR(u) &= \sum_{i \le L} \frac{relevance_i}{rank_i} \end{aligned} \tag{13}$$

*where, RR(u) is the reciprocal rank of a user u, and it is defined by the sum of relevance score of top L items weighted by reciprocal rank. MRR is simply the mean of all users in the test dataset.*

(6) *Normalized Distance-based Performance Measure (NDPM)* NDPM is an alternative ranking metric. It becomes applicable when users are not provided with a predetermined set of recommendations. In other words, if the user experience involves pag-

ination, where subsequent items are displayed sequentially or upon user request, NDPM can be employed [89, 160].

(7) *Gain* It is a metric used to assess the quality of rankings. Each recommendation produced by a RS has a validity score based on its relevance to the desired. For example, recommendations can be rated on a scale of 1 to 5, and a separate relevance score for each recommendation is determined. The Cumulative Gain (CG) is the sum of all graded relevance scores in a recommendation set [62, 134].

$$Cumulative\ Gain\ (CG) = \sum\nolimits_{i=1}^{n} relevance\ score_i \tag{14}$$

But one disadvantage of CG is that the order of the search results is not important, and the CG value may be the same even if the two separate listings have different relevance scores. Therefore, Discounted Cumulative Gain (DCG) has been proposed as a more precise measure. DCG is modelled to punish highly relevant items placed at the bottom [89].

$$DCG = \sum\nolimits_{i=1}^{n} \frac{relevance\ score_i}{\log_2 (i+1)}. \tag{15}$$

DCG seems to be a good measure initially as it considers sequencing importance; but it is still not enough. Depending on various factors, the number of recommendations offered may vary for each user. Accordingly, DCG will also change. Normalizing DCG allows the metric to be compared with ranking lists of different lengths.

$$NDCG = \frac{DCG}{iDCG} \tag{16}$$

In the above formula, iDCG stands for Ideal Discounted Cumulative Gain (iDCG). This metric is an extension of the DCG measurement and takes into account the position of relevant items in the ranked list. In the example below, we see that set B is ideally ranked. The ideal ranking is based on relevance scores from highest to lowest. The value of DCG for set B also gives its iDCG.

Finally, iDCG can be used in conjunction with the actual DCG to calculate Normalized Discounted Cumulative Gain (NDCG). Unlike DCG, NDCG can be used to evaluate datasets of different lengths. It is possible to assess the efficiency of RS solutions by calculating the average NDCG of the presented recommendations [62, 134].

The recommendation given sets A and B based on the relevance scores:

Set A = [2, 3, 3, 1, 2], Set B = [3, 3, 2, 2, 1], and the expected ideal ranking is [3, 3, 2, 2, 1]. Therefore, the CG, DCG and NDCG scores are:

$CG_A = 2 + 3 + 3 + 1 + 2 = 11$, $CG_B = 3 + 3 + 2 + 2 + 1 = 11$

$$\mathrm{DCG_A} = \sum_{i=1}^{n} \frac{relevance\ score_i}{log_2(i+1)} = \frac{2}{log_2(1+1)} + \frac{3}{log_2(2+1)} + \frac{3}{log_2(3+1)} + \frac{1}{log_2(4+1)} + \frac{2}{log_2(5+1)} \cong 6.6$$

$$\mathrm{DCG_B} = \sum_{i=1}^{n} \frac{relevance\ score_i}{log_2(i+1)} = \frac{3}{log_2(1+1)} + \frac{3}{log_2(2+1)} + \frac{2}{log_2(3+1)} + \frac{2}{log_2(4+1)} + \frac{1}{log_2(5+1)} \cong 7.14$$

$$\mathrm{NDCG_A} = \frac{\mathrm{DCG_A}}{\mathrm{iDCG}} = 6.6/7.14 = 0.92,\ \mathrm{NDCG_B} = \frac{\mathrm{DCG_B}}{\mathrm{iDCG}} = 7.14/7.14 = 1.0$$

where, since $\mathrm{NDCG_B} > \mathrm{NDCG_A}$, it can be seen that set B produces a more appropriate recommendation order when the ideal ranking is taken into account.

(8) *Spearman's Rho* It is a statistical criterion for ranking. [138]. It is a nonparametric measure that assesses the statistical dependence between the ranks of two variables [89]. In a monotonic relationship, if the value of the first variable changes, then the value of the second variable changes as well, but without a constant rate (is not linear). Compared to the PCC, the Spearman coefficient is rank-based and independent of non-parameters and size. PCC evaluates linear relationships between two variables, while Spearman evaluates their monotonic relationships [76].

*Non-accuracy metrics* These metrics don't compare the predictions against ground-truth values but instead evaluate the following properties of the recommendations.

(1) *Scalability* This metric refers to the ability of a system or model to maintain and cope well with increased workload or increased users' demands [89].
(2) *Novelty* This metric measures of how novel recommendation items are by calculating their recommendation frequency among users [89, 136].

$$\mathrm{nov(\ Ru\ )} = \sum\nolimits_{\mathrm{i} \in R_u} \min\nolimits_{\mathrm{j} \in H_u}^{\mathrm{d(class(i),class(j))}} \tag{17}$$

(3) *Diversity* This metric measures of how different items in a set are with respect to each other [12, 89, 134, 136, 163].
(4) *Serendipity* This metric measures how surprising recommendations are to a specific user by comparing them to the items that the user has already interacted with [12, 62, 134, 136].
(5) *Coverage* This metric measures related to the distribution of items recommended by the system [12, 113].

The above-mentioned evaluation metrics used in the RSs, their performance-comparative preference relationship, their abbreviations, and their scope are summarized in Table 1.

#### Challenges in setting up recommender systems

If an RS database is relationally and numerically weak, it may result in poor system performance for that RS. Therefore, it is logical to conclude that there are still many

**Table 1** Summary of the evaluation metrics reviewed in RSs

| Name | Symbol | Type | Pref | Scope | Rank | Length (L) |
|---|---|---|---|---|---|---|
| Mean squared error | MSE | Predictive | Small | Rating Accuracy | No | No |
| Root mean squared error | RMSE | Predictive | Small | Rating Accuracy | No | No |
| Mean absolute error | MAE | Predictive | Small | Rating Accuracy | No | No |
| Mean absolute user error | MAUE | Predictive | Small | Rating Accuracy | No | No |
| R squared | $R^2$ | Predictive | Large | Rating Accuracy | No | No |
| Explained Variance | EV | Predictive | Large | Rating Accuracy | No | No |
| pearson correlation coefficient | PCC | Predictive | Large | Rating Correlation | No | No |
| Accuracy | A(L) | Classification | Large | Classification Accuracy | No | Yes |
| Precision | P(L) | Classification | Large | Classification Accuracy | No | Yes |
| Recall | R(L) | Classification | Large | Classification Accuracy | No | Yes |
| Inverse precision | IP(L) | Classification | Small | Classification Accuracy | No | Yes |
| Inverse Recall | IR(L) | Classification | Small | Classification Accuracy | No | Yes |
| F1-Measure | F1(L) | Classification | Large | Classification Accuracy | No | Yes |
| Receiver operating characteristics | ROC | Classification | Large | Classification Accuracy | No | Yes |
| Logistic loss | LOG-Loss | Classification | Small | Classification Accuracy | No | Yes |
| Area under curve | AUC | Classification | Large | Classification Accuracy | No | No |
| Specificity | SPE | Classification | Large | Classification Accuracy | No | No |
| Precision @ k | P(k) | Rank | Large | Classification Accuracy | Yes | No |
| Average precision | AP | Rank | Large | Classification Accuracy | Yes | No |
| Mean average precision | MAP | Rank | Large | Classification Accuracy | Yes | No |
| Hit ratio | HitR | Rank | Large | Robustness | No | Yes |
| Mean reciprocal rank | MRR | Rank | Large | Ranking Correlation | Yes | No |
| Normalized distance-based performance measure | NDPM | Rank | Small | Ranking Correlation | Yes | No |
| Gain | G | Rank | Large | Satisfaction and Precision | Yes | Yes |
| Cumulative gain | CG (b, L) | Rank | Large | Satisfaction and Precision | Yes | Yes |
| Discounted cumulative gain | DCG (b, L) | Rank | Large | Satisfaction and Precision | Yes | Yes |
| Normalized discounted cumulative gain | NDCG (b, L) | Rank | Large | Satisfaction and Precision | Yes | Yes |
| Spearman's Rho | $\mathfrak{P}$ | Rank | Large | Rating Accuracy | Yes | No |
| Kendall's Tau | τ | Rank | Large | Rating Accuracy | Yes | No |
| Ranking score | RS | Rank | Small | Ranking Accuracy | Yes | No |
| Rank-biased precision | RBP (p, L) | Rank | Large | Satisfaction and Precision | Yes | Yes |
| Novelty | NOV | Non-Accuracy | Large | Popularity Ranking, Non-obviousness | Yes | No |
| Diversity | NA | Non-Accuracy | Small | Intra-list Similarity | No | Yes |
| Serendipity | SER | Non-Accuracy | Large | Novelty, Unexpectedness, Utility & Diversity, Non-Obviousness | Yes | No |
| Coverage | COV(L) | Non-Accuracy | Large | Coverage and Diversity | No | Yes |
| Hamming distance | H(L) | Predictive | Large | Intra-Diversity | No | Yes |
| Intra-similarity | I(L) | Predictive | Small | Intra-Diversity | No | Yes |
| Popularity | N(L) | Non-Accuracy | Small | Diversity and Novelty | No | Yes |
| Self-information | U(L) | Non-Accuracy | Large | Unexpectedness, Surprisal & Novelty | No | Yes |

The 3rd column indicates the type of metric, while the 4 th column indicates the preference of metric (e.g., smaller RMSE means higher rating accuracy). The 5 th column shows one of the categories in the evaluation metrics. The last two columns show whether the metric is obtained from a ranking and whether it depends on the length of the recommendation list (L)

problems with RSs that need to be solved. Some of these issues are addressed in this section.

(1) *Cold start* It refers to the difficulty of providing recommendations to a new user joining the system in an RS. It is among the primary issues with collaborative filtering in real-time systems. This can be avoided by showing popular items to that new user or by adding some rule-based filters [104].
(2) *Scalability* Scalability describes the capability of an RS to simultaneously process various types of data within a given period. As the number of users or items increases, the RS is expected to recommend the appropriate items to its users without degenerating into its performance and effectiveness. This challenge can be solved in several ways. For instance, when dealing with a very large data set, transitioning to big data systems such as Hadoop can be a solution. Another approach is to create models or rules on a sample set. Additionally, dividing the model into stages and running them is another potential solution [104].
(3) *Interpretability* In many sectors today, there is a high demand for the use of RSs and AI technologies. In several of these disciplines/sub-disciplines, it is crucial not only to achieve accurate predictions but also to provide a rationale behind the decision that can be transparent to end users. Therefore, a model integrated into the structure of RSs must be interpretable. Specifically, the model should elucidate the logic behind its predictions and offer insights into the underlying mechanism of user-item interactions. To address this, neural network architectures are among the choices to handle user-item interactions and preserve user-item properties. In addition, through the use of topology, it is possible to understand the individual contributions of the original input attributes to the decisions of the network, and by adapting on ideas of relevance propagation, which propagate backwards the relevance of the network function output through the network [56, 104].
(4) *Diversity* To ensure that the recommendation results generated by RSs cover as much of the product potential as possible and do not solely originate from the same group, it is essential to increase diversity [104].
(5) *Privacy* One of the major challenges in RSs is data privacy. To recommend suitable items based on the user's preferences/interests, RSs need to know some crucial information about their users in advance. While RS's keep this personal information of their users, user privacy must also be protected against unauthorized people [104].
(6) *Serendipity and Novelty* RSs can produce unusual recommendations. Not only can RSs produce unusual suggestions that the user has never heard of or been recommended before, but they can also have unexpected consequences [104].
(7) *Sparsity* Sparsity is another significant challenge encountered in RSs. This concern arises when a user has a very large matrix of preferred items (e.g., watched movies or songs listened to), and it occurs when the user does not rate his/her preferences. RSs rely on rating matrices to recommend users'preferences to others, making sparsity a notable problem in this context [7, 104].

The aforementioned issues will undoubtedly surface when RSs are incorporated into HR solutions. For example, when integrating a JRS into HR processes, ensuring the privacy of users who upload resumes to the HR system becomes essential. In addition to maintaining basic quality and performance criteria, preserving sensitive user data is crucial to protect against unauthorized attacks and maintain the trust of end users in the system. Privacy measures and robust security protocols can be implemented to create a secure environment for job seekers to upload and manage their resumes within the HR system. To achieve accurate matching results between job seeker(s)-job posting(s), HR professionals expect JRSs to be able to provide fast, diverse and reasonable recommendations. However, when it comes to the challenges mentioned above, it still seems that JRSs have not yet reached a sufficiently high level of maturity. Furthermore, with the rapid development of AI technologies, HR professionals or researcher believe that JRSs can efficiently match, sort, and tag uploaded resumes of job seekers with appropriate job descriptions [119, 137, 143, 151, 153, 159]. This associative linking and sorting can produce high volumes of rating matrices over time, enabling innovation and diversity in recommendations. According to the information we have obtained from our scientific studies and reviews, we can say that JRSs currently providing HR services are an active research area in terms of application, reliability, accuracy and the above-mentioned challenges and have not reached a fully mature level.

## Other similar literature review studies

In this section, the similarities and differences of existing SLR studies on the RSs and JRSs using a SLR are mentioned. While Sect."Survey Studies on Recommender Systems"analyses the SLR studies conducted on RSs in general and different fields, Sect."Survey Studies on Job Recommender Systems"analyses the SLR studies conducted on JRSs. Consequently, the evaluations and shortcomings of these SLR studies are discussed in Sect."Evaluations of the Survey Studies Reviewed".

### Survey studies on recommender systems

Guo et al. [64] conducted an SLR on RSs using Knowledge Graphs (KG), categorizing methods into embedding-based, connection-based, and propagation-based approaches. They explored various RS filtering methods, including Collaborative, Content, Hybrid, and KG-based, with a focus on KG-based systems. Two KG approaches, Item-based and User-Item based, were discussed. They summarized the issues addressed by each model, KG creation processes, and RS approaches. The study highlighted challenges in embedding-based methods and compared methods'advantages and limitations. Future research directions were proposed, but the evaluation methods, research questions, parameters, and tools were not discussed.

Portugal, Alencar, and Cowan [118] conducted a systematic literature review (SLR) to explore the use of ML algorithms in RSs, focusing on research questions related to their application. The study was divided into three sections: theoretical background, systematic review, and results. It categorized RSs based on AI algorithms, including collaborative, content-based, hybrid, trust-based, context-aware, and risk-aware recommendations. The ML section covered supervised, unsupervised, semi-supervised, and reinforcement learning. Out of 215 articles, 205 were included based on predefined

criteria, published between 2016 and 2018. Key findings highlighted MAE, Precision, Recall, and F-measure as common metrics, with coverage often used in ML. The study didn't mention the tools used and was limited to articles within the 2016–2018 period. Future work includes exploring SVM, ensemble methods, and clustering algorithms in RSs.

Lu et al. [88] conducted a literature survey focusing on recommendation techniques, as well as the current practices and applications of RSs. They categorized RS practices into eight main categories, encompassing areas such as e-government, e-business, e-commerce/e-shopping, e-library, e-learning, e-tourism, e-resource services, and e-group activities. Additionally, the researchers categorized filtering techniques of RSs, including collaborative, content-based, hybrid, computational intelligence-based, knowledge-based, social network-based, context awareness, and group recommendation filtering. The study delved into the practices and methods employed in RSs for each category, examining aspects such as RS software tools (e.g., BizSeeker), real-world application domains (e.g., e-business, e-tourism), and application platforms (e.g., mobile apps, Web portals). In their comprehensive review, the authors targeted other researchers and practical experts as their audience, aiming to disseminate recent developments in RS applications and methodologies.

Park et al. [115] conducted a literature review focusing on RSs, examining 210 selected articles published in 46 journals between the years 2001 to 2010. The researchers systematically categorized these articles based on factors such as publication year, journal source, application fields, and the specific data mining techniques employed. Additionally, the study encompassed eight application domains (books, documents, pictures, movies, music, shopping, TV shows, and others) and investigated the eight data mining techniques (association rule, clustering, decision tree, K-Nearest Neighbor (KNN), link analysis, neural network, regression, and other heuristics). The researchers not only classified but also analyzed these articles to provide a comprehensive understanding of the landscape of RS research during the specified period.

Martin et al. [95] published an article titled"The Big Promise of Recommender Systems"in AI magazine, focusing on the technological advancements and expertise of RSs. They analyzed vendors and insights from the first two waves and projected future trends for third-wave RSs. Key challenges discussed included integration, scalability, market competition, and the cold start problem. The authors emphasized that third-wave RSs should predict user behavior based on contextual preferences.

Bobadilla et al. [21] examined collaborative filtering methods in RSs, classifying application areas and trends while outlining pros, cons, and challenges. They analyzed 300 high-priority journal articles, evaluating key metrics (e.g., MAE, precision, recall) and RS evolution across three generations. Key challenges discussed included sparsity, cold start, scalability, and privacy. The study also highlighted ranking criteria, novelty, diversity, and reliability in RS evaluation.

Wu et al. [156] conducted an SLR on Graph Neural Network (GNN) techniques in RSs, categorizing them by information types and recommendation tasks. Reviewing 100 + articles, they identified five RS types: user-item collaborative, sequential, social, KG-based, and others. The study outlined GNN techniques, five GNN types,

and categorized user-item collaborative filtering methods. It also covered GNN applications in point-of-interest and multimedia recommendations, along with datasets and evaluation metrics, highlighting Precision, Recall, F1, NDCG, MAP, and AUC as key metrics.

Lü et al. [89] conducted a comprehensive review of RSs, covering research methodology, key RS products, and challenges affecting algorithm effectiveness. They explored fundamental RS concepts, evaluation metrics, and various recommendation algorithms, including similarity-based, dimensionality reduction (e.g., SVD, Bayesian Clustering), and diffusion-based methods (e.g., Heat Diffusion, Hybrid Spreading). Additionally, they categorized RSs based on external factors such as social relationships (e.g., trust-aware models) and temporal aspects (e.g., time-aware, tag-aware, hybrid methods).

#### Survey studies on job recommender systems

Vrontis et al. [148] conducted a SLR study on AI, robotics, and advanced technology-based HRM, examining 13,136 articles from databases like Business Source Ultimate and Science Direct. They selected 45 articles (24 empirical studies, 7 conceptual papers, 14 reviews) from 23 journals, employed a thematic analysis to develop a framework highlighting the impact of intelligent automation on HR processes. The framework encompassed three themes: AI, robotics, and other advanced technologies, for intelligent automation in HR. HRM was categorized into HR Activities (e.g. recruiting, training and job performance) and HR Strategies (e.g. organizations' decision-making processes, job replacement issues, enabling collaboration between robots, AI technologies and employees, offering learning opportunities for employees). In addition, the authors mentioned some of the limitations of their review and stated that they did not capture the full picture as they could only consider studies published in peer-reviewed journals. Lastly, the researchers mentioned the direction of their future studies as there may be valuable research studies published in non-peer-reviewed articles, books, and book chapters.

Freire and de Castro [55] conducted a SLR study focused on JRSs. Initially identifying 896 relevant articles, they narrowed down their study to 63 articles based on predefined inclusion and exclusion criteria and research questions. The selected papers, published between 2012 and 2020, were examined for their main approaches, including recommender type categories (e.g., content-based recommendation, collaborative filtering, knowledge-based recommendation, hybrid, and others), input types used, and an overview of the evaluation and development techniques applied. The study outlined the definition of each recommender type category, along with their main pros, cons, and challenges. However, specific details about evaluation metrics, assessment parameters, applied tools, and recently published papers were not provided in the review.

Azizi et al. [14] used a systematic review method to identify HRM challenges and strategies during the COVID-19 pandemic. They conducted a literature search across PubMed, Scopus, and WoS databases, gathering 797 articles. After removing duplicates, they screened 484 records and applied inclusion and exclusion criteria, ultimately selecting 15 articles for the final review. The focus was on HRM strategies for improving employee well-being, productivity, and motivation rather than recruitment processes. The study did not explore the structures, technologies, or evaluation methods of JRSs.

De Ruijt, and Bhulai [42] conducted a literature review of articles published on JRSs from 2011–2021. The researchers reviewed and discussed previously published databases on JRSs, CV generation systems, filtering techniques applied in JRSs, the selection of data method and its effect on the accuracy in JRS solutions, ethical considerations in JRSs, the effect of data availability on the choice of method and validation, and various proposed research studies on JRSs from reciprocal, temporal and ethical perspectives. The filtering techniques such as Collaborative, Hybrid, Content-based, Knowledge-based filtering approaches are discussed and compared. Furthermore, the researchers divided the large class of Hybrid RSs to obtain a better view on how these Hybrid RSs differ. The researchers classified the methods used in the Hybrid JRS as monolithic (e.g. model-based methods) and ensemble. The purpose of this was to obtain a new taxonomy on hybrid RSs by applying a multi-perspective perspective.

Hisham et al. [67] focused on job classification and categorization of job fields according to the skills in specific job descriptions. The researchers relied on the data, features and methods used in the classification of jobs. In this study, discussion and review of data used in classification, features, and various methods are mentioned. Firstly, information about the data was given. Accordingly, various areas in job classifications and JRSs were explained, for example E–recruitments/websites, social networks, students and graduates, applications (JRDP, FoDRA etc.). Secondly, in JRSs, the required features/criteria are mentioned for job description. Some features were highlighted such as CGPA, leadership, communication skills and experiences, age, gender etc. According to their results, CGPA is the most important feature for job classification. The last part of their study is methods. Methods are divided into 5 sections such as JRS, classifiers (Naive Bayes (NB), Decision Tree), Artificial Neural Networks (ANN), K-means clustering, KNN and are detailed. However, the researchers did not mention the evaluation method of the literature survey they conducted, the parameters used in the evaluation of the papers, filtering techniques, future work the tools used, and the recently published articles.

Brek and Boufaida [25] conducted a literature review on semantic-based JRSs. The researchers investigate the approaches of RSs, the comparative analysis of these approaches, the semantic methods and technologies used in JRSs, and the pros and cons of each technology mentioned in the literature. The authors select and review 41 articles in the literature to understand the contribution of the semantic technologies and approaches on the JRS solutions. However, for the article selection step, no information was provided about the publication date range applied, selection protocol, and other filtering details, etc.

Mishra and Rathi [103] conducted a literature survey to assess the efficiency and scalability of collaborative filtering methods in JRSs. Their study analyzed job seeker and job description profiling, comparing various algorithms (Casper, Proactive, Prospect) using datasets from CareerBuilder, Work4, and LinkedIn. The researchers identified common challenges in JRSs, such as sparse data, cold start, scalability, privacy, and item diversity. The researchers formulated six predefined research questions to guide their review and created a comparison table detailing the strengths, weaknesses, and drawbacks of different JRS approaches. Evaluation metrics like precision, recall, F1-score, Maximum Likelihood Estimation (MLE), Pointwise Mutual Information (PMI), and scalability

measurements were used to assess RS solutions' quality. Results indicated that Linear SVM outperformed cosine similarity, and Graph Based Approach (GPA) yielded the best results among SVM, KNN, and Deep Semantic Structured Models (DSSM) algorithms. The study also compared Matrix Factorization, Neural Network, KNN, NB, and SVM for various purposes. However, the study did not provide details about the future studies, well as information regarding the article search criteria and the number of articles reviewed.

Dhameliya and Desai [43] conducted a literature survey study on JRS, but they did not provide details on the article selection process, how many articles they reviewed, their research questions, inclusion and exclusion criteria, publisher details of articles included in the research, etc. Additionally, the authors did not consider newly published papers between 2017 and 2019. The researchers outlined their research approach and the benefits, drawbacks, and challenges of JRSs. In addition, the researchers mentioned the evaluation metrics, evaluation parameters, applied techniques, existing datasets used and the difficulties observed in the setup in JRS applications.

Tripathi et al. [143] provided an overview of JRSs and a partial review of recommendation techniques used in the field. They proposed an advanced job recommendation framework designed to handle big data, which includes the following steps: (1) transforming unstructured data into structured data using Hadoop, (2) preprocessing to remove noisy, duplicate, missing, and inconsistent data (Canonicalization), (3) clustering and association rules for reciprocal recommendations, (4) filtering results with personalized features, and (5) generating job and candidate lists for recommendations. However, the framework has not been implemented in practice, and empirical results regarding its accuracy, robustness, speed, and effectiveness are suggested as future research.

Al-Otaibi and Ykhlef [5] conducted a literature survey on JRSs, focusing on current solutions and electronic hiring procedures. They explored filtering categories such as collaborative, content-based, knowledge-based, and hybrid approaches, and compared them in terms of their application in e-recruitment systems. The study also examined system requirements, job recommendation information, and the general architecture model of JRSs. Similarity metrics like Cosine similarity, Euclidean distance, and the new Jaccard measurement were discussed. Each filtering category was evaluated in terms of its benefits, drawbacks, and challenges. However, the researchers did not provide specific details on aspects such as recently published articles, the publication date range, performance criteria, evaluation parameters, or development tools used in JRSs.

Siting et al. [137] explored the matching process between job seekers and job posts in RSs. The study outlined 3 stages: determining the target user, user profile analysis, and similarity identification. Resumes were segmented using Hidden Markov Modeling. Various similarity calculation methods like Constrained Pearson Correlation, PCC, Spearman Rank Correlation, Cosine Similarity, and MSE differences were discussed. The researchers categorized JRS recommendation technologies, including collaborative filtering, content-based, reciprocal, etc. Pre-implemented JRS systems like CASPER and Bilateral People-Job Recommender were investigated, but the study lacked a real application, detailed evaluation metrics, applied tools, and recent publications. Hybrid and knowledge-based filtering techniques in JRSs were not covered, and the study did not

discuss the benefits, drawbacks, and challenges of JRSs. In the next section, literature review studies conducted on both RSs and JRSs are examined and compared.

#### Evaluations of the survey studies reviewed

In this section, Table 2 compares the literature review studies on both RSs and JRSs mentioned in Sects."Survey Studies on Recommender Systems"and"Survey Studies on Job Recommender Systems"by considering various criteria such as review types, main ideas, article selection processes, tools applied, evaluation factors, future studies, and publication year for each study. According to Table 2 and the comparative discussion above, the literature review studies considered focused partially on literature review studies on RSs (8 articles) and JRSs (11 articles). Therefore, the missing parts in these studies were addressed in this study, and a more comprehensive SLR was conducted. The deficiencies of other review studies are listed below, and a more holistic and systematic review was conducted with this study:

- *Recent Articles Reviewed* Some of the studies did not address recently published articles. In contrast, our study includes 57 research papers published between January 1, 2010, and April 30, 2023, ensuring the inclusion of the latest developments in the field.
- *Future Enhancements* Some of the studies did not give any information about possible future enhancements to JRS solutions. However, our study discusses future research opportunities and suggests potential improvements to JRSs, including the adoption of innovative filtering techniques and advanced algorithms.
- *Evaluation Methodologies* Some studies did not explore all evaluation methodologies. While many studies focused on a limited set of evaluation methods, our study comprehensively examines a wide range of evaluation metrics, including predictive accuracy, classification-based accuracy, ranking-based accuracy, and non-accuracy-based metrics, contributing to a deeper understanding of JRSs evaluation.
- *Systematic Structure and Selection Processes* Some of the studies were not structured systematically, and article selection processes, as well as inclusion/exclusion criteria, were not specified. In our study, we present a clear, systematic structure and provide detailed information about our article selection processes and inclusion/exclusion criteria, enhancing the transparency and rigor of the review.
- *Comprehensive Classification* Some studies did not mention a detailed hierarchical classification of filtering mechanisms. This study developed a taxonomy based on the 19 literature reviews to categorize filtering techniques in RSs and JRSs. While collaborative, content-based, hybrid, and knowledge-based filtering are well-documented, demographic, reciprocal, popularity-based, and non-personalized filtering remain underexplored.
- *Simulation/Development Tools* Some studies did not provide explicit statistical information or details about tools for evaluating/developing JRS solutions. This study categorizes the tools used in JRS research based on filtering approaches. While collaborative, content-based, and hybrid filtering solutions frequently utilize frameworks like Spring and TensorFlow, tools for knowledge-based, demographic, and reciprocal filtering remain scarce.

**Table 2** Literature review studies have been conducted in both the field of RSs and JRSs

| Review type | Article | Publication year | Article selection process | Main topic | RS categories | Applied tools | Evaluation factors | Future studies | Covered year |
|---|---|---|---|---|---|---|---|---|---|
| JRS | Siting et al | 2012 | NA | A SLR is conducted on JRSs, and the researchers investigated matching criteria of the job descriptions to job seekers, recommendation technologies, and the architecture of a JRS solution | Collaborative, content-based, reciprocal filtering | Available | Similarity metrics (Cosine, PCC, Spearman Rank, MSE) | Available | NA |
| JRS | Al-Otaibi and Ykhlef | 2017 | NA | A SLR is conducted on JRSs, and the researchers investigated electronic recruitment processes through a set of existing JRSs | Collaborative, hybrid, content-based, knowledge-based filtering | NA | Similarity metrics (Cosine, Euclidean distance, New Jaccard) | Available | NA |
| JRS | Tripathi et al | 2016 | NA | Researchers provided an overview on the JRSs and proposed integrating an advanced job recommendation framework that handles big data | Collaborative, hybrid, content-based filtering | Available | NA | Available | NA |
| JRS | Dhameliya and Desai | 2019 | NA | A SLR is conducted on the benefits, drawbacks, and challenges of JRSs | Collaborative, hybrid, content-based, knowledge-based filtering | Available | MAE, RMSE, Precision, Recall, F1-score | NA | 2017–2019 |
| JRS | Mishra and Rathi | 2020 | NA | A SLR is conducted on the efficiency and scalability of collaborative filtering methods on JRSs | Collaborative, hybrid, content-based filtering | Available | Precision, Recall, F1-score, MLE, PMI, Scalability | NA | NA |
| JRS | Brek and Boufaida | 2020 | NA | A SLR is applied on semantic-based JRSs and 41 articles are reviewed | Collaborative, hybrid, content-based, knowledge-based filtering | Available | Semantic similarity techniques | Available | NA |

**Table 2** (continued)

| Review type | Article | Publication year | Article selection process | Main topic | RS categories | Applied tools | Evaluation factors | Future studies | Covered year |
|---|---|---|---|---|---|---|---|---|---|
| JRS | Hisham et al | 2021 | NA | Researchers classified jobs, used the classification methods, defined the data and features used in job descriptions for classification | NA | Available | Accuracy | NA | NA |
| JRS | De Ruijt and Bhulai | 2021 | Available | A SLR is applied on JRSs by considering CV generation, filtering techniques, accuracy assessment, ethical considerations in JRS solutions, to do these 87 articles are reviewed, and a new taxonomy of hybrid JRSs is developed | Collaborative, hybrid, content-based, knowledge-based filtering | Available | Similarity techniques | Available | 2011–2021 |
| JRS | Vrontis et al | 2022 | Available | A SLR is applied to HR activities (e.g. recruiting, training, job performance) and HR strategies and 63 articles are reviewed | NA | NA | NA | Available | NA |
| JRS | Freire and De Castro | 2021 | Available | A SLR is applied on e-recruitment RSs and 63 articles are reviewed | Content-based, collaborative, knowledge-based, hybrid, and others | NA | Expert validation, ML metrics, challenge metrics (e.g., user success), utility- metrics (e.g. click- rates) | Available | 2012–2020 |
| JRS | Azizi et al | 2021 | Available | The challenges facing HRM in COVID-19 (e.g., employees' mental well-being, satisfaction, productivity, motivation) are discussed and 15 articles are reviewed | NA | NA | NA | NA | 2020–2021 |

**Table 2** (continued)

| Review type | Article | Publication year | Article selection process | Main topic | RS categories | Applied tools | Evaluation factors | Future studies | Covered year |
|---|---|---|---|---|---|---|---|---|---|
| RS | Guo et al | 2020 | NA | A SLR is applied to KG based RSs | Collaborative, content-based, hybrid, KG-based filtering | NA | High meta-path-based similarity-based (user-user, user-item) | Available | NA |
| RS | Portugal et al | 2018 | Available | Several ML algorithms are discussed on RSs and 121 articles are investigated | Collaborative, content-based, hybrid, trust-based, context-aware, risk-aware | NA | MAE, Precision, Recall, F1-score | Available | 2016–2018 |
| RS | Lu et al | 2015 | NA | A SLR study is conducted on RSs, taking into account eight main filtering categories and 177 articles are investigated | Collaborative, content-based, hybrid, computational intelligence-based, knowledge-based, social network-based, context awareness, and group-based filtering | Available | Similarity metrics | Available | NA |
| RS | Park et al | 2012 | Available | A SLR study is conducted on RSs and 210 articles are reviewed. These articles are categorized into 8 application fields and 8 data mining techniques used | Collaborative and content-based filtering | NA | NA | Available | 2001–2010 |
| RS | Martin et al | 2011 | NA | RSs' technological advances and know-how so far | Examining the 3 main waves of collaborative filtering and content-based RSs that have changed and evolved over time | NA | NA | NA | NA |

**Table 2** (continued)

| Review type | Article | Publication year | Article selection process | Main topic | RS categories | Applied tools | Evaluation factors | Future studies | Covered year |
|---|---|---|---|---|---|---|---|---|---|
| RS | Bobadilla et al | 2013 | Available | Algorithms used in collaborative RSs | Collaborative, content-based, hybrid, and demographic filtering | NA | Quality of prediction-based (MAE, RMSE, Coverage), Quality set of recommendations (Precision, Recall, F1), Quality list of recommendations (HL, DCG), Novelty, Diversity, and others | Available | NA |
| RS | Wu et al | 2022 | Available | GNN-based RSs | User-item collaborative filtering, sequential graph-based, social graph-based, KG-based, and others | NA | Precision, Recall, F1, NDCG, MAP, AUC | Available | NA |
| RS | Lü et al | 2012 | NA | Algorithms used in RSs are based on traditional similarity approaches, dimensionality reduction techniques, and diffusion-based methods | Collaborative, content-based, hybrid, and social network filtering | NA | Accuracy Metrics (MAE, RMSE, PCC, NDPM, AUC), Rank-Weighted (HL, DCG, RBP), Diversity & Novelty (Distance-based), Coverage (COV(L)) | Available | NA |

*SLR* Systematic Literature Review, *NA* Not Available

By addressing these gaps, our study provides a more holistic and systematic review of both RSs and JRSs, contributing to a clearer understanding of the field and offering insights for future research and development. In the next section, we will discuss the research methodology and review processes applied in examining and identifying JRS solutions from the literature over the past 13 years in this SLR study.

## Research methodology

### Research methodology selected

Aromataris & Pearson [11] described the features of a well-defined and internationally accepted SLR. In our selection of SLR methodology, we applied an alternative SLR methodology approach proposed by Aromataris & Pearson [11] instead of the well-known PRISMA framework [120]. There are several reasons for preferring the Aromataris & Pearson methodology over the PRISMA framework. The Aromataris & Pearson methodology provides a comprehensive framework that addresses all stages of the systematic review process in detail, from research motivation to reporting.

On the other hand, the PRISMA framework (four-phase) focuses on the reporting and transparency phases of systematic reviews but does not cover the entire review process in detail. PRISMA's focus may not provide sufficient detail for a specific review process. The Aromataris & Pearson methodology (six-stages) offers a deeper and methodologically rigorous approach by addressing all critical aspects of the review process. It includes the stages of review protocol development, rigorous data extraction, and synthesis, which are key to ensuring the accuracy and reliability of the review. It also provides clearer guidelines for transparency and reproducibility, as it emphasizes not just the results, but also the entire process, including potential biases and limitations, which is crucial for building trust in the review outcomes.

Therefore, by adopting the SLR methodology of Aromataris & Pearson [11] in our study, we ensured a comprehensive and methodologically rigorous approach. The research methodology begins with identifying the research motivation for the systematic review and concludes with reporting the results obtained [11]. The key steps followed in this approach are:

*Stage 1–Determining the motivation of the research phase* In first step, the research objectives are clearly stated and the questions to be addressed through a preliminary study should begin to be defined.

*Stage 2–Specifying research questions phase* Research questions are defined in the second stage of the review study and are effective in validating and evaluating the review process, determining inclusion and exclusion criteria in filtering step. The aim is to find answers to these questions and to identify the current challenges in the performed study. It can also help identify innovative ideas and solutions in this regard. The research questions identified for our study are given in the following section.

*Stage 3–Establishing a search protocol phase* The research questions and the scope were obtained in the prior stage to define search terms and search directives in the SLR study.

*Stage 4–Paper selection phase* This phase starts with paper selections, followed by data extraction. The purpose of this phase is to describe the process of searching and selecting articles. At this stage, the quality criterias are determined in studies that should be included, the validity of the articles is evaluated, and reported if a quality-based exclusion has been made.

*Stage 5–Data extraction and synthesis phase* Using a systematic research guideline [11] the research studies are obtained from online academic databases and are expected to be presented in a systemic format according to their characterization features. A structured comparative analysis examines the limitations and potentials of the studied papers, as well as their implications for future work, enabling final research findings to be explored.

*Stage 6– Reporting phase* Transparent reporting of the methodology is ensured, after reporting the observations, threats and limitations to validity are discussed.

## Research methodology applied

### *Determining the motivation and specifying research questions (Stage 1 & 2)*

The motivation of our research, comparing and interpreting the existing JRS-focused literature reviews presented in the previous section, posed the following research questions. The next section discusses the steps considered in our systematic search protocol, the rationale for our SLR, and a comprehensive description of the methods applied.

RQ1: What are the approaches and methodologies applied in the JRSs?

RQ2: What are the main ideas focused on in JRSs?

RQ3: What are the quality and performance measurements applied in the evaluation of JRSs?

RQ4: What are the popular evaluation and modeling tools used in developing JRSs?

RQ5: What are the current challenges and future improvements in JRS implementations?

RQ6: What are the available standards applied in the JRSs?

### *Establishing a search protocol (Stage 3)*

In this study, a systematic search protocol was applied in the following order: (1) a research question and objective was identified, (2) inclusion/exclusion criterias (Table 3), article types, publishers, and interventions planned were identified, (3) a list of databases to be searched was chosen and irrelevant databases were eliminated, (4) a strategic search string was executed using a number of different relevant keywords or search terms accompanied by *and/or* logical operators, (5) data extraction and screening were provided according to a predefined publication year interval, and (6) the search protocol was completed after reporting results obtained.

**Table 3** Applied inclusion/exclusion criteria in our research

| Id | Inclusion Criterias | Exclusion Criterias |
|---|---|---|
| 1 | All studies related to JRSs that mention available and applied solutions, evaluations, or experience | Studies without or not found/available full text |
| 2 | Studies that are written in English | Studies that do not fit predefined research questions |
| 3 | Studies that are published between January 1, 2010 and April 30, 2023 | Survey papers, books, book chapters, editorial materials, unreviewed articles, dissertations, and short articles |
| 4 | Studies in related scientific categories such as Computer Science, Information Systems, Management, AI, Engineering Electrical Electronic, Interdisciplinary Applications, and Software Engineering | All published work outside the fields of Management and Computer Science (e.g., health, library, social media, etc.) |
| 5 | Studies **are published** online and easily accessible in popular/relevant academic databases such as Elsevier, Springer, Taylor & Francis, IEEE, Wiley, Sage, Emerald, Oxford, ACM, Hindawi, and IoS Press | Non-peer-reviewed research articles, non-English articles, and white papers |

***Papers selection stage (Stage 4)***

This section outlines the search process for identifying relevant research papers in this SLR. To ensure comprehensive and unbiased article selection, a strategic search query was formulated using search keywords and logical operators. Web of Science (WoS) academic database was used for the following search query in the papers' titles and author keywords: **"((Recommender < OR > Recommendation < OR > Recommending < OR > Matching < OR > Decision Support < OR > AI) [AND] (Job < OR > Skill < OR > Job Opportunities < OR > Recruitment))"**.

As shown in Fig. 4a, the search query is executed to find related papers containing the search terms selected in the titles of papers and author keywords in the online academic databases on the WoS. In addition, we considered only online articles published between Jan 2010 and September 2023. A total of **527** articles were returned regarding the query as seen in Fig. 4a. As a next step, **11** popular online academic databases such as Elsevier, Springer, Taylor and Francis, IEEE, Wiley, Sage, Emerald, Oxford, ACM, Hindawi, and IoS Press were chosen to filter some (as seen in Fig. 4b). In total, **360** papers were returned as relevant papers. The **360** papers were searched and continued to be screened according to inclusion/exclusion criteria. In this step, only papers in English (**356**) were selected (as shown in Fig. 4c). In the next step, paper types such as survey papers, books, book chapters, editorial materials, unreviewed articles, dissertations, and short articles were excluded, and **328** relevant papers were obtained (as shown in Fig. 4d). As shown Fig. 4e, WoS research categories such as Computer Science Information Systems (35), Computer Science Theory Methods (28), Management (28), Computer Science AI (25), Engineering Electrical Electronic (16), Computer Science Interdisciplinary Applications (13), Computer Science Theory Methods (28), Computer Science Artificial Intelligence (25), Business (10), Computer Science Software Engineering (8), Social Science Interdisciplinary (8), and Telecommunication (8) were selected and finally, **111** relevant articles remained (Fig. 4f). The **111** papers were then reviewed as full text. Of the remaining **111** articles screened in WoS, **54** articles were selected for our further review, which clearly covered the details of their review and the techniques they applied in the JRS field. In addition, **3** different publications were found on the Google Scholar search engine using our search query that were not screened on WoS, which were closely linked to our

**Fig. 4** The inclusion/exclusion criterias applied to the search query. **a** 527 research papers are found related from Web of Science Core Collection (WoS) after executing our search query: "*((Recommender < OR > Recommendation < OR > Recommending < OR > Matching < OR > Decision Support < OR > AI) [AND] (Job < OR > Skill < OR > Job Opportunities < OR > Recruitment))*", **b** 360 results returned from WoS after selecting some of online academic databases, **c** 356 results obtained after selecting English articles, **d** 328 results obtained from WoS after selecting articles, proceeding papers, and early access, **e** finally, 111 results retrieved after selecting a set of most related WoS field categories. Link: https://www.webofscience.com/wos/woscc/summary/ba50afde-acbb-4324-bf7a-2cf6d9a9ebf3-a7a2e1b1/relevance/1

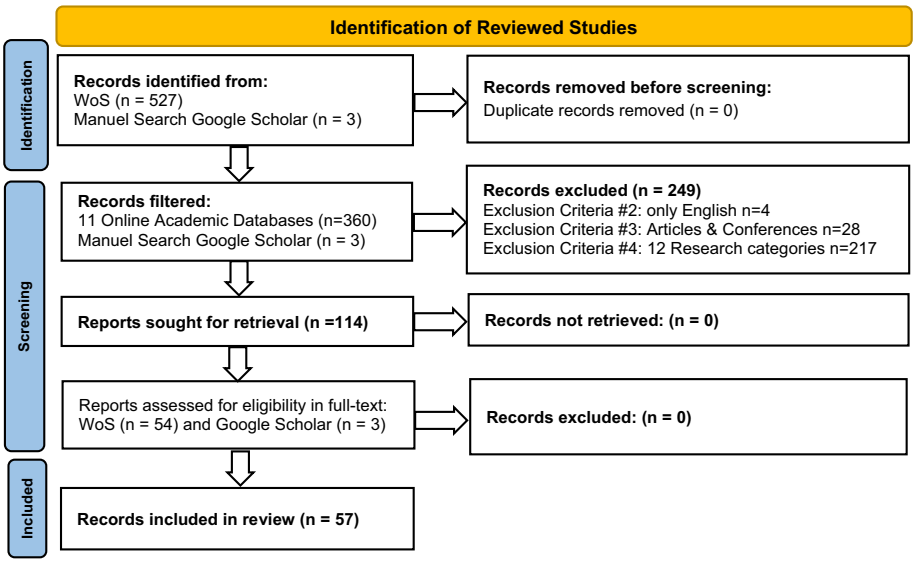


**Fig. 5** The flow diagram of the article selection process applied in this SLR

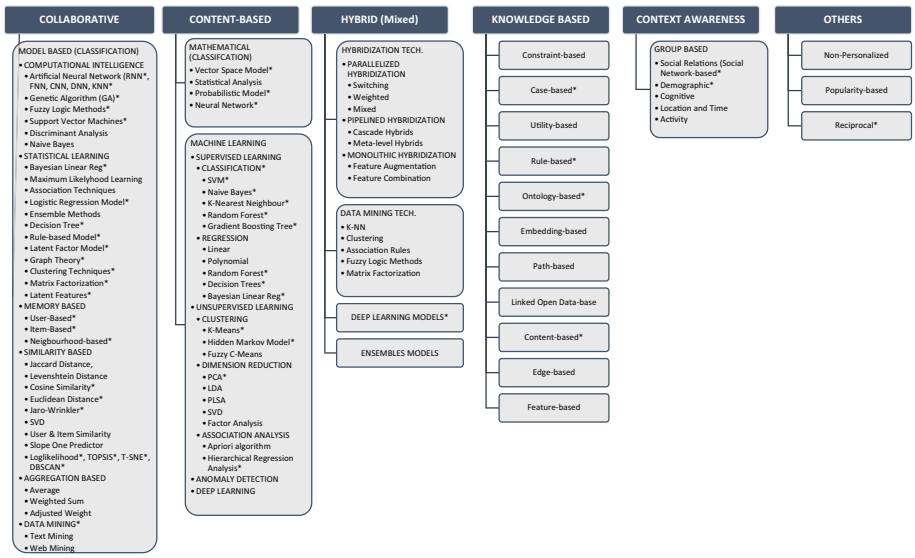


**Fig. 6** A taxonomy has been developed to categorize methodologies applied in the fields of RSs and JRSs. *Note: In addition to being applied in RSs, techniques tried and tested in JRS solutions are represented by '*'*

research topic and full content. As a result, all retrieved articles were evaluated qualitatively and **57** related articles on both WoS and Google Scholar academic databases in total were included in this SLR. Figure 5 illustrates the flow diagram of the article selection process applied in this research.

#### *Data extraction and synthesis and reporting (Stage 5 & 6)*

Following the outlined guidelines, we systematically collected and analyzed relevant WoS articles based on their characteristics. A structured comparative analysis identified limitations, potentials, and future research directions. Selected papers were rigorously reviewed, highlighting applied techniques, open issues, and challenges in JRS implementation. The findings are presented in comparative tables in the following sections.

## Review of the selected studies on JRSs

According to our findings from the literature review articles reviewed in Chapter 3 and many other related research studies, the taxonomy of filtering techniques applied across RS solutions is shown in Fig. 6. Basically, six filtering types in RSs has been determined and categorized: (1) Collaborative filtering, (2) Content-based filtering, (3) Hybrid based filtering, (4) Knowledge based filtering, (5) Context Aware filtering, and (6) others [6, 28, 63, 89, 95, 104, 112, 115, 117, 134]. Although the taxonomy developed in Fig. 6 is associated with these six categories, it is possible to say that there may be other taxonomies in the literature.

As outlined in Chapter 4, the 57 relevant articles selected in the"Paper selection stage"were examined in detail in the following subsections. All selected articles belonging to JRS solutions were examined and summarized in Sects. "Collaborative-based filtering JRSs" to"Other Types of JRSs"below according to the six

recommendation filtering categories, and a comparison of filtering techniques was made. Furthermore, Fig. 6 provides information on which of the wide range of techniques implemented in RSs have also been implemented and tested in JRSs.

In Fig. 6, the symbol'*'is utilized to indicate techniques commonly applied in the literature, encompassing both general RSs and JRSs. The taxonomy derived from our SLR study can facilitate easier categorization of articles on RSs in the literature, making the methodological investigation of studies covering subfields of RS solutions, such as JRSs, more accessible.

#### Collaborative-based filtering JRSs

Collaborative filtering-based RSs assume that users with similar choices and behaviors will make similar choices. These systems typically analyze existing data regarding the user's behavior/preferences (e.g., analysis of a job candidate's characteristics and preferences). It compares other similar users'preferences, then produces appropriate recommendations for that user (e.g., generates recommendations based on the characteristics of previously matched individuals with similar backgrounds). It recommends users who have made similar consumptions before will be able to consume similarly in the future. Thus, it can deduce suitable recommendations (e.g., suitable open job postings) through collaborative matching, even with elements that are complex or difficult to analyze (e.g., a resume). Collaborative filtering recommendations typically generate recommendations using the user-item rating matrix (e.g., users – resumes). This filtering can be divided into user-based and item-based.

(1) *User-based Collaborative Filtering* It creates recommendations by trying to establish relationships among similar users (e.g., similarity between resume holders). Even with an efficient algorithm that can handle multidimensional data, it requires a lot of data, which means it takes a long time for the JRS to work efficiently. Some of the studies based on this approach are discussed in these studies [28, 47].

(2) *Item-based Collaborative Filtering* Similar to the previous methodology, it focuses on similarities between items rather than between users. Item-item filtering analyzes product similarity based on users' ratings. A click/view rather than scoring can also be considered a variable. The purpose of this filtering is to ensure that similar items are recommended to other similar users by considering the similarity of the products that the user bought/viewed. In this methodology, the products (items) are considered as neighbors, and their similarities are found by similarity methodologies such as Pearson Correlation or Cosine Similarity. This type of filtering requires less data than the user-user filtering as it is based on item analysis [22, 28, 61, 161]. In this filtering model, item-based methods are suitable for offline calculation because it is easier to find similar items (e.g., items can be user resumes uploaded or job postings) than to find similar job seekers (e.g. resume holders), and the attributes of items (e.g. a particular segment in a resume such as user profile, work experiences, educational background, etc.) are more stable than users'preferences (e.g. job opportunity criteria defined by HR staff such as has driver license, married/not married, has at least 5 years of Java experience, etc.).

***JRS approaches and practices reviewed on collaborative filtering***

Collaborative-based approaches—comprising memory-based (e.g., user-based, item-based, and neighborhood-based methods), model-based (e.g., Bayesian models, neural networks), similarity-based (e.g., syntactic analysis), aggregation-based, and mining-based techniques—are frequently discussed in the literature and are systematically categorized in Fig. 6. The recent research studies outlined below have aimed to address JRS solutions by using these collaborative approaches.

Kwieciński et al. [81] proposed a JRS infrastructure that can provide real-time job recommendations and investigated the process and efficiency of producing real-time job suggestions using the RP3Beta model on this infrastructure. They developed the OLX application in 2 versions, batch and real-time, using the RP3Beta model. They conducted an online A/B test to compare batch and real-time RP3Beta approaches (for 200,000 OLX users). The results were that the real-time RP3Beta model increased the number of users responding to recommended job postings by at least 10%. The results show that real-time RSs can guide organizations in making informed decisions about whether to make an effort to move from a batch environment to a real-time recommendation environment. The researchers did not mention future plans for the current system that they proposed.

Aouadni and Rebai [9] introduced a "GMUSA System", which is a Decision Support System (DSS) and helps organizations evaluate and measure employee job satisfaction in instead of job recommendation. DSS is modeled based on MUSA method and Continuous Genetic Algorithm (CGA). Researchers proposed a new algorithm called GMUSA, which combines CGA and MUSA method to obtain an effective solution with good performance. The MUSA method has made significant contributions to preference discrimination modelling. It was initially developed to measure customer satisfaction and analyze customer needs [60]. Since the MUSA method depends on continuous variables, researchers preferred the CGA method. The system was tested at the University of Sfax to measure the job satisfaction of 100 instructors through a survey with 8 criteria. A five-point Likert scale was used to evaluate job satisfaction for each criterion and sub-criterion. The researchers evaluated the results based on several values, including: (1) response frequencies in each criterion, (2) global satisfaction index, (3) weights of each criterion, (4) satisfaction index, (5) demand index. The average global satisfaction index was identified as very low (28%) because instructors were dissatisfied with the most important criteria such as salary (30.4%) and resources offered (40%). The findings show that, in contrast to the MUSA approach,"Salary"is the most significant criterion in the GMUSA method—rather than"Student Interaction".

Dorn et al. [47] stated that creating a productive team of employees requires a balance between maintaining a sufficiently high skill range and the interconnectedness of sufficient team members. The researchers proposed an approach, which uses two suitable heuristics methods such as Genetic Algorithm (GA) and Simulated Annealing to address the multi-objective and appropriate team composition. The aim of the developed approach is to achieve team configurations that provide desirably low team distance while maintaining consistently high skill coverage. Relations between experts in this model typically emerge based on previous interactions in context of collaborative tasks. Researchers focused on 3 main contributions in their study: (1) Team Composition

Discovery Heuristics, (2) Development of the Skill-Based Suggestion Model, and (3) Development of the Self-Adjusted Swap Model. Researchers have found that using both methods can lead to significantly better solutions than a simple group of top experts. In addition, comparative studies have shown that GA produces better and more stable results, especially for small networks.

Boccuzzo et al. [22] discussed the interaction between graduates' job potential and their educational performance and social background. Researchers have explored the correlation between work-study mismatch and job satisfaction, aiming to evaluate whether the time and effort students invest during their college semesters are adequately rewarded with a satisfactory job. In their study, the data collected from a group of graduates (2046 graduates) 36 months after their graduation were analyzed using univariate and multivariate methodologies, and a structural equation modeling from graduates' social capital to job satisfaction was attempted. Analysis studies were conducted on the answers of the participants in a longitudinal questionnaire based on three predefined research questions. For example, this survey includes many questions regarding general job satisfaction. Additionally, survey respondents were asked to rate their level of satisfaction with various specific aspects of the job. Accordingly, during the evaluation phase, firstly the results regarding the levels of harmony between work and education were evaluated. Researchers then evaluated pathways to job satisfaction, moving from graduates'social capital to high school and college performance. Direct, indirect and total standardized effects of graduates'characteristics on job satisfaction were evaluated. The researchers did not present any JRS and used filtering approach in their research in the matching between the job and graduates.

Maurya and Telan [98] introduced two novel unsupervised Bayesian Multiview models, namely BayesMatch and NpBayesMatch. These models were designed to match members with similar members and correlate jobs based on shared features. They also proposed a collaborative filtering-based skill RS that exploits the gap between skills that members supply and those that the labor market demands. The skills developed by the RS can help its members acquire the skills demanded by the labor market and thus achieve strategic advancement in their professions. The researchers collected the data of 414 million profiles, 58 million jobs, 29 thousand skills, 7.7 million companies, 98 thousand unique job titles, 95 thousand unique schools, 158 unique degrees, and 7.6 thousand unique majors from LinkedIn. Their evaluation with cosine similarity showed that personalized skill recommendations made by their skill RS using skill demand can be indicatory of skills that remain in high demand in future jobs. In future studies, researchers have planned to use spectral approaches that afford guaranteed parameter recovery for latent variable models.

Guan et al. [61] designed a JRS that brings job seekers and businesses together on a common platform in the Web and mobile environments. The main aim of the JRS is to process and rank the resumes of job seekers and businesses using fuzzy mathematics. The JRS proposed involves 2 modules: the capability module and the enterprise module. The first module is divided into sub-modules such as the quantification module, job-seeking intention module, community discussion module. The enterprise module is divided into sub-modules such as the enterprise culture module and the talent

recommendation module. Using this JRS, job seekers can find more suitable jobs and also enable enterprises to find more qualified talents.

Ghosh et al. [57] have developed a novel and interpretable monotonic nonlinear state-space model that can analyze online user profiles, provide feedback and suggestions to users to improve their career goals. In their inference method, the researchers used a series of binary-valued and non-decreasing latent states representing each user's skill set that expands and changes over time. In their experimental studies, 2 large real-world datasets were used. It was claimed that their proposed model produced better results than other available methods in the tasks of company, job title, and skill prediction. The system is recommended as a tool in the career path planning process for the users who want to improve their careers. The researchers conducted a series of case studies in experimental studies, and they exemplified that their model suggests appropriate paths for a set of users who aim to reach their career goals.

Okfalisa et al. [108] selected the prisoners as the user group who wanted to receive job education in prison. They have developed a JRS to help the users find the appropriate training activity, taking into account some criterias such as age, recent education, work experience, abilities and interests. The researchers developed an integrated multi-attribute decision making Fuzzy Analytical Hierarchy Process (Fuzzy-AHP) system which is designed as a decision tool and compromises the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS). It was developed as a ranking tool while using the Fuzzy-AHP weighting criterion. According to user preferences, various educational job offers are offered, such as cooking, sewing, construction, furniture, farming and animal husbandry. 50 prisoners were successfully offered various training jobs in experimental studies.

Jarrett and Blake [74] proposed an interoperability standard and computation method that supports scalability and facilitates the collection of ever-increasing jobs data entered into systems, capable of providing an automated recommendation of workers.

Yi et al. [161] have developed an item-based collaborative filtering algorithm for JRS solutions. The aim is to find suitable job postings that meet the job seekers'needs and make it easy for them to reach their goals without using just the right keywords. Researchers combined the position explanations and users' resume information in their proposed algorithm and then optimized it. The algorithm uses an optimized job preference prediction formula which takes into account historical delivery weight calculated by the position explanations and similar user weight derived from resume information. The historical delivery weight calculated with the position descriptions on the resume and the similar user weight calculated with the person's resume information were added as two impact factors in the preference estimation process. The researchers tested their optimization method with a real-world dataset to compare it with the original algorithm and stated that it performed significantly better.

Ortega et al. [109] have studied the applicability of the Learning from Interpretation Transition (LFIT) method on an AI application scenario by using Inductive Logic Programming (ILP) and LFIT techniques, which are sub-fields of AI. To this end, LFIT processes fair recruitment soft biometric information (gender and ethnicity) based on an automated tool built with ML methods to incorporate accurate declarative annotations into classical ML and resume ranking. The researchers aimed to present the

expressiveness of LFIT for the recruitment domain and have proposed a scheme that can be applicable to other domains (e.g. e-health, e-learning, etc.).

Prafajar et al. [119] have presented a comparison research study of the methods used in the JRSs in the field of Information Technology. Researchers first proposed an approach that includes classification with the KNN algorithm and prediction processes with SVM algorithm. In experimental studies, an accuracy assessment process was carried out with Google Collab on a dataset containing job postings. The results of the classification model based on KNN algorithm, RBF model of DVM and Poly model of DVM were compared. As a result of experimental studies, it has been predicted that the model using KNN will provide better performance in JRS solutions than the model using SVM.

Wang et al. [151] addressed the advantages and disadvantages of the models implemented in three existing online JRS solutions. In addition, the researchers proposed a new JRS model based on Gradient Boosting Regression Tree and Time Factors (T-GBRT). This model aggregates time factors in GBRT to predict personal preferences. It then adds time factor weighting to top K rankings with a neighbor-based filtering trick that reduces the amount of computation. The researchers applied a four-criteria experiment to compare their newly developed model with the other three models, and they showed that the newly developed model gave the best performance, despite the others. In conclusion, the proposed model integrates the interaction history, neighbor, time and performs better in each criterion.

Upadhyay et al. [144] proposed a JRS that can match users'profile to the most relevant job postings and state the reason with a textual explanation after the matching task. In the proposed system, KG structure was applied in modeling the pairs (job seeker and job posting) to create a single homogeneous structure in the matching process. Graphical relations between the job seekers and job postings were obtained through NLP techniques of text content from job postings and user profiles. In addition, every recommendation extracted by the system through NER is presented to the user in the form of a human-readable textual explanation. The explanation consists of the processed factors and their values in the matching process. The explanations are a direct result of the embedded contextual knowledge about users and job-postings in the KG. The system involves 4 main steps: (1) data collection and processing, (2) document embedding generation through NLP techniques, (3) knowledge graph construction, where job-postings and user profiles are modeled to infer user recommendations, (4) explanation construction, where a customized named entity classifier is applied to obtain the values for the user textual explanation. In experimental studies, researchers prepared a dataset consisting of user profile information and job postings gathered from online social networking job recommendation platforms and tested the proposed system on their dataset. BELU and Rouge-L scores were used to generate the explanations of users.

In the context of identifying and recommending appropriate job opportunities for job seekers, leveraging artificial intelligence and Semantic Web technologies to match user-entered data on job search platforms with the criteria outlined in job advertisements can significantly enhance the effectiveness of this process. This approach diverges from traditional methods by necessitating data processing that encompasses both syntactic and semantic dimensions. For instance, research by Çelik and Elci [33] has provided substantial insights into ontology-based matching and the integration of business processes,

which are applicable to HR automation services. In a related study, Çelik and Elci [31] concentrated on the discovery and evaluation of semantic web services based on user data through a semantic search agent. Further studies in the literature also explore semantically enhanced concept matching and inferencing systems [30, 32, 114].

Moreover, the integration of RSs with Semantic Web technologies has been proposed to enhance the efficacy of JRSs. Bernabe-Moreno et al. [18] introduced an innovative approach to standardize job information and subject specializations by employing collaborative filtering and semantic techniques, alongside a method to structure the inference of the tasks and skills required for specific occupations. Additionally, word embedding methods have been utilized to analyze the skills listed in existing job postings, with semantic matching techniques applied to meaningfully contextualize these skills. This methodology has been implemented to evaluate the complexity of professions, such as data scientists, by analyzing job titles, key phrases, and frequently used terms. Future research aims to investigate the evolution of skills, and the scope associated with identified job roles.

Saito and Sugiyama [125] proposed a system designed to display job postings and assist users in finding relevant employment opportunities. The system generates job recommendations based on individual user behavior data. The study utilized real data from the job search website glit.io, employing metrics such as Mean Reciprocal Rank (MRR) and normalized Discounted Cumulative Gain (nDCG) to evaluate system performance. In their approach, users'preferences and behavioral data were graphically represented within the model through metric learning. The authors introduced a learning method specifically designed for handling overlapping multiple behaviors, assessing its effectiveness on the dataset. The proposed model includes modules such as Multi-Behavior Learning (MBL), Explicit Preference Learning (EPL), Recommendation (comprising Multi-Behavior Recommendation and Graph-based Recommendation), as well as Training and Inference, and Time Complexity. Evaluation results indicated improved recommendation accuracy, although occasional irrelevant suggestions were remarked as a limitation. The authors acknowledged the need for refinement, particularly in handling negative behaviors such as dislikes or hides, where the proposed method showed room for improvement.

As a result, collaborative filtering-based JRSs are traditionally collectively trained and designed to generate personalized recommendations for large numbers of JRS users at the same time. Additionally, in many industries, the aim is to generate real-time recommendations, taking into account the latest user interactions [81]. For the collaborative methods, the estimation and classification criteria generally are based on a matching score, ranking users/skills or binary output predictions.

*An overview and comparison of collaborative based JRSs* In Table 4, the objectives, evaluation methods used, development methods and tools used, advantages and disadvantages of collaborative based JRSs of the reviewed articles are examined and compared. Also, according to the articles reviewed above on collaborative based JRSs, the evaluation metrics used in collaborative based JRS approaches are summarized in Table 5.

**Table 4** Overview and comparing of the reviewed collaborative based JRS approaches

| Article | Year | Category | Approaches used | Main ideas | Dataset/evaluation methods | Applied tools | Limitations | Future studies |
|---|---|---|---|---|---|---|---|---|
| Kwiecinski et al | 2023 | Model-based | A/B tests analysis, collaborative filtering, RP3Beta model | A JRS that can provide real-time job recommendations and investigate the efficiency of the production process is proposed with the RP3Beta model | Online A/B test is done to compare batch RP3Beta approach and real-time RP3Beta approach for 200,000 OLX users | RP3Beta model, A/B test, OLX platform | They only examined the RP3Beta model in depth and could also examine whether the performance of the system was degraded by ongoing changes in user preferences | Since the experiments were carried out using the RP3Beta model, the integration of the proposed infrastructure to other RSs domain can be investigated |
| Aouadni & Rebai | 2017 | Model-based | GMUSA algorithm is developed by combining the CGA and MUSA methods to obtain an effective DSS | A DSS is proposed that measures employee satisfaction | The researchers tested their system at the University of Sfax to measure 100 instructors' job satisfaction through a survey which consists of 8 criteria | Survey, GMUSA tools, five-point Likert-type scale approach | The researchers didn't include a filtering approach or address the problem of matching job opportunities with suitable candidates in their study | The researchers planned to analyze employee satisfaction from a new perspective using the combination of the CGA and MUSA methods |
| Dorn et al | 2011 | Model-based | GAs and Simulated Annealing approaches | An approach using GAs and Simulated Annealing is proposed to create a versatile and appropriate team composition | Evaluated their model based on multiple configurations of a simulated collaboration network that features close resemblance to real-world expert networks | Coverage, Interaction Distance, and Recommendation Distance assessment were done | Not Mentioned | Additional research is planned, incorporating a more extensive range of employee skills and utilizing various real-world datasets |
| Boccuzzo et al | 2016 | Model-based | Structural equation model, Univariate and multivariate methods | Analyzed the interaction of graduates'job potential by their past educational performance and social background with recent job | Data were collected from 2046 graduates of the University of Padua in Italy regarding current professional status 36 months after graduation | The data was analyzed utilizing SAS software, employing both univariate and multivariate methods | Only job-major match and job satisfaction the variables were investigated | Not Mentioned |

**Table 4** (continued)

| Article | Year | Category | Approaches used | Main ideas | Dataset/evaluation methods | Applied tools | Limitations | Future studies |
|---|---|---|---|---|---|---|---|---|
| Maurya & Telang | 2017 | Model-based | Unsupervised Bayesian Multi-View Modelling, Ontologies and POS tagging are used to extract features from job posts | A Collaborative Filtering-based Skill RS and two novel unsupervised Bayesian multi-view models were created. The models, leveraging Dirichlet prior and Dirichlet process, aim to match members with others to identify suitable skills | A dataset from LinkedIn, consisting of 414 M profiles, 58 M jobs, 29 K skills, 7.7 M companies, 98 K unique job titles, 95 K unique schools, 158 unique degrees, and 7.6 K unique majors, was collected and analyzed | LinkedIn is used | Evaluation did not use MAP or Precision@K. Challenge in extracting features from a large number of job postings in unstructured text format | Planned to use spectral approaches that afford guaranteed parameter recovery for latent variable models |
| Guan et al | 2019 | Model-based | Matching is applied between job seekers' resumes and enterprises through grading using fuzzy mathematics | An intelligent system has been developed that effectively matches job seekers and businesses in both directions | Not Mentioned | WXML, WXSS, JDBC, MySQL, Java, Redis, Spring, Bootstrap, LayUI, etc | Not Mentioned | Not Mentioned |
| Ghosh et al | 2020 | Model-based | Monotonic Nonlinear State-Space (MNSS) model, RNN-type model (Monotonic GRU —MGRU), Series of Stochastic (Binary-Valued Latent States), LSTM, K-Fold Cross-Validation | Proposed an interpretable monotonic nonlinear state-space model for career path modeling and analyzed two large-scale online-user professional profile datasets | Two large datasets were collected from LinkedIn and Indeed | They implemented all their methods in PyTorch and train all models on a single NVIDIA Titan X GPU | Factor in real-world constraints like family, transportation, and geography are crucial when making career choices. Also, considering job durations and user ideologies can provide a comprehensive perspective | The researchers plan to enhance the development of the mentioned limitations by incorporating related taxonomies in the future |

**Table 4** (continued)

| Article | Year | Category | Approaches used | Main ideas | Dataset/evaluation methods | Applied tools | Limitations | Future studies |
|---|---|---|---|---|---|---|---|---|
| Okfalisa et al | 2021 | Model-based | Fuzzy-AHP, TOPSIS | Created an integrated Fuzzy-AHP system for multi-attribute decision-making, serving as a tool to determine suitable training activities for prisoners seeking job education | Age, recent education, work experience, abilities and interests' data of 50 prisoners in Bankinang city, Indonesia were used. Case study-based verification is done on the dataset | Not Mentioned | Not Mentioned | Not Mentioned |
| Jarrett & Blake | 2017 | Model-based | Low Rank Matrix Factorization Algorithm, Gradient Descent Optimization Algorithm, Linear Least Squared Error | A collaborative-based JRS was developed. Its responsiveness and scalability were evaluated for expanding pools of workers, considering recruitment dynamics, and incorporating employer feedback | # of Workers (10–1000), number of jobs (500), three major crowdsourcing labor markets data were used (Amazon Mechanical Turk, UpWork and Microworkers) | Not Mentioned | Not Mentioned | Not Mentioned |
| Yi et al | 2016 | Model-based | Item-based Collaborative Filtering, Log-likelihood Similarity, Similarity ranking | An optimization algorithm is proposed to increase the accuracy of job recommendation | In 2013, 737 positions, 100 resumes, and 1072 job-delivery records were collected from a real recruitment website. The data was then used to evaluate the proposed optimization algorithm | Not Mentioned | Not Mentioned | Planned to verify the feasibility of the proposed optimization method on more datasets |

**Table 4** (continued)

| Article | Year | Category | Approaches used | Main ideas | Dataset/evaluation methods | Applied tools | Limitations | Future studies |
|---|---|---|---|---|---|---|---|---|
| Ortega et al | 2021 | Model-based | LFIT, General Usage LFIT Algorithm (GULA), PRIDE | The researchers aimed to check whether PRIDE and LFIT are useful in providing declarative explanations through neural networks | In experimental studies, FairCVdb containing 24,000 synthetic resume profiles were used. Each resume includes 12 job seeker-related features, gender and three ethnic groups features, and a face photo | GULA and PRIDE are particular implementations of the LFIT framework were used | Not Mentioned | In the experiments, facial image data was extracted for simplicity in the processes and included in further studies |
| Prafajar et al | 2022 | Model-based | Multiclass AI, KNN, SVM | A comparative study was conducted for JRSs in the field of Information Technologies using KNN and SVM methods with different kernels | Not Mentioned | Python, Google Collaboratory application, Pandas, NumPy, Matplotlib, and Skit-Learn libs | The absence of reference code for SVM and KNN Multi-Class models in Jupyter Notebook/Google Colab hindered the improvement of each model's performance during JRS development | Planned to develop a JRS by using textual data from interviews or descriptions of applicants with NLP |
| Wang et al | 2016 | Neighborhood-based | T-GBRT, CART Algorithm, Cosine Similarity | Proposed a new JRS model based on T-GBRT | The time span is from the 34 th week to the 45 th week in 2015. The training set is the data from the 34 th to the 44 th week. The total number of job seekers is 741650 and the number of jobs is 1500000 | The data is collected from www.xing.com job finder website. GraphLab computation tool is used in implementation of the GBRT algorithm | Not Mentioned | Not Mentioned |

**Table 4** (continued)

| Article | Year | Category | Approaches used | Main ideas | Dataset/evaluation methods | Applied tools | Limitations | Future studies |
|---|---|---|---|---|---|---|---|---|
| Upadhyay et al | 2021 | Model-based | KGs, NLP, NER, Paragraph Vector-Distributed Memory (PV-DM), SVM, Word2 Vec, Cosine Similarity, Dijkstra's Shortest Path, BELU, Rouge-L scores | A JRS is developed to match users'profile to the most relevant job postings and produce its reason with a textual explanation to users after the matching task | Experiments were conducted with anonymous 50 user profiles and 1606 IT-and engineering-related job postings collected from Monster job-hunting website | Not Mentioned | Visual representation aspect, other user factors can be taken into account | Planned to improve user evaluation by adding a questionnaire to evaluate other factors (e.g. voting, explanation satisfaction, etc.) |
| Bernabe-Moreno et al | 2019 | Model-based | Collaborative Filtering, Semantic Matching, Skills Modelling, Similarity (T-SNE, DBSCAN), GloVe, fitTransform, Density Clustering | A new approach to homogenize job information and subject specializations using collaborative filtering and semantic techniques | The dataset comprises data scientist job postings published in English-speaking countries from the beginning to the end of 2018 | Python, Google Cloud Entity POS Tagger, GloVe algorithm | Not Mentioned | It aims to analyze the changes in skills and scopes of the identified job |
| Saito & Sugiyama | 2022 | Model-based | Collaborative Filtering, MBL, EPL, and Recommendation Module | The system for both displaying job postings and helping users find the jobs they are looking for | Job search website (glit.io) is used as the data set | Not Mentioned | The system occasionally offers irrelevant suggestions, indicating a need for improvement, especially in addressing negative behaviors | The researchers have planned to enhance the explainability of recommendations |

**Table 5** An overview of the evaluation metrics used in the collaborative based JRS approaches reviewed

| Article | Year | Category | Methodology | Accuracy | Recall | Precision | Cost | Reliability | Coverage | F-Score | MAE/MRE | Security | Robustness | Scalability | Time | MSE/RMSE | Hit-Rate | NDCG | AUC | Loss-Func | Others |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Kwiecinski et al | 2023 | Model-based | User-to-Item, Item-to-Item, RP3Beta model | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | A/B Testing |
| Aouadni & Rebai | 2017 | Model-based | User-Based, CGA, MUSA Metod | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Average Global Satisfaction Index |
| Dorn et al | 2011 | Model-based | User-Based, GAs and Simulated Annealing approaches | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Coverage, Interaction Distance, Recommendation Distance |
| Boccuzzo et al | 2016 | Model-based | Structural Equation Modelling, ML (FIML) Algorithm | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Standard Error Analysis on Direct, Indirect, and Total Standardized Effect Variables |
| Maurya & Telang | 2017 | Model-based | Unsupervised Bayesian Multi-View Modelling, Ontologies and POS tagging | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Cosine and Jaccard similarity dependencies' comparison between skill sets and jobs |
| Guan et al | 2019 | Model-based | User-Based, Fuzzy Mathematics | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Weighted Pessimism Distance |
| Ghosh et al | 2020 | Model-based | User-Based, Monotonic Nonlinear State-Space Model | – | – | + | – | – | – | – | – | – | – | – | – | – | – | – | – | – | MPR, P@10, P@100 |

**Table 5** (continued)

| Article | Year | Category | Methodology | Accuracy | Recall | Precision | Cost | Reliability | Coverage | F-Score | MAE/MRE | Security | Robustness | Scalability | Time | MSE/RMSE | Hit-Rate | NDCG | AUC | Loss-Func | Others |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Okfalisa et al | 2021 | Model-based | User-Based, Fuzy Logic, Multi-Factor Analysis | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Consistency Ratio |
| Jarrett & Blake | 2017 | Model-based | User-Based, Matrix Factorization, Gradient Descent Optimization, Least Squared Error | – | – | – | – | – | – | – | – | – | – | + | – | + | – | – | – | – | |
| Yi et al | 2016 | Neighborhood-based | Item-based Collaborative Filtering, Loglikelihood Similarity | + | + | + | – | – | – | + | – | – | – | – | – | – | – | – | – | – | |
| Ortega et al | 2021 | Model-based | User-Based, Neural Networks | + | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | |
| Prafajar et al | 2022 | Model-based | User-Based, KNN, SVM | + | – | – | – | – | – | + | – | – | – | – | – | – | – | – | – | – | |
| Wang et al | 2016 | Neighborhood-based | User-Based, GBRT model based on time factor | + | + | + | + | – | – | – | – | – | – | – | – | – | + | – | – | – | |
| Upadhyay et al | 2021 | Model-based | User-Based, Job-Based, KGs, NLP, NER | + | + | + | – | – | – | + | – | – | – | – | – | – | – | – | – | – | Max/Median/Min Similarity Scores, BELU & Rouge-L Scores, Variance |
| Bernabe-Moreno et al | 2019 | Model-based | User-Based, Skill Modelling, Similarity and Semantic Matching | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Not Mentioned |
| Saito & Sugiyama | 2022 | Model-based | User-Based, Collaborative Filtering, JRS | – | – | – | – | – | – | – | – | – | – | – | – | – | – | + | – | – | MRR, Time Complexity |

#### Content-based filtering JRSs

A content-based RS solutions essentially suggests items similar to those the user has previously preferred, consumed, viewed, or liked. Over time, content-based RSs improve in accuracy as users respond to or consume recommended items [4, 28, 159].

(1) *User-Item profile* In content-based RSs, user data (e.g., ratings or clicks) is collected and analyzed to create a user profile. This profile is then utilized to generate personalized recommendations for the user. Content-based RS solutions typically use vectors to represent user preferences within these profiles.
(2) *Item profile* Item profiles are established to encapsulate the crucial features of each item. For instance, in the context of a resume item, key features such as the resume owner's name, age, occupation, driver's license type, and previous work experience are considered crucial features.
(3) *Utility Matrix (User-Item Relationship)* Utility matrices are often used to associate a user profile with item profiles. Vectors are often used to relate users'preferences (e.g. like or dislike items) for certain items.

Content-based solutions can be divided into learning-based and mathematics-based classification models (Fig. 6). In this section, the learning-based models are Supervised Learning [15, 44, 46], Unsupervised Learning [24], and Reinforcement Learning (RL) models while Mathematical-Based models involve Vector Space models [72] and Statistical Analysis-Based models [37, 44] are discussed and compared.

##### *JRS approaches and practices reviewed on content filtering*

Content-based JRS can use the defined criteria of a job posting description to predict suitable resumes for it [3, 4]. Additionally, it recommends other resumes that share similar characteristics with the matched resumes that an HR staff has interacted with before for the job description. These features can be demographic data of people, availability and class of driver's license, past work experiences, education degrees, etc. In addition, current open job postings with similar features can be recommended based on a job seeker's profile and previous work experiences. In a content based JRS, assume that a job posting is an item, and the user entity is a resume. As an example, the keywords of a job posting item are"software engineering graduate","at least 3 years of SQL experience","class B driver's license"etc. Assume that a user has previously clicked and assessed this job posting. Other similar job postings containing the same keywords are presented to that user as possible job opportunity recommendations.

Martinez-Gil et al. [96] addressed the challenge of effectively matching job offers with candidate profiles using ML-based adaptive approaches. Their focus was on accurately predicting matches between job descriptions and candidates'details and skill sets. They employed a weighting method to evaluate different elements in job posts based on their relative importance and calculated conversion costs to convert profiles into job posts. For this, the researchers calculated a cost value for each resume segment (e.g., education, language, skills, etc.). The researchers used shared terminologies to overcome the issue of heterogeneous representations of skills/competences in different profiles. Average Means and Variance analysis applied on both candidate offers and employers requests

for 4 different sector fields. In addition, Spearman rank correlation is used as a measurement factor. As future work, the researchers plan to design a solution that will automatically map information contained in applicant profiles to DISCO [45], ISCO (ISCO [70]) and ISCED [69]. For each applicant's profile, the researchers only focused on education, skills and languages.

Almalis et al. [3] enhanced the Minkowski distance algorithm to evaluate the suitability of employees for specific job positions, using it to compare an organization's needs with the skills of potential employees. The research involved a three-step process: content analysis, algorithm refinement, and execution with results. The content analysis step utilized a structured form, including job position attributes, attribute values, and attribute weights. The output was mapped into a 9-dimensional vector representing job positions. The researchers applied their refined algorithm to a dataset of 52 CVs (candidate employees) obtained from LinkedIn. It's worth noting that the researchers published another paper in 2015, evaluating the same algorithm with eight skill sets and 100 different job categories, providing additional details on the topic.

Almalis et al. [4] introduced the"FoDRA-Four Dimensions Recommendation Algorithm,"a content-based recommendation algorithm based on a neighborhood-based approach. The algorithm updates the Minkowski metric to enhance the task of matching job seekers and job posts. It calculates a suitability score between each candidate and job, suggesting a list of the top k jobs with the highest scores. The Minkowski metric is considered a generalization of Euclidean and Manhattan distance metrics in a normed vector space. In the structured form of the job domain, job attributes represent required skills, and attribute values reflect the skill levels, taking forms such as an exact value, a lower-limit range, an upper-limit range, or both. In order to verify FoDRA, 100 different job categories from the IT category (analyst, health IT, project management, system designer, programming, software engineering, etc.) were considered in their test runs. They also focused on 8 skills: education, past experience, language skill, programming knowledge, network knowledge, technical support knowledge, development experience, and project management experience. The researchers plan to further validate FoDRA by testing it on datasets from various job sectors and using different similarity measures (e.g., Cosine similarity, Pearson correlation) in future studies. The tools used in the research were not specified.

Ju and Li [75] investigated how education, job duration and training-job and skill-job matching affect employees'turnover intention over 10 years. The relationships among training, job tenure, education-job match, skills-job match and turnover intention were assessed using hierarchical regression analysis. Accordingly, the hierarchical regression analysis consists of five parts: time, control variables, training predictor variables, job tenure, and education-job and skills-job matches. The variables such as on-the-job training, distance education, working time, off-the-job training, training-job, and skill-job matching were taken into account by researchers as time and control variables. Moreover, the study shows the effect of job duration on education-job and skill-job mismatch trends. Therefore, skills-job mismatch trajectory analysis was performed. Accordingly, a negative relationship was found between length of stay at work and education-job and skill-job mismatches. It has been determined that education-job and skill-job matchups

negatively affect employees'intention to leave, and tenure positively affects education-job and skill-job matching.

Heap et al. [66] introduced a career transition JRS that suggests suitable jobs based on user profiles. The system, organized as a cascaded system of recommendation modules, merges recommendations from various modules using a multiplexer. The researchers compared this approach with a baseline method using cosine similarity to measure suitability. The researchers argued that the business transition model provides better performance than cosine similarity. The study incorporated a model considering job transitions and argued for its superior performance. The analysis, based on data from 2,400 LinkedIn users over two different periods in 2013 and 2014, assessed the system's ability to predict users'current job positions. They evaluated how well their system was able to predict users'current job positions from the users'profiles and previous job position history. Using the Apache OpenNLP library, user-typed nouns and compound noun phrases were extracted for improved matching. The system calculates cosine similarity scores for job postings, generating recommendations sorted by similarity score. They suggested that incorporating time-based job filtering could enhance the efficiency of the recommendations.

Bachtiar et al. [15] proposed a Profile Matching Algorithm to select the best candidate by assessing based on ranking criteria. The ranking criteria were assessed from two aspects. The first aspect is resume that consists of Educational Background, Work Experience, and Video. The second aspect is interview that consists of Attitude, Public Speaking, Appearance, Intelligence, and Sensitivity. The main factor in the resume aspect is Education Background and Video, and the secondary factor is Work experience. Main factor of the interview aspect are attitude, public speaking, and sensitivity, and the secondary factor of it are public speaking and appearance. The researchers used Naive Bayes to predict the appropriate employee placement according to the characteristics of the employees. During the validation process, bivariate correlation was performed to assess Profile Matching. In addition, accuracy was obtained to assess the suitability of employee placement. The data set used is limited in variation to train the model (only 15 candidates had all necessary data for assessment). The robustness of the system should be assessed with more variety and more data.

Chou and Yu [37] introduced a JRS framework facilitating the selection of suitable job seekers'resumes for companies and identifying fitting job postings for seekers. The framework includes three analysis steps: competitiveness analysis, personality trait analysis, and job recommendation analysis. The competitiveness score is derived from education, experience, skills, and personality traits, utilizing Word frequency analysis, DISC personality analysis, and talent analysis. System assessment involved analyzing job applicants'feedback using the JRS part. The position-talent recommendation task employs term frequency matrix, document vector, and similarity calculation. The system generates an online report visualizing personality trait analysis, competitiveness analysis, and job recommendation analysis for job applicants. On the company side, talent recommendations suitable for a specific job post are displayed. To evaluate system performance, job seekers were surveyed to assess the suitability of job vacancies generated based on their data.

Malherbe et al. [93] proposed a method to formalize the textual content of job offers in response to the growing volume of job data in e-recruitment solutions. The aim was to identify key information in job offers and use this formalization for job recommendation and profile matching in social networking environments. Their method was tested on Multiposting and Work4, leading e-recruitment platforms. The representation considered three classes: job offers, job categories, and user profiles of potential job seekers. These classes contain specific textual fields. The solution aims to maintain matching performance while reducing the cost of matching operations. By using a cross-matching approach, it performs dimensionality reduction and allows dynamic definition of text fields for each class. Experimental results claim that the method significantly reduces computational costs in both classification and recommendation processes, improving system performance.

Diaby and Viennet [44] developed a taxonomy-based JRS targeting Facebook and LinkedIn users, providing relevant job recommendations to job seekers. Users can view suitable jobs by granting access to parts of their profile data. The researchers used the O*NET-SOC taxonomy to create a new taxonomy-based vector model for social network users and job descriptions. They proposed two functions based on fuzzy logic operators,"AND and OR,"aligned with the vector model. Performance comparison with the traditional TF-IDF model demonstrated that the proposed taxonomy vector model outperformed it. SVMs were then employed to learn similarity functions in the data and process unbalanced datasets, with the learned models yielding superior results to heuristic similarity measures. The researchers argued that the proposed taxonomy-based vector model allows for efficient dimensionality reduction during the recommendation task in JRSs.

Mirza et al. [102] proposed a model that suggests appropriate job opportunities and trainings to individuals working in the IT sector by focusing on their technical/personal skill characteristics and their levels. Holland Codes were used to identify job roles and the individual characteristics (Realistic, Investigative, Artistic, Social, Enterprising and Conventional). The researchers also predicted personality analysis using the reliable Big 5 model. The model involves 5 dimensions which are Agreeableness, Neuroticism, Extraversion, Openness, and Conscientiousness. ONETOnline tool is used to get potentially suitable jobs.

Jain and Kakkar [72] proposed a JRS. This system provides a list of the most suitable jobs according to the abilities of the people and the most suitable job seeker proposal according to a job description. ML, Android API, MongoDB and RESTful Web Server application were used for this system, which makes job recommendation by examining the resumes. Using the API, data is sent and processed into the database. These data are analyzed by the RS and produce results. TF-IDF was used to measure the frequencies of the words in the RS. Thus, the most used words will be found. Data set contains 147 categories, 2232 skills and 50 occupational knowledge. In the future, it is planned to use a joint program with LinkedIn, to search the web for location, and to use NER for more accurate results.

Rutkauskiene and Gudoniene [124] stated that women without compulsory education face employment challenges. To address this, a digital online employment system was proposed to provide specialized training and guidance. The study aimed to assess

job seekers'skills, competencies, and experiences based on their work history and match them with suitable job vacancies and training programs. This approach ensures job offers align with individual goals, competencies, and age.

Bothmer and Schlippe [24] proposed a JRS tailored for educational institutions, employers, and job seekers using NLP techniques. A product is developed that vectorizes and compares the skills of people working in educational institutions, employers, and job seekers using the techniques; (1) Sentence-BERT, (2) UMAP, (3) DBSCAN, and (3) K-means clustering. The process involves resume processing, skill extraction, skill comparison with the dataset, and reporting missing skills in the job market using mathematical formulas. The results are intended to be shared with employers and job seekers. As a future study, the authors aim to expand the project by applying it to other business areas.

Dong et al. [46] proposed an online JRS to address the challenges posed by the large volume of items in recruitment networks and the unique characteristics of job seekers. They employed the Monte-Carlo Tree Search (MCTS) structure to estimate the most suitable results by determining similarity between objects. Comparing their algorithm (based on Work4 database) with existing literature, they demonstrated superior performance. Their proposed algorithm, Sublinear regret limit, Sublinear Space Complexity calculations, and Adaptive Context Partition (ITFACP) algorithm were used for estimation via data mining. The researchers evaluated their algorithms against various approaches like UCB, HCT, OLCR, ACR, and ITFACP based on average regret and item size.

Li and Lioma [83] introduced a JRS that matches job positions to job seekers by identifying competencies in resumes and job postings. They focused on aligning open job positions with suitable workers by extracting competencies from both job postings and resumes. The use of BERT during the classification and extraction processes contributed to the high efficiency of the system.

#### *An overview and comparison of content filtering JRSs*

In Table 6, the objectives, evaluation methods used, development tools and methods used, advantages and disadvantages of content based JRSs from the reviewed articles are examined and compared. Also, according to the articles reviewed above on content based JRSs, the evaluation metrics used in content based JRS approaches are summarized in Table 7.

### Hybrid filtering JRSs

In hybrid-based RSs, the aim is to combine two or more recommendation strategies in different ways to take advantage of complementary benefits [27]. For example, Spotify releases weekly recommendation lists. The goal is to find similar songs to the music a user listens to, and find the songs that similar users listen to, as a result, combine them all and present them to the user as a list of suggestions. In hybrid filtering types of RSs, different RS techniques are identified, as seen in Fig. 6. These are hybridization techniques (parallelized hybridization, pipelined hybridization, and monolithic hybridization), datamining-based models, deep learning-based models, and ensembles models [5, 27, 157].

**Table 6** Overview and comparing of the reviewed content based JRS approaches

| Article | Year | Category | Approaches Used | Main Ideas | Dataset/Evaluation Methods | Applied Tools | Limitations | Future Studies |
|---|---|---|---|---|---|---|---|---|
| Martinez-Gil et al | 2020 | Model-based | Item-based | An automatic matching learning method between job offers and suitable candidate profiles is proposed | The approach has been tested with real recruitment scenarios. The results were compared with the OKAPI BM25 algorithm, which did not implement the learning capability | Recruiting taxonomies were used (ISCO, DISCO, and ISCED) in relating skills | Some fields are considered in resumes | They planned to adopt their solution on the DISCO, ISCO and ISCED frameworks and to analyze more features such as photographs, personal addresses, former employers, etc. in a resume |
| Almalis et al | 2014 | Model-based | Item-based | Minkowski distance algorithm was revised as a new RS algorithm to evaluate an employee candidate's suitability for a particular job position | The proposed algorithm was evaluated on the dataset of 52 resumes collected from LinkedIn. A three-step experiment was conducted (content analysis, refinement of the algorithm, execution and obtain results) | Not mentioned | Needs to improve the performance, in terms of reliability and time response and tested on other datasets from various branches of the enterprise sector | Future research is planned to include the reverse procedure |
| Almalis et al | 2015 | Model-based | Item-based | A content-based recommendation algorithm called "FoDRA" was proposed, and they revised the Minkowski metric to perform job seeker-job postings match | FoDRA is tested taking into account 100 different job categories and 8 skills belonging to IT sector found in resumes | Not mentioned | There is a need to evaluate the algorithm in more different business sectors and different skills | Test the algorithm on other similarity measures and apply it to different job sectors |

**Table 6** (continued)

| Article | Year | Category | Approaches Used | Main Ideas | Dataset/Evaluation Methods | Applied Tools | Limitations | Future Studies |
|---|---|---|---|---|---|---|---|---|
| Ju & Li | 2019 | Model-based | Hierarchical Regression Analysis | The relationships among training, job tenure, education-job match, skills-job match and turn over intention were investigated | For the years 2003–2014, by the Korea Labor Institute, some questions were asked to 1,531 employee participants about the training given at the workplace, the training given at an out-of-work institution, and the distance education provided by their organizations. Then the answers of the participants were evaluated | STATA 14.0 was used for clustering the standard error for each participant through regression | Despite the longitudinal nature of the data, more reliable data are required as there may be inherent bias from the self-report method | The relationship between education types in different countries and employee turnover intention will be investigated. In addition, the relationship between turnover intention and organizational factors that provide it, will be investigated |
| Heap et al | 2014 | Model-based | Cosine Similarity | A career transition JRS is proposed that suggests suitable jobs to users based on their profiles | 2,410 candidate profiles from LinkedIn consisting of 6,534 unique job positions | Apache OpenNLP library to extract user-typed singular nouns as well as compound noun phrases in job titles | Not Mentioned | Information about a user's qualifications and skills, as well as the length of time they have worked at a particular job level, can be integrated into the proposed model to generate better recommendations |
| Bachtiar et al | 2019 | Neighborhood-based | Naive Bayes | Profile Matching Algorithm has been developed to find the best candidate by evaluating user profiles and applying ranking criteria | 15 out of 15 candidates were correctly classified in job placement with 100% accuracy | Not Mentioned | The dataset used was limited (only 15 candidates had all necessary data for assessment) | The robustness of the system will be evaluated with more data |

**Table 6** (continued)

| Article | Year | Category | Approaches Used | Main Ideas | Dataset/Evaluation Methods | Applied Tools | Limitations | Future Studies |
|---|---|---|---|---|---|---|---|---|
| Chou & Yu | 2020 | Model-based | 1) Term Frequency Matrix, 2) Document Vector, and 3) Similarity Calculation (NLP, Count Vectorizer, TF–IDF, Cosine) | A JRS framework that enables both the selection of suitable CVs for companies and the finding of suitable job postings for candidates is engineered | A 5-question survey was conducted with 40 people | Jieba Kit, a self-developed dictionary | Assessed only from job applicant's side | Increasing accuracy, scoring and weighting of frequently used keywords in job postings, word extraction and labeling |
| Malherbe et al | 2014 | Model-based | TFIDF, Field to Field Weighting (FtFw), SVM, Forest, Average Field to Field Similarity (AFtF), Field Weighting (Fw), Basic Cosine (BC) | Developed a methodology for formalizing the text content of job posts to identify key information and defined fields. Applied this formalization for JRS and profile matching in social network environments | Facebook users: Work, Education, Quote, Bio, and Interests fields. For LinkedIn users: Headline, Educations, Positions fields. For Job offers: Title, Description, Jobs fields were applied on the proposed models during evaluations | Not Mentioned | Some models (e.g. FtFw, Fw) may have information loss. Many e-recruitment sites do not divide their job postings into more well-structured fields | It may be possible to improve the computation time according to advancements on the models proposed |
| Diaby & Viennet | 2014 | Model-based | O*NET vector model, TF-IDF vectors, cosine similarity, PCC, fuzzy logic, SVMs | Four heuristic-based and one SVM-based JRSs using a novel taxonomy-based vector model developed were proposed | The developed methods were compared with two well-known methods (a Matrix Factorization-CTR and Collaborative Subject Regression-CR). The proposed rating-based system developed better results than these two methods in terms of AUC | O*NET-SOC taxonomy, Facebook and LinkedIn platforms | Only English and French | Planned to extend the proposed taxonomy-based vector model to languages other than English and to improve it by including metadata about users'jobs |

**Table 6** (continued)

| Article | Year | Category | Approaches Used | Main Ideas | Dataset/Evaluation Methods | Applied Tools | Limitations | Future Studies |
|---|---|---|---|---|---|---|---|---|
| Mirza et al | 2015 | Model-based | Holland Codes, Big 5 Model | Analyzing the individual characteristics of those seeking employment in the IT sector to recommend fitting jobs and essential skills | Not Mentioned | ONETOnline | Not Mentioned | Not Mentioned |
| Jain & Kakkar | 2019 | Model-based | TF-IDF, Data Mining, ML, Job Recommendation | Generates a list of the most appropriate jobs based on individuals'capabilities and suggests the most fitting job seekers in alignment with a provided job description | Data set contains 147 categories, 2232 skills and 50 occupational knowledge | SQL, MongoDB, API, RESTful Web Server, JAX-RS Framework, Tomcat7 | Not Mentioned | It is planned to use a joint program with LinkedIn, to search the web for location, and to use NER for more accurate results |
| Rutkauskiene & Gudoniene | 2019 | Model-based | Matching, Automated System, European Skills, Competences, Qualifications and Occupations Classification Tool (ESCO) | The objective is to assess job seekers'skills, competencies, experiences, and attitudes based on their work history. Next, the aim is to identify job openings that align with the individual's goals and age | Google Analytics | ESCO, DigComp, and e-CF Taxonomies | Not Mentioned | Implementing the proposed job profiling tool and using it for different purposes is targeted in the future |

**Table 6** (continued)

| Article | Year | Category | Approaches Used | Main Ideas | Dataset/Evaluation Methods | Applied Tools | Limitations | Future Studies |
|---|---|---|---|---|---|---|---|---|
| Bothmer & Schlippe | 2022 | Model-based | AI, NLP, Sentence-BERT, UMAP, DBSCAN, and K-means clustering, GloVe, Word2 Vec, Silhouette, PCA | Created an NLP-powered JRS for employers and job seekers in educational institutions that utilizes skill vectorization to compare the capabilities of individuals employed in educational institutions | The dataset, BeautifulSoup, comprises 2,633 job postings and 21.5 K bullet points. Evaluation methods include Sentence-BERT, UMAP, DBSCAN, and K-means | Not Mentioned | Not Mentioned | As a future study, it is aimed to develop this project by applying it in other business areas |
| Dong et al | 2017 | Model-based | ITFACP Algorithm, MCTS, Big data, Data Mining | An online job and job seeker RS due to the large number of items in the big data quantity in recruitment networks and the individual characteristics of job seekers | Work4 database used. Space and Time complexity, Regret are evaluated | Local Smoothness, Near-Optimality dimension, ITFACP Algorithm, MCTS | Not Mentioned | It will be on object space for cases where smoothness information that cannot be obtained |
| Li & Lioma | 2023 | Model-based | BERT, Data Mining, classification, and extraction | A system that matches job-to-job seekers based on competencies present in resumes and job postings | The efficiency of the system was found to be high | Not Mentioned | Not Mentioned | Not Mentioned |

**Table 7** An overview of the evaluation metrics used in the content based JRS approaches reviewed

| Article | Year | Category | Methodology | Accuracy | Recall | Precision | Cost | Reliability | Coverage | F-Score | MAE/ MRE | Security | Robustness | Scalability | Time | MSE/ RMSE | Hit-Rate | NDCG | AUC | Loss-Func | Others |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Martinez-Gil et al | 2020 | Model-based | Item-based | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Average Means and Variance Analysis |
| Almalis et al | 2014 | Neighbor-hood-based | Case-based Analysis | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Suitability Value Analy-sis using Manhat-tan and Euclidian Distances |
| Almalis et al | 2015 | Model-based | Case-Based Analysis | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Suitability Value Analy-sis using Manhat-tan and Euclidian Distances |
| Ju & Li | 2019 | Model-based | User-Based, Hierarchical-Regression Analysis | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Mean, Standard Deviation, Standard Error of Coefficient |
| Heap et al | 2014 | Model-based | User-Based Profile Match-ing | – | + | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | |
| Bachtiar et al | 2019 | Neighbor-hood-based | User-Based, Profile Match-ing | + | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | |
| Chou & Yu | 2020 | Neighbor-hood-based | User-Based | + | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | |
| Malherbe et al | 2014 | Model-based | User-Based, Text Mining | – | – | – | + | – | – | – | – | – | – | – | – | + | – | – | + | – | Mean ROC-AUC |

**Table 7** (continued)

| Article | Year | Category | Methodology | Accuracy | Recall | Precision | Cost | Reliability | Coverage | F-Score | MAE/MRE | Security | Robustness | Scalability | Time | MSE/RMSE | Hit-Rate | NDCG | AUC | Loss-Func | Others |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Diaby & Viennet | 2014 | Model-based | User-Based, Taxonomy | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | + | – | |
| Mirza et al | 2015 | Model-based | User-Based, Software Skills | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Not Mentioned |
| Jain & Kakkar | 2019 | Model-based | Data Mining, ML, Job Rec., Content-Based Filtering | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Not Mentioned |
| Rutkauskiene & Gudoniene | 2019 | Model-based | Matching, Digital Jobs | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Not Mentioned |
| Bothmer & Schlippe | 2022 | Model-based | User-Based, AI, NLP, Clustering | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Not Mentioned |
| Dong et al | 2017 | Model-based | Content-Based, Big Data, Data Mining | – | – | – | – | – | – | – | – | – | – | – | + | – | – | – | – | – | Space Complexity, Regret |
| Li & Lioma | 2023 | Model-based | BERT, Data Mining, Classification and Extraction | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Efficiency |

***JRS approaches and practices reviewed on hybrid filtering***

In this section, usage of the hybridization techniques [23, 90, 158], data mining-based models [41, 73, 140], deep learning-based models [17, 85, 140], and ensembles models in the JRS solutions are investigated.

Text mining technology integrates various disciplines, including data mining, NLP, mathematical statistics, and visualization. Recruitment data on websites best reflects current market demands. By analyzing recruitment information through methods such as mining, NLP, and mathematical approaches, potential sectors with the highest market demand can be identified. Liang and Wan [85] crawled and analyzed recruitment data using data analysis methods in their research. They employed NLP, text mining, and multi-criteria decision making to analyze data posts and extract core requirements. During data collection, additional information such as salary, position qualifications, work experience, and the number of recruits was gathered. The analysis focused on capabilities to quickly derive market demand. Results showed that the internet/technology, education, real estate, and media sectors had the highest to lowest demand. The researchers did not propose a recommendation filtering approach or address job-candidate matching. Their goal was to provide individuals with a comprehensive understanding of the market to improve job selection.

Yang et al. [159] proposed a hybrid JRS using both content-based filtering and collaborative filtering techniques. The researchers adopted Statistical Relational Learning (SRL) approach and considered precision and recall factors in their experimental studies. The aim of this study is to create a JRS and prove how effective it is to combine collaborative filtering and content-based filtering with real-world JRS data. Relational Functional Gradient Boosting (RFGB) is used to gain recommended jobs' confidence. Some information from CVs and published job postings has been mined. These are: *JobSkill, UserSkill, JobClass, UserClass, PreAppliedJob, UserJobDis, UserCity, MostRecentCompany,* and *MostRecentJobTitle*. The proposed model was evaluated first as content-based filtering and then by combining content-based filtering and collaborative filtering. The performance of the proposed model is evaluated by FPR, FNR, precision, recall, accuracy and AUC-ROC. The researchers did not mention any planned future developments of the current system they proposed. According to the evaluation results, it was seen that the hybrid model proposed was very efficient and also reduced false positive predictions.

Lu et al. [90] proposed a hybrid RS for job seeking and recruiting website. The system uses job and user profiles and processes user-generated actions to create personalized recommendations. Researchers have tried to describe the interaction-based relations (e.g., Post, Apply, Favorite, Like, Visit) between entity pairs, such as entities of a different group (e.g. similarity between a candidate's resume content and a job posting) or two entities of the same group (e.g. resume—resume similarity, job posting—job posting similarity). Four recommendation situations are considered in their system: (1) recommending employers to candidates, (2) jobs to candidates, (3) recommending candidates to employers, and (4) recommending candidates to jobs. With their content-based approach, the researchers achieved profile similarity between the two entities and integrated it into a graph-based ranking approach, producing a hybrid solution to provide personalized recommendation. The data collected from the website is modeled using a directed, weighted, and multi relational graph, and the 3 A ranking algorithm is used to

rank items according to their relevance to the target user. They argued that their system outperforms collaborative filtering on content-based profile matching and recommendation certainty and user coverage.

Dave et al. [41] developed three types of information networks of historical job data for JRSs: (1) job transition network, (2) job-skill network, and (3) skill co-occurrence network. The researchers proposed a representation learning model which can employ the information from the 3 networks to collectively learn the representation of the jobs and skills in the joint k-dimensional latent space. In their empirical work, the researchers argued that their model provides better recommendations for both jobs and skills by learning jointly the representation of jobs and skills. In addition, the researchers discussed several case studies to validate their proposed model. Researchers also used 2 loss functions based on ranking. The first is based on the Bayes personalized ranking, the second is based on the Margin order, both selected for the appropriate job recommendation task. The proposed representation learning model is transductive. Learns representation vectors of jobs and skills available in input graphs. But the model needs to be retrained each time for new job titles and skills in sector. Therefore, an inductive learning model is needed. The researchers planned to develop an inductive learning model in their future studies.

Sun et al. [140] proposed a sustainable job skill recommendation approach aimed at providing cost-effective, long-term, and explainable recommendations for individuals seeking new job skills. A data-driven JRS was built using deep reinforcement learning. The researchers proposed an environment to predict the benefits of skill learning by mining large job post data, including a skill-matching-based salary estimator and a frequent itemset-based learning difficulty estimator. A new Skill Proposal Deep Q-Network (SRDQN) with a multitasking structure was designed to predict long-term skill learning tools based on the environment. The proposed system was argued to help individuals earn higher salaries while reducing learning effort, considering their existing skills. Numerous experiments were conducted on a real-world dataset from the IT industry, collected from the Lagou recruitment website in China, to verify the effectiveness and interpretability of SRDQN. Each job posting in the dataset included salary and a description of skill requirements. Future studies will focus on improving SRDQN performance and utility estimation, with the aim of applying it to additional applications, such as curriculum recommendation tools.

Janusz et al. [73] proposed a recruitment support system that helps HR officials find candidates who are likely to be interested in a particular job offer. A data processing chain model with domain information is used to extract meaningful features between a job seeker's profile and a job posting. In total, they identified 374 features under three groups (job offers features, relationships between candidates and offers, candidates'features). They also used a word2vec model to find vector representations of the job postings. The developed system (as a toolbox in HR department) is used in an RPO company operating in Warsaw, Poland. To make an empirical assessment of the system, the researchers collected data on 2288 profiles, corresponding to a total of 76 job offers. After the textual descriptions of job offers are translated into vectors using word-2vec, the system finds similar case examples from previous recruitment processes to find out a set of"new candidate-job offer"pairs. It uses these case examples as a training set to

calculate a scoring model for a new case. They used the skip-gram model with the noise contrastive training to train the model on available textual descriptions of these historical job offers. As a result of the empirical evaluations regarding the performance of the system, the researchers claimed that more effective results can be obtained with this system than the decisions of the HR officials.

Fang et al. [52] aimed to predict expert employee candidates in line with new skills and skill concepts/tags. They used collaborative filtering, content-based and hybrid recommendation approaches to avoid the cold start issue and process different data sources. The data includes employee evaluation data, free-text skill description data, and employee internal social media tags. The researchers conducted an empirical study of real-world enterprise data, as it is difficult to combine all the different data sources effectively. According to the results, it was emphasized that employee tags alone did not provide benefits, and it was emphasized that skill tags should be used in order to make expert recommendations more effective. Therefore, the estimation using skill similarity produced a smaller bias but a larger variance result. Plus, the hybrid model with skill similarity for prediction produced a higher recall rate. They stated that they will consider other models that deal more directly with user and skill tag interactions in their future studies.

Kenthapadi et al. [79] have presented a research article discussing the practical challenges and technical experiences they encountered in developing their personalized JRS on LinkedIn. The architecture of the system involves two subsystems as online system and offline system. While the online system includes job recommendation task services, the offline system includes services that provide process flow to update different ML and statistical prediction models. The details of the ML and statistical forecasting models in the offline system are given in the researchers'other article [23]. Researchers have discussed some important challenges and considerations when developing a personalized JRS in their article. For instance, real-time computation and prediction of personalized job recommendations for users, difficulty incorporating different types of user interaction signals, determining if a typical job posting is suitable for hiring only one or a few potential suitable employees (e.g., in book RSs, a book may be suitable for millions of users, while a job posting may fit the profiles of only a few individuals).

Ouyang et al. [110] investigated the suitability of new recruits for their recent new job position and proposed a framework for this. In addition, the opinions of the experts on the performance of these new employees were also included in the evaluation. In the evaluation process, the Discrete Interval Type-2 Fuzzy Set method was used on the time-varying universe. BP neural networks were used to obtain the result of the predicted value. The researchers then used the Markov approach to revise this value result and generate a final prediction result. As a result of the experimental studies, it was found that the results obtained from the Markov model were obtained more accurately than the BP neural network.

Borisyuk et al. [23] have focused on the job application redistribution problem to make JRSs (such as LinkedIn's job recommendations product) more efficient and to balance the agglomerations of too many or too few job applicants for open positions. The purpose of the system is to try to distribute it evenly while recommending

relevant job positions to suitable job seekers with the same relevance. The researchers applied a dynamic prediction model to estimate the expected number of applicants by the job posting deadline. The researchers also applied various algorithms to encourage or penalize open job postings, according to the output of the prediction model. In their research, LiJAR (Linkedln's Business Applications Prediction and Redistribution system) was developed, which works both offline and online environment. In test experiments, LiJAR was evaluated with the A/B method. Developed as part of LinkedIn's JRS engine, the system resulted in a 6.5% increase in job seekers applying to job postings from unpopular job providers.

González-Briones et al. [59] have developed a JRS, a multi-agent system model. The JRS aims to offer job offers to job seekers on an employability focused social network. The proposed JRS was based on a hybrid approach. The system consists of a CBR inference module and an argumentation framework. In their proposed system, a decision is made whether a job offer can be proposed to a job seeker/user based on a set of metrics and similar cases. In addition, the CBR, which is the inference process of the system, has been expanded with the argumentation framework. It also includes system factors. There is a process of interaction between agents in the system, taking into account the different suggested implications and experience from past cases. Based on the different solutions proposed by the agents and the experience gained from past cases, a discussion process was established between the system agents.

Benabderrahmane et al. [17] proposed a JRS solution to facilitate the recruitment processes of HR departments. The system stores click-through rates and time series of job postings posted online in the database. To predict click streams in future, Deep Neural Network (DNN) architecture is used, and time series were used for mathematical calculation of clicks. In addition to measuring the number of clicks, prediction of symbols can also be calculated by applying N-grams. Thus, job postings with the highest clickstreams can be listed. They tested it on real dataset. The performance of the system was measured by applying Jaccard Index, SAX, F-Score and RMSE. As future work, it is envisaged to test it by applying the Multiple Resolution Vector Quantification (MVQ) method. In addition, Word2 Vec and Doc2 Vec will be used.

Yang et al. [158] focused on the person-job matching problem that is very important to effectively carry out HR allocation and management processes. The researchers emphasized the need for an accurate analysis of the need for person-job matching in the process of promoting and using overseas talents abroad. In addition, the researchers are of the opinion that the preferences and interests of two-sided should be taken into account in the person-job matching problem. For this, the researchers proposed two-sided matching approach on the person-job fit problem and used heuristic Best–Worst Method (BWM) and TOPSIS methods, taking into account Large-Scale Group Decision Making (LSGDM) and evaluative Social Network Relationships (SNRs). In the evaluations, methods such as BMW, Intuitionistic Best Worst Method (I – BMW), eigenvector centrality, ordered weighted averaging, TOPSIS, and degree centrality were used. Finally, the result of the case study revealed that their proposed method with I-BTM and LSGDM gave better results.

#### *An overview and comparison of hybrid filtering JRSs*

In Table 8, the objectives, evaluation methods used, development tools and methods used, advantages and disadvantages of hybrid based JRSs from the reviewed articles are examined and compared. Also, according to the articles reviewed above on hybrid based JRSs, the evaluation metrics used in hybrid based JRS approaches are summarized in Table 9.

### Knowledge-based filtering JRSs

Knowledge-based RSs usually provide recommendations based on user queries and the domain knowledge base of a system, not based on the user's rating history. It may require some information from its users such as a set of user directives, rules, constraints, or an example of an item. The system then searches the item database and returns similar items as a result to the user [26, 27]. Knowledge-based RSs are described by the advantage of improved reliability since they usually have less noise. On the other hand, a KB system requires considerable knowledge acquisition work for setup and maintenance during its lifetime [3].

#### *JRS approaches and practices reviewed on knowledge-based filtering*

Knowledge-based filtering in Recommendation System (RS) solutions can take various forms such as constraint-based, case-based, rule-based, utility-based, ontology-based, embedding-based, feature-based, and more. Observations indicate that some of these approaches are implemented in JRS solutions, with certain instances being rule-based [29, 34], ontology-based [29, 34, 106, 122, 126, 141], case-based [126], and content-based [106].

Fernández-Sanz et al. [53] developed a professional competency and skills framework for the ICT domain named eSkills Match. Researchers referred to three crucial systematic conceptual maps (ESCO, e-CF, and BOK) that include skills, occupations, occupational profiles, and related concepts in the field of ICT. This framework facilitates the development of future support systems, enabling self-assessment for job seekers, recommender modules to guide training, and options for validating knowledge aligned with specific skills, knowledge, and competences. It encompasses concepts related to skills, occupations, competences, qualifications, knowledge areas, and proficiency levels in the ICT field. The current version of the framework has limitations, acknowledging the potential emergence of new occupations and concepts related to competence and skills. The researchers did not discuss a recommendation filtering approach or address the problem of matching candidates to job opportunities in their study. Nevertheless, they successfully developed a knowledgebase scheme for the ICT field, encompassing occupations, qualifications, skills, knowledge areas, and competence levels.

Gutiérrez et al. [65] introduced the"Labor Market Explorer,"a user-centered system designed to offer personalized job recommendations for job seekers based on their skills and competences. The tool explores open job positions, knowledge, and skills, aiming to address potential competence gaps. It provides competency overviews through a map of open positions, a distance-to-job slider, an open job positions table, and detailed competency information. The tool was based on client–server software architecture

**Table 8** Overview and comparison of the reviewed hybrid based JRS approaches

| Article | Year | Category | Approaches Used | Main Ideas | Dataset/Evaluation Methods | Applied Tools | Limitations | Future Studies |
|---|---|---|---|---|---|---|---|---|
| Liang & Wan | 2022 | Model-based | NLP, text mining, and multi-criteria decision making, Word2 Vec, K-means, TF-IDF | The purpose of this study is to screen recruitment information through data collection and data analysis of the recruitment website | Combining qualitative evaluation and quantitative evaluation | Not mentioned | Not Mentioned | Future goals are not stated |
| Yang et al | 2017 | Model-based | Model-based, Hybrid Filtering, SRL Model, RDNS, RFGB | Proposed a hybrid JRS using both content-based filtering and collaborative filtering techniques | 707,820 total job postings are analyzed | FitRel Regression Tree | Not Mentioned | Not Mentioned |
| Lu et al | 2013 | Model-based | RS, multi relational graph, Hybrid RS, Interaction-based relations, Content-based Similarity, Ranking Algorithm | Directed, weighted, and multi relational graph, and the 3 A ranking algorithm were considered | A preliminary evaluation based on dataset collected from production website is done | Not mentioned | Need to evaluate the accuracy, usability, and other features of the proposed system though online data | More online data-driven experimental work is planned |
| Dave et al | 2018 | Model-based | Use latent embedding approach to combine the job and skill and learns | A Representation learning based JRS approach is proposed to solve proper job and skill recommendation | CareerBuilder dataset and case studies were done to compare with several existing state-of-the-arts | Carotene, CareerBuilder | The proposed model needs to be retrained each time for new job titles and skill descriptions in the sector | Inductive learning model is planned for representing new job titles and skills |
| Sun et al | 2021 | Model-based | Deep Reinforcement Learning | A sustainable job skill recommendation approach is proposed which aims to provide longsighted, cost-effective, and explainable recommendations for career seekers | A total of 805,182 job postings were collected from Lagou, a Chinese online recruitment site, in a 36-month period (from 2016.07 to 2019.06) | RMSProp optimizer, Lagou recruitment web site | The necessity of incorporating data from other industries to improve utility estimation and assess SRDQN performance in potential applications | Designing and evaluating the model on potential applications such as the curriculum recommendation tool |

**Table 8** (continued)

| Article | Year | Category | Approaches Used | Main Ideas | Dataset/Evaluation Methods | Applied Tools | Limitations | Future Studies |
|---|---|---|---|---|---|---|---|---|
| Janusz et al | 2018 | Model-based | Vector representations is used for job offers. In matching, standard skip-gram model, k-NN, decision trees built with the recursive partitioning algorithm and a XGBoost model, Euclidean distance are used | Candidate-job offer pairs were analyzed for matching | Data of job seekers who contacted between November and December 2017 were collected. Used BIZON to feed the FARM database with data. About 2288 candidates' profile corresponding to 76 job posts were used in total | Google Cloud Datastore NoSQL database, R environment, TensorFlow, XGBoost library | Sufficient good quality data is needed to construct more reliable scoring models. Integration of more expert knowledge into the system would be beneficial | Deep learning-based models need to be applied and validated on the proposed system |
| Fang et al | 2013 | Model-based | Predictive models Taxonomy, Collaborative, Content and Hybrid filtering, A novel Matrix Factorization model, Media data processing | A RS predicts expert employee candidates in line with new skills and skill concepts/tags | The data in the employee rating system of the multinational Fortune 500 company, where employees are rated between 1 and 5 according to their skills, were used in experimental processes. Of these data, 10739 unique employees, 8529 unique skill concepts and 954324 relationship data were used | Matrix factorization for skill Assessment, Prediction Based on Cosine skill similarity, hybrid method (combines matrix factorization and skill similarity-based prediction) | Not Mentioned | Not Mentioned |

**Table 8** (continued)

| Article | Year | Category | Approaches Used | Main Ideas | Dataset/Evaluation Methods | Applied Tools | Limitations | Future Studies |
|---|---|---|---|---|---|---|---|---|
| Kenthapadi et al | 2017 | Model-based | A/B testing, CaSMoS, Decision Trees, and Generalized Linear Mixed (GLMix) models | The challenges of developing a JRS product on LinkedIn, the overall architectural design of the proposed JRS product, and lessons learned during the distributed deployment of JRS solutions on LinkedIn were discussed | Not Mentioned | LinkedIn's Galene search platform, LiJzAR | Not Mentioned | Planned to encourage further research and collaboration with the RS community |
| Ouyang et al | 2014 | Model-based | Back Propagation NN, Markov Model, Interval Type-2 Fuzzy Set, Time-Varying Universe | By combining the BP neural network and the Markov model, a model is proposed that predicts whether new staff will match their jobs | Example analysis has been done. The dataset is not mentioned | Not Mentioned | Not Mentioned | Not Mentioned |
| Borisyuk et al | 2017 | Model-based | Boosting and Penalization strategy to obtain the job score | A statistical job application prediction model was developed to provide confidence intervals of job applications at the expiration time. This model has been used to punish or increase confidence intervals for jobs believed to receive too few or too many job applications, respectively | Performance of the proposed model was investigated by using the entire 2015 job application data as the training set and the September-December 2016 data as the test set on LinkedIn | LinkedIn's job recommendations product | Not Mentioned | In the future, it is planned to integrate important measures such as customer satisfaction and member loyalty into the jobs ranking model |

**Table 8** (continued)

| Article | Year | Category | Approaches Used | Main Ideas | Dataset/Evaluation Methods | Applied Tools | Limitations | Future Studies |
|---|---|---|---|---|---|---|---|---|
| González-Briones et al | 2019 | Model-based | Case Base Reasoning and Multi Agent System | CBR uses the contents of available job offers on a social network to recommend best offers to suitable users. A series of metrics were used to measure affinity between job offers and system users. An argumentation process was also considered to establish a dialogue between the agents | The JRS was evaluated on a sample of active job seekers in the last 15 days prior to the study. Recommendation results were obtained as follows; Accepted, Rejected, Ignored rates | Not Mentioned | Not Mentioned | Not Mentioned |
| Benabderrahmane et al | 2017 | Model-based | Hybrid-Collaborative Filtering, DNN, LSTM, RNN, N-gams, | The JRS to facilitate the recruitment processes of HR departments. The system stores click-through rates and time series of job postings posted online in the database | Evaluation metrics are RMSE, F-Score, Jaccard Index, SAX. Real data is used from the industrial partner | Not Mentioned | Not Mentioned | To test it by applying the Multiple Resolution Vector Quantification (MVQ) method. In addition, Word2 Vec and Doc2 Vec will be used |
| Yang et al | 2021 | Model-based | Fuzzy sets, BWM, TOPSIS and LSGDM | Proposed two-sided matching method with I-BTM and LSGDM applied to high level overseas talent and job matching problem | In the evaluations, the methods such as BMW, I – BMW, eigenvector centrality, ordered weighted averaging, TOPSIS, degree centrality were used | Not Mentioned | Not Mentioned | Since the matching relationship between overseas talent and jobs is dynamic over time, the researchers planned to improve their matching task by adding a time variable |

**Table 9** An overview of the evaluation metrics used in the hybrid based JRS approaches reviewed

| Article | Year | Category | Methodology | Accuracy | Recall | Precision | Cost | Reliability | Coverage | F-Score | MAE/MRE | Security | Robustness | Scalability | Time | MSE/RMSE | Hit-Rate | NDCG | AUC/AUC-ROC | Loss-Func | Others |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Liang & Wan | 2022 | Model-based | NLP, Text Mining, Multi-Criteria Decision Making, Word2 Vec, K-Means, TF-IDF | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Not Mentioned |
| Yang et al | 2017 | Model-based | Hybrid Filtering, SRL Model based | + | + | + | – | – | – | – | – | – | – | – | – | – | – | – | + | – | FNR, FPR |
| Lu et al | 2013 | Model-based | RS, multi relational graph, Hybrid RS | – | – | + | – | – | + | – | – | – | – | – | – | – | – | – | – | – | Ranking |
| Dave et al | 2018 | Model-based | Embedding Learning | – | – | – | – | – | – | – | – | – | – | – | – | – | + | + | + | | |
| Sun et al | 2021 | Model-based | Deep Reinforcement Learning | – | – | – | – | – | – | – | – | – | – | – | – | + | – | – | – | + | |
| Janusz et al | 2018 | Model-based | User-Based | + | + | + | – | – | – | – | – | – | – | – | – | – | – | – | + | – | |
| Fang et al | 2013 | Model-based | User-Based, Skill Matching | + | + | + | – | – | – | + | – | – | – | – | – | – | – | – | – | – | |
| Kenthapadi et al | **2017** | Model-based | User-Based, IR, ML, Statistical Modeling | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Not Mentioned |
| Ouyang et al | 2014 | Model-based | User-Based, Markov Model, Interval Type-2 Fuzzy Set | – | – | – | – | – | – | – | + | – | – | – | – | – | – | – | – | – | Relative Error Analysis |
| Borisyuk et al | 2017 | Model-based | User-Based, Dynamic Prediction Model | – | + | + | – | – | – | – | – | – | – | – | – | + | – | – | – | – | |

**Table 9** (continued)

| Article | Year | Category | Methodology | Accuracy | Recall | Precision | Cost | Reliability | Coverage | F-Score | MAE/MRE | Security | Robustness | Scalability | Time | MSE/RMSE | Hit-Rate | NDCG | AUC/AUC-ROC | Loss-Func | Others |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| González-Briones et al | 2019 | Model-based | User-Based, Case Base Reasoning and Multi Agent Sys | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Accept/Reject/Ignore Rates |
| Benabderrahmane et al | 2017 | Model-based | Collaborative Filtering, Time Series, Deep Learning, DNN | – | – | – | – | – | – | + | – | – | – | – | – | + | – | – | – | – | |
| Yang et al | 2021 | Model-based | Fuzzy sets, BWM, TOPSIS and LSGDM | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Criterion-Weight, Novel Distance Measure, Node Degree & Eigenvector Centralities |

that utilizes the MyCareer API to collect job seekers'profile data in XML format, and the ELISE component employs a knowledge-based recommendation approach to match user profiles with suitable job positions. GeoServices are utilized for location information, and MySQL is used for storing data on previous user searches. Experimental studies using mixed methods revealed positive outcomes. Evaluation results demonstrated that the prototype enables job seekers to discover, understand, and find suitable open positions, regardless of their background and age. However, it's important to note that the prototype has yet to be linked to actual job positions. Broader user evaluations are recommended for a more comprehensive assessment. Additionally, the interface was evaluated within a specific job recommendation domain. To generalize the findings, exploring the design process in a different area may yield new discoveries.

Rácz et al. [121] examined how well a set of skill information required for a specific job posting matched individuals'profile information. Filtering and matching methods were applied using hierarchical data on ontologies. A model was then obtained that addresses the similarities between different skills not related by the subsume relationship. Finally, probabilistic matching analysis based on maximum entropy was used to evaluate the model. The results showed that the algorithms could produce completely different orderings of job applicants for the same job posting. Therefore, the algorithm that provides the most accurate ordering depends heavily on the field.

Rimitha et al. [122] have designed an ontology based JRS, which recommends a set of suitable jobs to the users using their profile information. Personal job preferences such as favorite jobs list and recently viewed jobs are collected as user characteristics. The researchers considered two sequential algorithms in the system. In the first step, job postings that a particular user viewed and favorited recently are kept in user profile data. In the next step, the profile data collected is processed to find a set of suitable jobs for the user. In experimental studies, the number of jobs recommended by the system was less than the normal search and more relevant results were returned based on user's preferences.

Nguyen et al. [106] targeted newly graduated individuals as the user group in their research. The researchers proposed an ontology based JRS to recommend the most suitable job positions for users based on the demands of the job and the abilities of the students to positively develop their careers. Ontologies have been used to model industrial job positions and students'abilities in their study. The researchers used a set of functions in the matching process that provides match between the job modeling and the student modeling for job recommendation. Experimental studies show a high correlation between the recommendations of the proposed JRS and the choices of recruiter.

Salazar et al. [126] proposed a case-based Multi-Agent System (MAS) with an ontology knowledge base to facilitate matching between the information of the job seekers and employers. The HR-XML standard was used to standardize knowledge representation through ontologies and to map resumes. The proposed system was designed according to the Prometheus methodology and then a prototype of the system was engineered. The researchers performed experimental studies to verify their system by conducting a case study.

Çelik et al. [34] proposed an ontology-based JRS engine that can convert automatically the free-formatted resumes uploaded by job seekers through the websites of online

recruitment companies into a structured form via the JRS and transfer extracted information to the relevant fields in the system's relational databases. Researchers shared the outputs and experiences of their research conducted jointly by Kariyer.net (Turkey's largest online recruitment site) and TÜBİTAK (Turkish Scientific and Technological Research Council). In the system, the ontology knowledge base was used to transform the free-formatted resumes uploaded by job seekers into an OWL form (structured). The system was tested on 250 free-formatted/unstructured Turkish and English resumes and the expected meaningful parsing from the résumés was achieved with %80,00 success.

Tamburri et al. [141] proposed using the DataOps model and AI to localize skills in the labor market and extract skills from open job positions using ML models from resumes. Real dataset is used (actual data from Flemish employment agencies in the Netherlands and Belgium). Results were obtained with ML on these data. Main purposes are to match the inferred skills with the work and occupation ontologies (ISCO, O*NET, ESCO) within the ontology. DataOps has been used to combine Big Data and ML models to process resumes with suitable job opportunities, skill extraction and matching, and to find phrases that contain skills. BERT Cloud, BERT Local, and Simple-NN pre-training models were used for model training. According to the accuracy result, BERT Cloud was the best performing model. As a future work, they planned to collect more data and integrate it with standard ontologies in order to better develop the proposed prototype.

A limited number of solutions have been found for knowledge-based JRSs in literature. Generally ontology, rule, case and content-based approaches have been discussed and observed.

#### *An overview and comparison of knowledge-based JRSs*

In Table 10, the objectives, evaluation methods used, development tools and methods used, advantages and disadvantages of knowledge-based JRSs from the reviewed articles are examined and compared. Also, according to the articles reviewed above on knowledge-based JRSs, the evaluation metrics used in knowledge-based JRS approaches are summarized in Table 11.

### Context-awareness based filtering JRSs

According to the RSs and JRSs survey researches reviewed and presented in Sect."Other Similar Literature Review Studies"and our own observations, context-aware filtering in JRS can be characterized by the specific domain it focuses on. For instance, a context may relate to a user's current state or changes in preferences and activities. In e-commerce, a context might involve information helping identify user classes based on his/her purchase intentions. Contextual information can also include data accessible via mobile devices, such as location, contacts, and nearby places of interest. Consequently, context-aware based filtering JRS solutions can be categorized as Demographic, Social Relationships (Social Network based), Cognitive, Place and Time (e.g., month, hour, weekday), and Activity-based. However, according to our literature review, JRS solutions using a context-aware filtering approach appear to be limited to only two demographic JRSs found in the literature, as presented below.

**Table 10** Overview and comparing of the knowledge-based JRS studies reviewed

| Article | Year | Category | Approaches used | Main ideas | Dataset/evaluation methods | Applied tools | Limitations | Future studies |
|---|---|---|---|---|---|---|---|---|
| Fernández-Sanz et al | 2017 | Model-based | Competences Skills | Developing a comprehensive matching framework to benefit job seekers and employers | Not mentioned | AXELOS Skills Framework | Recognized limitations cover the potential emergence of new professions and novel concepts of competence and skills associated with these evolving profession | In the future, the details of the study will also include recommendation modules that will guide the training of professionals |
| Gutiérrez et al | 2019 | Model-based | User-centered tool based on a client–server architecture | A user-centered"Labour Market Explorer"tool is proposed that facilitates job seekers to find personalized job recommendations based on their skills and competencies | The researchers asked some survey questions to these participants and then evaluated the results obtained to answer their main research questions. Participants: a group of job seekers and a group of job mediators | XML, ELISE developed by Public Employment Service VDAB, MySQL | Identified limitations include the necessity for a connection with real open job positions, the requirement for more comprehensive user data for testing, and the focus on a very specific domain of job recommendations | Additional research is planned to investigate job mobility scenarios and explore further"what-if"scenarios |
| Rácz et al | 2016 | Probabilistic Model-based | Semantic matching, Ontology matching, Lattice filters, Probabilistic Maximal Length Matching (Maximum Entropy) | A JRS model is designed that matches the profile information of a job seeker's skills and the skills required by a particular job posting | Similarity metrics (Matching Using Ontology Edges vs Maximum Entropy) | Not Mentioned | Not Mentioned | Not Mentioned |
| Rimitha et al | 2019 | Model-based | GitHub Jobs API, JSON, Building User Profiles and Getting Job Recommendations Algorithms | A conceptual model and a desktop-based prototype for the ontology-based JRS for job seekers were developed | Case study-based verification. The dataset used was not detailed | Java, Eclipse IDE, GitHub Jobs API, MySQL | Not Mentioned | Their plan involves employing ML algorithms in the recommendation task and extending the job ontology to other domains |

**Table 10** (continued)

| Article | Year | Category | Approaches used | Main ideas | Dataset/evaluation methods | Applied tools | Limitations | Future studies |
|---|---|---|---|---|---|---|---|---|
| Nguyen et al | 2013 | Ontology-based user model | Ontology Knowledge base, Job Matching technique | An ontology-based JRS has been developed to facilitate finding the most suitable job positions for a student based on the job demands and the student's abilities | Euclidean Distance | SoNITS | Not Mentioned | Not Mentioned |
| Salazar et al | 2015 | Model-based | Ontologies, HR-XML Standard, Case-based reasoning, Multi Agent Systems (MAS), Negotiation support systems | Researchers focused on HR-XML for resume representation in XML files and developed an ontology-based MAS using semantic HR-XML descriptions. The system aims to match job seekers with employers'postings and offer recommendations | A case study has been done. The dataset is not mentioned | JADE Framework, HR-XML standard, JAVA, OWL language, Protégé ontology editor, SPARQL query language | The limited features of the HR-XML standard have been taken into account | More features of the HR-XML standard are planned for further studies to make better recommendations |
| Çelik et al | 2013 | Model-based | Ontologies, N-Gram Algorithm, Jaro-Wrinkler Similarity Algorithm | A JRS engine is proposed with the goal of automatically transforming free-formatted resumes submitted by job seekers through online recruitment company websites into a structured format using ontologies. The parsed information is subsequently forwarded to the system's database | The proposed JRS verified with 250 free formatted Turkish resumes during case studies and achieve average 80% of information extraction from the resumes in the dataset | OWL, Protégé editor, OWL API, Java, SWRL, | JRS can still extract a few fields from CVs (e.g. work experience, education information, general information, etc.) and some sections are ignored | SWRL rule-based inferencing mechanism is planned to be enhanced |
| Tamburri et al | 2020 | Model-based | O*NET, ESCO, ISCO vector models, Ontology Knowledge base, Decision-making, ML, BERT Cloud and BERT Local, Simple-NN | Using the DataOps model and AI to localize skills in the labor market and extract skills from open job positions using ML models from resumes | BERT Cloud, BERT Local and Simple-NN are performed as evaluation method | Not Mentioned | Not Mentioned | Planned to collect more data and integrate it with standard ontologies in order to better develop the proposed prototype |

**Table 11** An overview of the evaluation metrics used in the knowledge-based JRS approaches reviewed

| Article | Year | Category | Methodology | Accuracy | Recall | Precision | Cost | Reliability | Coverage | F-Score | MAE/MRE | Security | Robustness | Scalability | Time | MSE/RMSE | Hit-Rate | NDCG | AUC | Loss-Func | Others |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Fernández-Sanz et al | 2017 | Model-based | Competences Skills | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Not Mentioned |
| Gutiérrez et al | 2019 | Model-based | User-Based, Interactive Visualization Appr | + | – | – | – | – | – | – | –– | – | – | – | – | – | – | – | – | – | |
| Rácz et al | 2016 | Model-based | User-Based, Clustering | + | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | |
| Rimitha et al | 2019 | Model-based | User-Based | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Not Mentioned |
| Nguyen et al | 2013 | Model-based | User-Based | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Not Mentioned |
| Salazar et al | 2015 | Model-based | User-Based, Ontologies, HR-XML Standard | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Not Mentioned |
| Çelik et al | 2013 | Model-based | User-Based, Ontologies, SWRL Rules | + | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | |
| Tamburri et al | 2020 | Model-based | User-Based, Ontologies, ML, Big data | + | + | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | |

***Demographic JRSs***

Demographic RSs are designed to provide recommendations to users based on their demographic attributes [6, 27]. These systems use demographic classifications to make predictions, such as suggesting products tailored for female users or specific age groups. Unlike other RSs, demographic RSs do not consider users'past actions, such as ratings, likes, or preferences. Instead, they typically involve posing a set of questions to users to classify them based on demographic characteristics, and recommendations are then generated based on similarity results obtained from this classification process [40].

*JRS approaches and practices reviewed on demographic based filtering* Arita et al. [10] proposed a JRS tailored for senior employees in the community to address aging-related issues. This solution aims to match the competencies of senior employees with current job postings, enhancing their quality of life by safeguarding and promoting health, strength, and skills. Categorized as a Demographic JRS, it targets senior employees in the community. The researchers developed the web-based application"Gathering Brisk Elderly in the Region-GBER,"functioning as an online platform for consumer-to-consumer/provider matching, allowing senior workers to find local jobs. GBER includes four user interfaces to extract user information and label jobs, utilizing an algorithm to match senior employees to available jobs and update their profiles. In a pilot study conducted over seven months with ninety-two senior employee users in Kashiwa, Japan, the GBER system collected daily data, which was analyzed. The employee-job matching considered contextual data such as Time, Place, and Skills, taking into account factors like time availability, commuting details, and reduced physical strength or IT skills. The system successfully retrieved users'time, location, and skill information through proposed interfaces, demonstrating its usability and the ability to promote senior local activities. Matching results related to time were found to be particularly suitable, showcasing the effectiveness of the system.

Zhao et al. [164] focused on high school students and devised a JRS for finding summer jobs, emphasizing person-occupation fit and comprehensive assessment algorithms. Various factors, including quantitative and qualitative elements, were considered. The researchers employed the well-known Holland career interest test model, categorizing personality into six dimensions: investigative (I), Entrepreneurial (E), Traditional (C), Artistic (A), Social (S), and Realistic (R). Users utilized the Holland rating scale to create a personalized professional interest map based on scores in these six factors. Matching was calculated using the geometric mean. The researchers evaluated their model using a job posts dataset and virtual characters, applying similarity measurement approaches commonly used in demographic-based JRSs.

*An overview and comparison of demographic based JRSs* In Table 12, the objectives, evaluation methods used, development tools and methods used, advantages and disadvantages of demographic-based JRSs from the reviewed articles are examined and compared. Also, according to the articles reviewed above on demographic filtering JRSs, the evaluation metrics used in this approach are summarized in Table 13.

**Table 12** Overview and comparison of the demographic filtering JRS studies reviewed

| Article | Year | Category | Approaches Used | Main Ideas | Dataset/Evaluation Methods | Applied Tools | Limitations | Future Studies |
|---|---|---|---|---|---|---|---|---|
| Arita et al | 2017 | Model-based | Matching the competencies of senior employees with current job postings using similarity approaches | Discusses a ranking-based JRS focused on matching the competencies of elderly senior employees in the community with current job postings. For ranking, the inner product of 72-dimensional interest vectors and 72-dimensional vectors of job labels was calculated | Data from 92 users aged over 60 were gathered through Q&A cards, with most participants being volunteer members of local senior groups. A total of 130 jobs were registered, resulting in 874 matches | Jquery mobile widgets were utilized for the development of the Q&A tool, accessible on both PC and smartphone platforms | Not Mentioned | Implementing the proposed JRS, expanding its services to other cities, thus providing incentives to the workforce by strengthening the skills of elderly senior employees |
| Zhao et al | 2021 | Model-based | Holland's Self-Directed Search Model, Holland Assessment Scale, Geometric Mean, Fuzzy Logic, Consistency Index, Random Index and Consistency Ratio | Proposed a JRS tailored for high school students during summer vacation, incorporating person-job fit and comprehensive assessment algorithms | 100 occupations were specified, then 3–5 specific positions were collected for each type of occupation from popular job sites in the US and CareerBuilder sites. In total, close to 70 profile information were analyzed | Not Mentioned | A limited dataset was used, resulting in an oversight of several aspects of job seekers'needs. It may be necessary to consider additional crucial factors for a more comprehensive assessment | Planned to include more factors (e.g. location) to further optimize the proposed decision model |

**Table 13** An overview of the evaluation metrics used in the demographic filtering JRS studies reviewed

| Article | Year | Category | Methodology | Accuracy | Recall | Precision | Cost | Reliability | Coverage | F-Score | MAE/ MRE | Security | Robustness | Scalability | Time | MSE/ RMSE | Hit-Rate | NDCG | AUC | Loss-Func | Others |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Arita et al | 2017 | Model-based | User Characteristics Matching | + | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | |
| Zhao et al | 2021 | Model-based | User-Based Multi-Factor Analysis | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | Consistency Ratio |

### Other types of JRSs

#### *Popularity-based JRSs*

Popularity-based RSs by nature do not analyze user behaviors; rather, they recommend products or items based on general popularity, trends, or wide consumption [80]. These systems do not take into account a user's individual preferences. While they suggest popular or trending items, they can serve as an opportunity to broaden users'interests and promote diverse application preferences. For instance, during the Covid-19 pandemic, an e-commerce site employing a popularity-based RS might recommend masks or disinfectants, even when a user is viewing unrelated products. It's worth noting that, as per the literature reviewed, a popularity-based JRS solution has not been identified.

#### *Non-personalized JRSs*

Non-personalized RSs are often the primary approach for many recommendation sites. In these systems, there is no need to collect information about users'past preferences, which simplifies implementation and operation. Non-personalized RSs bear a striking resemblance to popularity-based RSs. Unlike personalized RSs, which take into account a user's past interactions (e.g., purchase history), non-personalized RSs recommend items based on their popularity among all users in a certain time period [117]. As a result of our research, a non-personalized JRS solution has not been described in the literature.

#### *Reciprocal JRSs*

In traditional RSs, the success of a recommendation item is determined solely by the end user's response to the recommendation items produced by the system. However, in a Reciprocal Recommendation System (RRS) this is the opposite. Users interact with other users through recommendation items and a match/approval is expected at the end [112]. RRS solutions have become more common in various online matching services such as finding friends/partners, finding a job, etc. In RRSs, for a recommendation item to be considered successful, the opinions or satisfaction of users on both sides about the recommendation item are taken into account. In other words, for an RRS recommendation item to be considered successful, both parties must be satisfied with it, and a mutual agreement must be reached. Additionally, in RRS solutions, if two users agree to connect with each other and are satisfied with the match done, RRS may no longer recommend the recommended user to other users [112].

In addition, RRS-based JRS solutions have been mentioned in the literature, and it has been claimed that a recruitment process based on mutual consent can be carried out. The steps in this match are:

(1) A job seeker's profile is matched with active job postings and the matched job postings are prioritized for that job seeker.
(2) In order to recommend the most suitable job candidates for a particular job opportunity, appropriate candidates are identified, ranked and recommended to the organization's HR personnel.

Therefore, the recommendations generated in an RRS-based JRS solution are evaluated by both parties and the recommendations must be satisfactory to both job seekers and the organization [82, 153, 162].

*JRS approaches and practices reviewed on reciprocal JRSs* According to our investigation, JRS solutions based on RRS are quite limited in the literature. Recommended JRS solutions typically either recommend candidate resumes to HR personnel or list suitable job opportunities recommended to job seekers. In general, JRS solutions in the literature support one-way matching. Only three RRS-based JRS solutions were found in the literature between January 1, 2010 and April 30, 2023 are presented below.

Li and Li [82] introduced a model called MEET, based on bipartite graph modeling, designed to preserve users'correlations and global"reciprocal"utilities within a RRS-based JRS. The model was evaluated using real-world datasets on online dating and online recruitment, comparing its performance with existing RSs. The study highlighted challenges in reciprocal RSs, including issues related to reciprocity, limitedness, passivity, and sparseness. A new quality metric, Set Vitality, was proposed to assess the recommendation list, taking into account users'interaction history. The paper discussed Gradient Boosted Decision Trees (GBDTs), and online meeting (RECON) and content-collaborative recommendation systems (CCR) in the context of reciprocal recommendation. Performance comparisons of MEET alternatives were made using metrics such as precision, recall, NDCG, and F1-score. The authors outlined future plans to further develop MEET for handling more reciprocal tasks.

Wenxing et al. [153] developed Reciprocal-based a mobile JRS named iHR +. The system, focusing on job seekers and job characteristics, calculated bi-directional preferences based on cross-similarity to achieve reciprocal recommendations. Cosine Similarity was employed for similarity calculations within iHR +. Text features were processed using Lucene IK Analyzer for Chinese Word Segmentation and TF-IDF transformations. The implementation utilized SQL Server 2008, Ajax, and jQuery Mobile as tools.

Yu et al. [162] introduced a reciprocal-based recommendation algorithm based on two-way matching to address the challenge of providing bi-directional recommendations between users and recommendation items. The algorithm incorporates methods for calculating users'preferences and measuring similarity. By combining explicit and implicit user preferences, the proposed algorithm accurately identifies user characteristics and interests. A calculation method is devised to extract job preference information from resumes. The algorithm considers the differential impact of explicit and implicit preferences on user similarity calculation and integrates both types of preference information. The study observed an increased accuracy rate in recommendations with the proposed algorithm. The TF-IDF algorithm was employed to calculate the weight of terms.

*An overview and comparison of reciprocal JRSs* In Table 14, the objectives, evaluation methods used, development tools and methods used, advantages and disadvantages of RRS based JRSs from the reviewed articles are examined and compared. Also, according to the articles reviewed on RRS-based JRSs, the evaluation metrics used in this approach are summarized in Table 15.

**Table 14** Overview and comparison of the reviewed reciprocal filtering JRS approaches

| Article | Year | Category | Approaches Used | Main Ideas | Dataset/Evaluation Methods | Applied Tools | Limitations | Future Studies |
|---|---|---|---|---|---|---|---|---|
| Li & Li | 2012 | Bipartite Graph Modeling | MEET model, Gradient Boosted Decision Trees (GBDTs), RECON, CCR | Preserves user correlations and global reciprocal utilities within RRS-based JRS. Introduced Set Vitality metric | Real-world datasets (online dating & recruitment) | Not Mentioned | Challenges such as reciprocity, limitedness, passivity, and sparseness for RRs have been identified | Extending MEET for broader reciprocal tasks |
| Wenxing et al | 2015 | A mobile Reciprocal-based JRS is proposed | iHR +, Bi-directional preference, Cosine Similarity | Focus on job seekers and job characteristics, utilizing cross-similarity for reciprocal recommendations | Not Mentioned | SQL Server 2008, Ajax, jQuery Mobile, Lucene IK Analyzer, TF-IDF | Limited dataset and focus on Chinese job market | Not Mentioned |
| Yu et al | 2011 | A reciprocal-based recommendation algorithm based on two-way matching | Explicit & implicit user preference integration, TF-IDF, Similarity Scoring | Enhances accuracy by combining explicit and implicit user preferences for bidirectional recommendations | Not Mentioned | TF-IDF Algorithm | Potential bias in preference weighting | Further refinement of user similarity calculations |

**Table 15** An overview of the evaluation metrics used in the reciprocal filtering JRS approaches reviewed

| Article | Year | Category | Methodology | Accuracy | Recall | Precision | Cost | Reliability | Coverage | F-Score | MAE/MRE | Security | Robustness | Scalability | Time | MSE/RMSE | Hit-Rate | NDCG | AUC | Loss-Func | Others |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Wenxing et al | 2015 | Model-based | TF-IDF, Cosine Similarity | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | – | |
| Yu et al | 2011 | Model-based | Similarity, Ranking Analysis | + | + | + | – | – | – | – | – | – | – | – | – | – | – | – | – | – | |
| Li and Li | 2012 | User-Oriented | A new Reciprocal-based JRS alg | + | + | + | – | – | – | + | – | – | – | – | – | – | – | – | – | – | Reciprocity, Limitedness, Passivity & Sparseness |

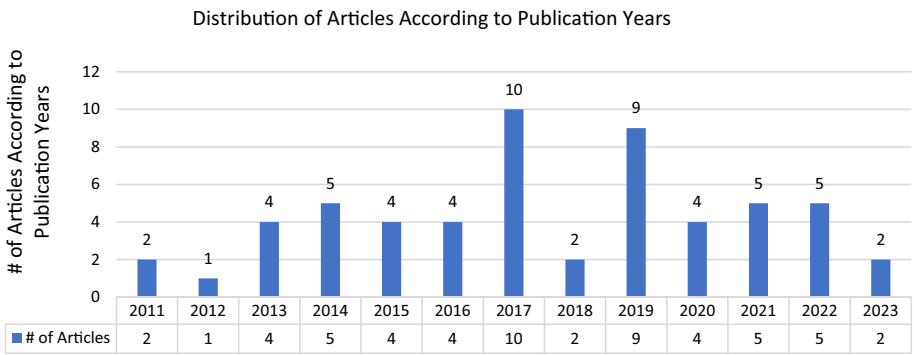


**Fig. 7** Distribution of reviewed articles according to publication years

## Results and comparisons

In this section, the results of our SLR study by considering our search protocol mentioned earlier are evaluated. This section is divided into two main parts (1) a summary of the reviewed articles after the screened process done is presented, and (2) a comparative presentation of the results and achievements obtained in response to the research questions is summarized below.

RQ1: What are the approaches and methodologies applied in the JRSs?
RQ2: What are the main ideas focused on in JRSs?
RQ3: What are the quality and performance measurements applied in the evaluation of JRSs?
RQ4: What are the popular evaluation and modeling tools used in developing JRSs?
RQ5: What are the current challenges and future improvements in JRS implementations?
RQ6: What are the available standards applied in the JRSs?

## Summary of the selected studies

The following research and discovery questions have been selected to explore current technology and applications from the recent reviewed studies in JRS:

- How many articles about JRS were published in which year?
- Which publishers have published the most articles about JRSs?
- Which filtering technique do researchers focus on most in JRS solutions?

Figure 7 presents the distribution of all articles examined between 2010 and 2023 according to their publication years. While the highest number of articles (10) were published in 2017, the smallest number of articles (1) were published in 2012. Figure 8 shows the distribution of the reviewed articles according to their publication associations. 47% of articles are published in IEEE, 19% in Springer, 18% in ACM, 5% in Elsevier, 3% in Emerald, 4% in Nature, and 2% in Hindawi and others.

As mentioned before, 6 filtering types of RSs have been discussed in this study: (1) Collaborative filtering, (2) Content-based filtering, (3) Hybrid based filtering, (4) Knowledge

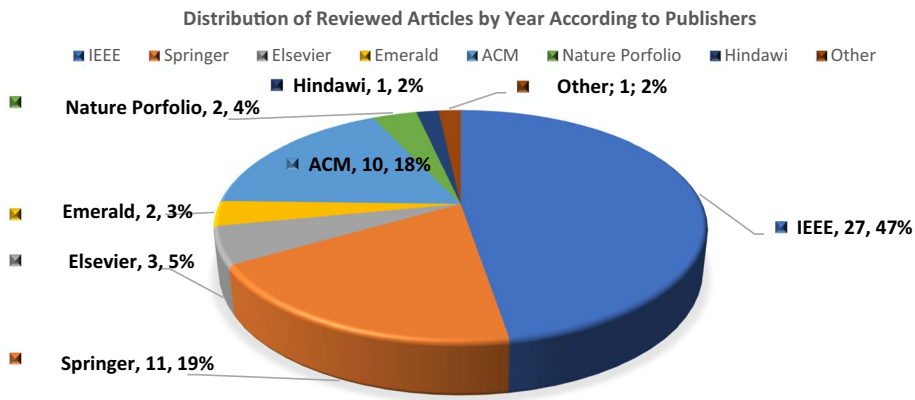


**Fig. 8** Distribution of reviewed articles by year according to publishers

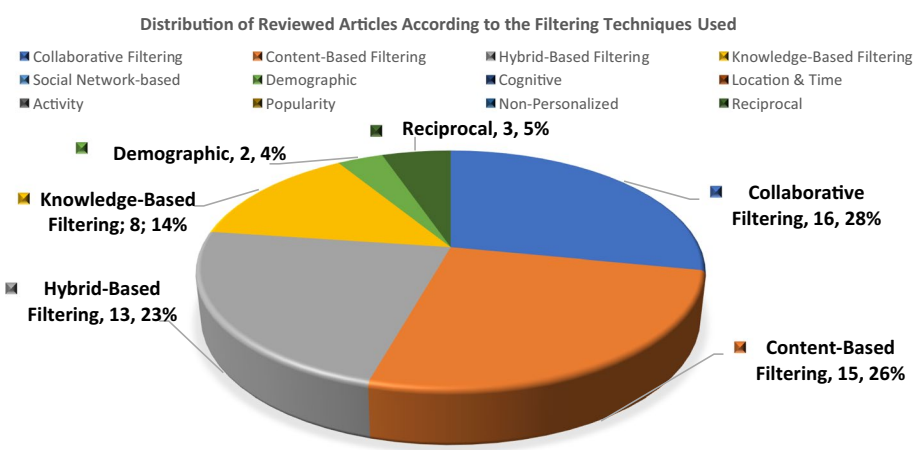


**Fig. 9** Distribution of reviewed articles according to the filtering techniques used

based filtering, (5) Context Aware filtering, and (6) others. The filtering approaches used of the reviewed JRS solutions are presented in Fig. 9.

It provides information about the distribution according to preferred filtering approaches in literature: 28% of articles used Collaborative filtering, 26% of articles used Content based filtering, 23% of articles used Hybrid based filtering, 14% of articles used Knowledge based filtering, 2% of articles used Demographic based filtering, and 2% of articles used Reciprocal based filtering. In addition, the reviewed JRS solutions were classified into two categories as journal or conference publication types that are presented in Table 16 according to their publication type, year, filtering approach, publisher, and article title.

### Answers to research questions based on the findings and achievements obtained

As mentioned before, in this study, filtering techniques used in RSs are investigated under six categories: (1) Collaborative filtering, (2) Content-based filtering, (3) Hybrid based filtering, (4) Knowledge based filtering, (5) Context Aware filtering, and (6) others.

**Table 16** The distribution of the reviewed articles according to publication type, year, filtering approach used, publisher, and article title

| # | Type | Year | Filter | Publisher | Article Title |
|---|---|---|---|---|---|
| 1 | C | 2019 | COLLABORATIVE | ACM | Recruitment and Recommendation System Based on Intelligent Computing |
| 2 | J | 2011 | COLLABORATIVE | ELSEVIER | Interaction mining and skill-dependent recommendations for multi-objective team composition |
| 3 | C | 2019 | COLLABORATIVE | ELSEVIER | An automatic skills standardization method based on subject expert knowledge extraction and semantic matching |
| 4 | J | 2016 | COLLABORATIVE | EMERALD | Job-major match and job satisfaction in Italy |
| 5 | C | 2016 | COLLABORATIVE | IEEE | A Job Recommendation Method Optimized by Position Descriptions and Resume Information |
| 6 | C | 2016 | COLLABORATIVE | IEEE | The Analysis and Design of the Job Recommendation Model Based on GBRT and Time Factors |
| 7 | C | 2017 | COLLABORATIVE | IEEE | Bayesian Multi-View Models for Member-Job Matching and Personalized Skill Recommendations |
| 8 | C | 2017 | COLLABORATIVE | IEEE | Interoperability and Scalability for Worker-Job Matching across Crowdsourcing Platforms |
| 9 | C | 2020 | COLLABORATIVE | IEEE | Skill-based Career Path Modelling and Recommendation |
| 10 | C | 2021 | COLLABORATIVE | IEEE | Explainable Job-Posting Recommendations Using Knowledge Graphs and Named Entity Recognition |
| 11 | C | 2021 | COLLABORATIVE | IEEE | Symbolic AI for XAI: Evaluating LFIT Inductive Programming for Fair and Explainable Automatic Recruitment |
| 12 | C | 2022 | COLLABORATIVE | IEEE | Multiclass Job Recommendation System in the IT Field between Classification and Prediction Method |
| 13 | C | 2022 | COLLABORATIVE | IEEE | Job Recommendation Based on Multiple Behaviours and Explicit Preferences |
| 14 | J | 2023 | COLLABORATIVE | IEEE | Comparison of Real-Time and Batch Job Recommendations |
| 15 | J | 2017 | COLLABORATIVE | SPRINGER | Decision support system based on genetic algorithm and multi-criteria satisfaction analysis (MUSA) method for measuring job satisfaction |
| 16 | C | 2022 | COLLABORATIVE | SPRINGER | Job Training Recommendation System: Integrated Fuzzy AHP and TOPSIS Approach |
| 17 | J | 2019 | CONTENT | EMERALD | Exploring the impact of training, job tenure, and education-job and skills-job matches on employee turnover intention |
| 18 | C | 2014 | CONTENT | IEEE | Taxonomy-based Job Recommender Systems on Facebook and LinkedIn Profiles |
| 19 | C | 2014 | CONTENT | IEEE | A Content Based Approach for Recommending Personnel for Job Positions |
| 20 | C | 2014 | CONTENT | IEEE | Field Selection for Job Categorization and Recommendation to Social Network Users |
| 21 | C | 2015 | CONTENT | IEEE | FoDRA—A New Content-Based Job Recommendation Algorithm for Job Seeking and Recruiting |
| 22 | C | 2015 | CONTENT | IEEE | Generating Personalized Job Role Recommendations for the IT Sector through Predictive Analytics and Personality Traits |
| 23 | C | 2017 | CONTENT | IEEE | Job and Candidate Recommendation with Big Data Support: A Contextual Online Learning Approach |
| 24 | C | 2019 | CONTENT | IEEE | Employee Recruitment Recommendation Using Profile Matching and Naive Bayes |
| 25 | C | 2019 | CONTENT | IEEE | Job Recommendation System based on Machine Learning and Data Mining Techniques using RESTful API and Android IDE |
| 26 | C | 2019 | CONTENT | IEEE | Engineering Solutions on Multimodal Profiling Tool for Digital Jobs Analysis and Matching of Requirements Competences Frameworks |
| 27 | C | 2020 | CONTENT | IEEE | Based on the application of AI technology in resume analysis and job recommendation |
| 28 | J | 2020 | CONTENT | SPRINGER | A Novel Approach for Learning How to Automatically Match Job Offers and Candidate Profiles |

**Table 16** (continued)

| # | Type | Year | Filter | Publisher | Article Title |
|---|---|---|---|---|---|
| 29 | C | 2014 | CONTENT | SPRINGER | Combining Career Progression and Profile Matching in a Job Recommender System |
| 30 | C | 2022 | CONTENT | SPRINGER | Investigating Natural Language Processing Techniques for a Recommendation System to Support Employers, Job Seekers and Educational Institutions |
| 31 | C | 2023 | CONTENT | SPRINGER | Joint Extraction and Classification of Danish Competences for Job Matching |
| 32 | C | 2017 | DEMOGRAPHIC | ACM | GBER: A Social Matching App which utilizes Time, Place, and Skills of Workers and Jobs |
| 33 | C | 2021 | DEMOGRAPHIC | ACM | Summer Job Selection Model Based on Job Matching and Comprehensive Evaluation Algorithm |
| 34 | C | 2013 | HYBRID | ACM | A Recommender System for Job Seeking and Recruiting Website |
| 35 | C | 2017 | HYBRID | ACM | LiJAR: A System for Job Application Redistribution towards Efficient Career Marketplace |
| 36 | C | 2017 | HYBRID | ACM | Personalized Job Recommendation System at LinkedIn: Practical Challenges and Lessons Learned |
| 37 | C | 2018 | HYBRID | ACM | A Combined Representation Learning Approach for Better Job and Skill Recommendation |
| 38 | C | 2021 | HYBRID | ACM | Cost-Effective and Interpretable Job Skill Recommendation with Deep Reinforcement Learning |
| 39 | J | 2022 | HYBRID | HINDAWI | Job Matching Analysis Based on Text Mining and Multicriteria Decision-Making |
| 40 | C | 2013 | HYBRID | IEEE | Quantifying and Recommending Expertise When New Skills Emerge |
| 41 | C | 2014 | HYBRID | IEEE | A Model to Forecast the Matched degree between Staffs and Jobs |
| 42 | C | 2017 | HYBRID | IEEE | Predicting the Users Clickstreams Using Time Series Representation, Symbolic Sequences, and Deep Learning: Application on Job Offers Recommendation Tasks |
| 43 | J | 2017 | HYBRID | NATURE | Combining content-based and collaborative filtering for job recommendation system: A cost-sensitive Statistical Relational Learning approach |
| 44 | J | 2021 | HYBRID | NATURE | Two-sided matching based on I-BTM and LSGDM applied to high-level overseas talent and job fit problems |
| 45 | C | 2018 | HYBRID | SPRINGER | How to Match Jobs and Candidates—A Recruitment Support System Based on Feature Engineering and Advanced Analytics |
| 46 | C | 2019 | HYBRID | SPRINGER | Case-Based Reasoning and Agent Based Job Offer Recommender System |
| 47 | C | 2019 | KNOWLEDGE | ACM | Explaining and Exploring Job Recommendations: A User-driven Approach for Interacting with Knowledge-based Job Recommender Systems |
| 48 | J | 2017 | KNOWLEDGE | ELSEVIER | e-Skills Match: A framework for mapping and integrating the main skills, knowledge and competence standards and models for ICT occupations |
| 49 | C | 2013 | KNOWLEDGE | IEEE | Towards an Information Extraction System based on Ontology to Match Resumes and Jobs |
| 50 | C | 2019 | KNOWLEDGE | IEEE | Improving Job Recommendation Using Ontological Modelling and User Profiles |
| 51 | C | 2020 | KNOWLEDGE | IEEE | DataOps for Societal Intelligence: a Data Pipeline for Labor Market Skills Extraction and Matching |
| 52 | C | 2013 | KNOWLEDGE | SPRINGER | Supporting Career Counselling with User Modelling and Job Matching |
| 53 | C | 2015 | KNOWLEDGE | SPRINGER | A Case-Based Multi-Agent and Recommendation Environment to Improve the E-Recruitment Process |
| 54 | C | 2016 | KNOWLEDGE | SPRINGER | Semantic Matching Strategies for Job Recruitment: A Comparison of New and Known Approaches |

**Table 16** (continued)

| # | Type | Year | Filter | Publisher | Article Title |
|---|---|---|---|---|---|
| 55 | C | 2012 | RECIPROCAL | ACM | MEET: a generalized framework for reciprocal recommender systems |
| 56 | C | 2015 | RECIPROCAL | IEEE | iHR +: A mobile reciprocal job recommender system |
| 57 | J | 2011 | RECIPROCAL | OTHER | Reciprocal recommendation algorithm for the field of recruitment |

In addition, a taxonomy based on these six categories is developed and shown in Fig. 6. In the 57 JRS articles reviewed, research findings such as filtering approaches applied, main ideas, applied techniques/methods, performance evaluation metrics, and tools used were researched, compared and presented below.

RQ1: What are the approaches and methodologies applied in the JRSs?

- As presented in Table 4, a total of 16 of the 57 JRS articles reviewed applied the collaborative filtering approach, and 4 of them were published as journal publications and 12 as conference publications. Various methods such as graph-based models (e.g., RP3Beta, KGs, Dijkstra's algorithm, MBL, Bayesian Networks), statistical hypothesis testing (e.g., A/B test), multicriteria decision analysis (e.g., MUSA, GMUSA, TOPSIS) with various optimization techniques (e.g., Simulated Annealing, Gradient Descent, GA), neural networks (e.g., Fuzzy logic, RNN, LSTM, Bayesian), series of stochastic (e.g., binary-valued latent states), predictive models (e.g., K-fold cross-validation, PRIDE, Gradient Boosting, T-GBRT, CART), dimension reduction models (e.g., Low Rank Matrix Factorization), regression analysis (e.g., Linear Least Squared Error), similarity learning and methods (e.g. Log-Likelihood Ratio, KNN, SVM, Cosine Similarity, T-SNE, DBSCAN), NLP techniques (e.g., Word2 Vec, GloVe, POS tagging, PV-DM model, BELU, Rouge-L scoring), inductive logic programming (e.g., LFIT, GULA), Semantic Web (ontology knowledgebase, inferencing through semantic rules), density-based clustering, etc. have been applied in the development of collaborative JRS solutions.
- As presented in Table 6, a total of 15 of the 57 JRS articles reviewed applied the content filtering approach, and 2 of them were published as journal publications and 13 as conference publications. Various methods such as computational intelligence-based classification approaches (e.g., Fuzzy logic, Naive Bayes, SVM), probabilistic and heuristic driven algorithms (e.g., Monte-Carlo tree search), dimension reduction models (e.g., UMAP, PCA), similarity-based clustering algorithms (e.g., DBSCAN, K-means, Silhouette method, ITFACP), regression analysis (e.g., hierarchical-regression), fresh Cosine similarity methods in matching (e.g. Basic Cosine, Field Weighting, Field to Field Weighting, Average Field to Field similarity), NLP-based techniques (e.g., GloVe, Word2 Vec, Term Frequency Matrix, Document Vector, Count Vectorizer, TF-IDF, Cosine similarity, BERT, O*NET Vector Model), data mining techniques and big data approaches, etc. have been applied in the development of content based JRS solutions.

- As presented in Table 8, a total of 13 of the 57 JRS articles reviewed applied the hybrid filtering approach, and 3 of them were published as journal publications and 10 as conference publications. Various methods such as deep learning approaches (e.g., Deep Reinforcement Learning, Latent Embedding Approach, LSTM, RNN, NGAMS, CaSMoS, Multi-Agent Reinforcement Learning, Back Propagation NN), data mining approaches (e.g., K-NN, Clustering, Type-2 Fuzzy Set, Matrix Factorization, Recursive Partitioning Algorithm), NLP-based approaches (e.g., Word2 Vec, TF-IDF, Skip-Gram Model), statistical hypothesis testing (e.g., A/B test), mathematical and predictive models (e.g., SRL Model, GLMix models), graph-based approaches (e.g., RDNs, RFGB, Multi Relational Graphs), multi-criteria-based decision-making (e.g., BWM, LSGDM, TOPSIS), similarity-based clustering algorithms (e.g. K-Means, Euclidean Distance, Ranking Algorithm, Markov Model), statistical learning and regression-based approaches (e.g., Decision Trees, XGBoost, Boosting and Penalization), etc. have been applied in the development of hybrid based JRS solutions.
- As presented in Table 10, a total of 8 of the 57 JRS articles reviewed applied the knowledge-based filtering approach, and 1 of them were published as journal publication and 7 as conference publications. Various methods such as developing a cross-approach frameworks (e.g., skill-based competencies matching approach, negotiation support systems, multi-agent based job matching systems, case-based reasoning systems), Semantic Web approaches (ontologies, semantic rules using SWRL, HR-XML, etc.), adaptive filtering approaches (e.g., Lattice filters), probabilistic based matching analysis (e.g., Probabilistic Maximal Length Matching or Maximum Entropy), NLP techniques (e.g., Jaro-Wrinkler matching, N-Gram matching, O*NET, ESCO, ISCO vector models, BERT Cloud, BERT Local, Simple-NN), have been applied in the development of knowledge-based JRS solutions.
- As presented in Table 12, a total of 2 of the 57 JRS articles reviewed applied the demographic-based filtering approach, and all of them were published as conference publications. Various methods such as similarity-based clustering algorithms, computational intelligence-based classification approaches (e.g., Holland's Self-Directed Search model, Holland Assessment Scale, Geometric Mean, Fuzzy Logic, Consistency Index, Random Index and Consistency Ratio) have been preferred in the development of demographic-based JRS solutions.
- As presented in Table 14, a total of 3 of the 57 JRS articles reviewed applied the reciprocal-based filtering approach, and 1 of them were published as journal publication and 2 as conference publications. Various methods such as graph-based models (e.g., Bi-partite graph modeling), probabilistic and optimization-driven algorithms (e.g., Expectation Maximization Algorithm), NLP-based techniques (e.g., Lucene IK Analyzer for Chinese Word Segmentation, TF-IDF, Cosine similarity), regression and classification analysis (e.g., Gradient-Boosted Decision Trees) and ranking analysis have been mostly preferred in the development of reciprocal JRS solutions.

The distribution of the number of articles according to the filtering approaches used and the techniques applied in the 57 JRS articles reviewed is presented in Table 17.

RQ2. What are the main ideas focused on in JRSs?

**Table 17** Distribution of the number of articles reviewed according to the techniques and filtering types used

| Research Focus | Collaborative | Content | Hybrid | Knowledge | Context-Aware | Reciprocal |
|---|---|---|---|---|---|---|
| TF-IDF | | 4 | 1 | | | 1 |
| K-means, Markov Model, Bayesian Multi-View Modelling | 1 | 1 | 2 | | | |
| Text mining technology | | | 1 | | | |
| Word2vec/Skip-Gram/nGram/GloVe/Silhouette | 2 | 3 | 3 | 1 | | |
| Multi Relational Graph | 1 | | 1 | | | 1 |
| Ranking Algorithm | 1 | | 1 | | | |
| Similarity Methods (Content-based, Euclidean distance, Jaro-Wrinkler, Cosine, Loglikelihood, T-SNE, DBSCAN, UMAP, TOPSIS) | 4 | 5 | 3 | 2 | 1 | 1 |
| Interaction-based relations | | | 1 | | | |
| Deep/Machine Learning | | 3 | 1 | 1 | | |
| Matrix Factorization Model | 1 | | 1 | | | |
| KNN, RNN, Simple-NN, SVM, Naive Bayes, Random Forest, Decision tree | 2 | 4 | 5 | 1 | | |
| XGBoost, LSTM, BERT Cloud and BERT Local, Relational Functional Gradient Boosting (RFGB), POS tagging, K-Fold Cross-Validation | 3 | 2 | 3 | 1 | | |
| A/B testing | 1 | | 1 | | | |
| Generalized Linear Mixed (GLMix) model | | | 1 | | | |
| Semantic/Ontology matching | 1 | | | 6 | | |
| Probabilistic Maximal Length Matching (Maximum Entropy) | | | | 1 | | |
| O*NET, ESCO, ISCO vector models | | 2 | | 1 | | |
| RP3Beta model | 1 | | | | | |
| GMUSA algorithm, Learning from Interpretation Transition (LFIT), CART Algorithm | 3 | | | | | |
| NER | 1 | 2 | | | | |

The distribution of the number of articles according to the filtering approaches used and the main ideas discussed in the 57 JRS articles reviewed is presented in Table 18As a result of our SLR study, it has been observed that the goal of JRS solutions is not only to identify competent individuals who match the skill set required by HR in a shorter time but also to support other relational HRM processes. For instance, processes such as creating pre-employment candidate assessments, predicting an individual's potential performance before hiring, designing cognitive ability tests, or generating assessment materials for job seekers can be easily facilitated through JRS.

Furthermore, it is understood that JRS solutions can be highly useful in socially responsible or unbiased recruitment processes, such as determining the qualifications of

**Table 18** Distribution of the number of articles reviewed according to main ideas and filtering approaches used

| Research Focus | Collaborative | Content | Hybrid | Knowledge | Context-Aware | Reciprocal |
|---|---|---|---|---|---|---|
| Employee job satisfaction | 2 | | | | | |
| Finding proper employee and team composition | 1 | 2 | 1 | | | |
| Demographic/Social background and skillset assessment/Prediction | | 1 | 2 | 1 | 2 | |
| Educational background assessment and Competences Gaps in education field | 2 | 1 | | 1 | | |
| JRS development to match job seeker with job opportunity (Online/Offline/Real-time) | 5 | 7 | 7 | 5 | | |
| Career path planner | 1 | 1 | | | | |
| A novel job matching algorithm/model development | 3 | 1 | 1 | | | |
| Relationships between job tenure and turnover intention assessment | | 1 | | | | |
| A comparing JRSs solutions for different methods | 1 | 1 | | | | |
| An approach to homogenize business knowledge and subject specializations using semantic web | 1 | | | 1 | | |
| Proposing the most demanded sector in the market according to the data collection and data analysis results of the recruitment website | | | 1 | | | |
| Creating JRS and proving how effective it is to combine collaborative filtering and content-based filtering with real-world JRS data | | | 1 | | | |
| Two sided recommendations (Recommend both job seeker with job opportunity) | | | | | | 3 |

inmates seeking job training in prison and suggesting the most suitable training activities. Additionally, developing domain-specific knowledge bases and taxonomies for job-candidate matching to enhance the effectiveness and accuracy of JRS applications is another research area. Focusing on smaller-scale JRS solutions for specific professional fields (e.g., IT, education, healthcare) may be more practical. Developing effective online/offline JRS solutions by converting data into high-dimensional vectors and processing them with various AI methods (e.g., ML models, deep learning, big data analysis, semantic analysis) to calculate exponential correlations and determine the appropriate match between the growing number of job postings and the competency characteristics of job seekers is another popular topic in this field.

RQ3: What are the quality and performance measurements applied in the evaluation of JRSs?

Various evaluation metrics and methodologies to assess the performance success and efficiency of RSs are discussed in Sect."Metrics Used in the Evaluation of Recommender Systems". These metrics are classified in four main categories: (1) Predictive Accuracy Metrics (7 metrics), (2) Classification Accuracy Metrics (10 metrics), (3) Rank Accuracy Metrics (8 metrics), and (4) Non-Accuracy Metrics (5 metrics) [35, 62, 63, 147]. In this section, the distribution of the JRS studies reviewed is examined according to the preferred evaluation metrics and filtering approaches (as seen in Fig. 6).

Table 5 shows that 16 different evaluation metrics were used in the 16 collaborative JRS solutions examined and are presented in Fig. 10, which are:

- *Predictive accuracy metrics* MSE (1 article), RMSE (1 article), similarity assessment using Cosine similarity (1 article) and Jaccard similarity (1 article),
- *Classification accuracy metrics* Accuracy (5 articles), Precision (4 articles), Recall (3 articles), F-Score (3 articles),
- *Rank accuracy metrics* MRR (1 article), NDCG (1 article), MPR (1 article), Hit Rate (1 article),
- *Non-accuracy accuracy metrics* Scalability (1 article), Coverage (1 article), Cost (1 article), Time complexity (1 article).

Table 7 shows that 14 different evaluation metrics were used in the 15 content-based JRS solutions examined and are presented in Fig. 10, which are:

- *Predictive accuracy metrics* MSE (1 article), RMSE (1 article), ROC (1 article), similarity assessment using Cosine (1 article), DBSCAN (1 article), Euclidean Distance (1 article), Manhattan Distance (1 article), and PCC (1 article),
- *Classification accuracy metrics* Accuracy (2 articles), Recall (1 article), AUC (2 articles),
- *Rank accuracy metric* None,
- *Non-accuracy accuracy metrics* Time complexity (1 article), Cost (1 article), Space complexity (1 article).

Table 9 shows that 17 different evaluation metrics were used in the 13 hybrid-based JRS solutions examined and are presented in Fig. 10, which are:

- *Predictive accuracy metrics* MSE (3 articles), RMSE (3 articles), MAE (1 article), MRE (1 article), and similarity assessment using Cosine (1 article),
- *Classification accuracy metrics* Accuracy (3 articles), Precision (4 articles), Recall (4 articles), F-Score (2 articles), AUC (3 articles), FPR (1 article), FNR (1 article), and Logistic Loss (1 article),
- *Rank accuracy metric* NDCG (1 article) and Hit Rate (1 article),
- *Non-accuracy accuracy metrics* Coverage (1 article) and Novelty (1 article).

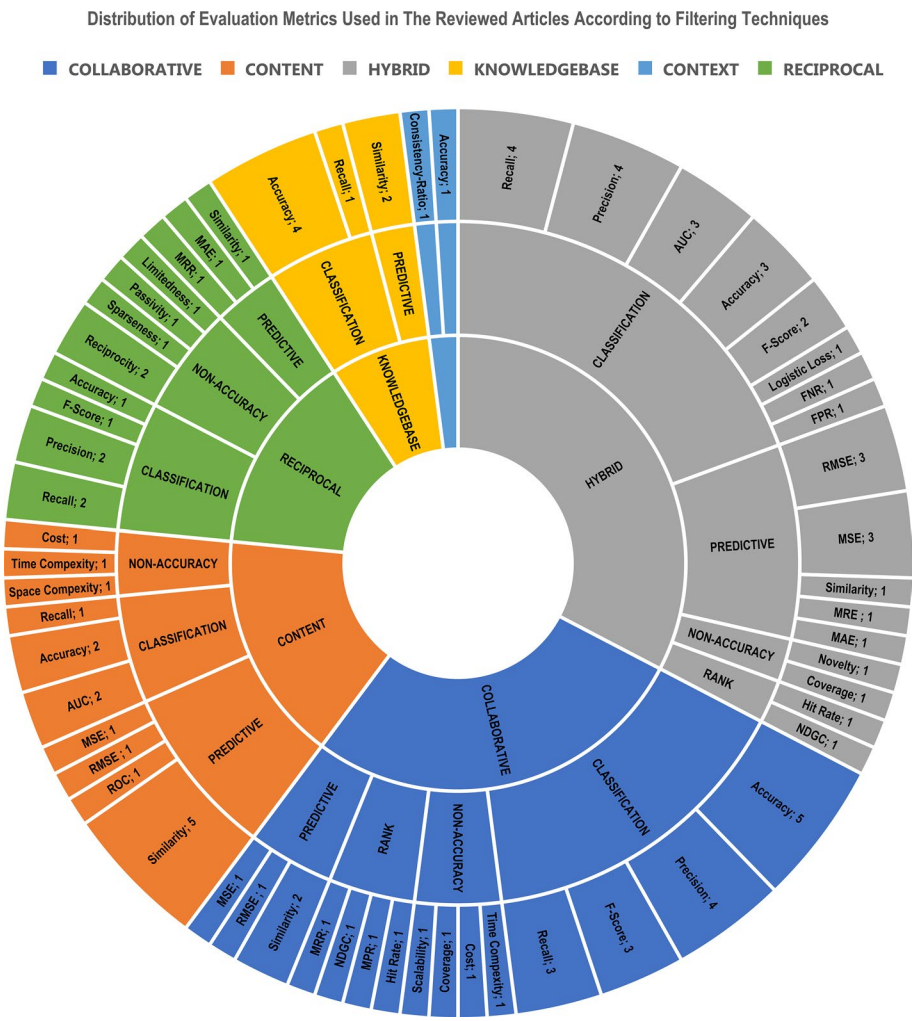


**Fig. 10** Distribution of the number of articles using evaluation metrics according to the preferred filtering type. (1) The inner layer shows the JRS filtering categories, (2) the middle layer shows the evaluation metric categories, and (3) the outer layer shows the number of articles evaluated by metric types

Table 11 shows that 4 different evaluation metrics were used in the 8 knowledge-based JRS solutions examined and are presented in Fig. 10, which are:

- *Predictive accuracy metrics* Similarity assessment using Jaro-Wrinkler (1 article) and Euclidean Distance (1 article),
- *Classification accuracy metrics* Accuracy (4 articles) and Recall (1 article),
- *Rank accuracy metric* None,
- *Non-accuracy accuracy metrics* None.

As far as we know and have scanned in the literature, it is possible to say that JRS solutions based on context-based filtering approach are quite limited. As mentioned before, only 2 demographic research studies of context based JRSs have been found in

the literature. Table 13 shows that only 2 different evaluation metrics are used in the 2 demographic filtering JRS solutions reviewed, and are presented in Fig. 10, which are:

- *Predictive accuracy metrics* None,
- *Classification accuracy metrics* Accuracy (1 article),
- *Rank accuracy metric* Consistency-ratio (1 article),
- *Non-accuracy accuracy metrics* None.

Table 15 shows that 7 different evaluation metrics are used in 3 the reciprocal JRS solutions reviewed, and are presented in Fig. 10, which are:

- *Predictive accuracy metrics* Similarity assessment using Cosine (1 article), MAE (1 article), and MRR (1 article),
- *Classification accuracy metrics* Accuracy (1 article), Precision (2 articles), Recall (2 articles), and F-Score (1 article),
- *Rank accuracy metric* None,
- *Non-accuracy accuracy metrics* Reciprocity (2 articles), Limitedness (1 article), Passivity (1 article), and Sparseness (1 article).

RQ4: What are the popular evaluation and modeling tools used in developing JRSs?

In this section, the tools used in the JRS research studies were examined according to the filtering types applied and were categorized as follows: (1) Frameworks (FW), (2) Database tools (DB), (3) Knowledge Base/Markup Language tools (KB), (4) Components & Libraries tools (CL), (5) Software tools (SW), (6) Data Collection & Platforms tools (DC), (7) Development tools (DEV), (8) Data Analysis & Classification tools (DA), and (9) Ranking tools (RT).

The distribution of tools used in the reviewed JRS studies, according to the tool categories given above and the filtering approaches used, is presented in Table 19. Collaborative, content-based, and hybrid filtering-based JRS solutions are particularly widespread in literature; consequently, tools supporting these filtering types are also abundant. Frameworks such as Spring and Bootstrap, databases like MySQL and Redis, software platforms such as MySQL, and development platforms such as PyTorch and TensorFlow are frequently applied in these filtering based JRS solutions. Tools for knowledge-based, demographic, and reciprocal types, however, are less commonly encountered. This is likely due to the limited presence of JRS solutions based on these filtering types in the literature.

RQ5. What are the current challenges and future improvements in JRS implementations?

*Disability, justice, data privacy, and algorithmic bias.* The integration of human characteristics, such as reasoning, collaboration, creativity, social sensitivity, and problem-solving, into AI and robotic systems poses both current challenges and

**Table 19** The tools used in the selected articles are categorized according to the type of recommendation filtering used

| Filtering Types | Collaborative | | | | | | | | Content-Based | | | | | | | Hybrid | | | | | | | Knowledge-Based | | | | | Demographic | | | Reciprocal | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Tools Used** | **FW** | **DB** | **KB** | **CL** | **SW** | **DC** | **DEV** | **DA** | **RT** | **FW** | **DB** | **CL** | **DC** | **DEV** | **DA** | **FW** | **DB** | **CL** | **DC** | **DEV** | **DA** | **RT** | **DB** | **KB** | **DC** | **DEV** | **DA** | **FW** | **DC** | **DA** | **FW** | **DB** | **DC** | **DA** |
| Spring | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Bootstrap | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| PyTorch | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| LFIT | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| GraphLab | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| MySQL | | + | | | | | | | | | + | | | | | | | | | | | | + | | | | | | | | | + | | |
| Redis | | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | | + | | | |
| FairCVdb | | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| WWXML | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| WXSS | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| LAYUI | | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Matplotlib | | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Scikit-Learn Lib | | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| SAS | | | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| OLX Platform | | | | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Linked in | | | | | | + | | | | | | | + | | | | | | + | | | | | | | | | | | | | | | |
| Google Cloud Entity | | | | | | + | | | | | | | | | | | | | + | | | | | | | | | | | | | | | |
| GLIT.IO | | | | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Monster Job-Hunting | | | | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Java | | | | | | | + | | | | | | | | | | | | | | | | | | | + | | | | | | | | |
| Python | | | | | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Panda | | | | | | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | |
| NumPy | | | | | | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | |

**Table 19** (continued)

| Filtering Types | Collaborative | | | | | | | | Content-Based | | | | | | | Hybrid | | | | | | | Knowledge-Based | | | | | Demographic | | | Reciprocal | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Tools Used** | **FW** | **DB** | **KB** | **CL** | **SW** | **DC** | **DEV** | **DA** | **RT** | **FW** | **DB** | **CL** | **DC** | **DEV** | **DA** | **FW** | **DB** | **CL** | **DC** | **DEV** | **DA** | **RT** | **DB** | **KB** | **DC** | **DEV** | **DA** | **FW** | **DC** | **DA** | **FW** | **DB** | **DC** | **DA** |
| A/B Testing | | | | | | | | + | | | | | | | | | | | | | + | | | | | | | | | | | | | |
| GMUSA | | | | | | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | |
| GULA-PRIDE Alg | | | | | | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Google Collab | | | | | | | | + | | | | | | | | | | | | | | | | | | | | | | | | | | |
| RP3 Beta Model | | | | | | | | | + | | | | | | | | | | | | | | | | | | | | | | | | | |
| TOPSIS | | | | | | | | | + | | | | | | | | | | | | | + | | | | | | | | | | | | |
| JAX-RS | | | | | | | | | | + | | | | | | | | | | | | | | | | | | | | | | | | |
| Dig Comp | | | | | | | | | | + | | | | | | | | | | | | | | | | | | | | | | | | |
| e-CF | | | | | | | | | | + | | | | | | | | | | | | | | | | | | | | | | | | |
| MongoDB | | | | | | | | | | | + | | | | | | | | | | | | | | | | | | | | | | | |
| Facebook | | | | | | | | | | | | | + | | | | | | | | | | | | | | | | | | | | | |
| Beautiful Soup | | | | | | | | | | | | | + | | | | | | | | | | | | | | | | | | | | | |
| O*NET | | | | | | | | | | | | | + | | | | | | | | | | | | + | | | | | | | | | |
| Stata 14.0 | | | | | | | | | | | | | | | + | | | | | | | | | | | | | | | | | | | |
| ISCO | | | | | | | | | | | | | | | + | | | | | | | | | | | | + | | | | | | | |
| ESCO | | | | | | | | | | | | | | | + | | | | | | | | | | | | + | | | | | | | |
| ISCED | | | | | | | | | | | | | | | + | | | | | | | | | | | | | | | | | | | |
| DISCO | | | | | | | | | | | | | | | + | | | | | | | | | | | | | | | | | | | |
| Big5 Model | | | | | | | | | | | | | | | + | | | | | | | | | | | | | | | | | | | |
| Jieba.io | | | | | | | | | | | | | + | | | | | | | | | | | | | | | | | | | | | |
| Apache OpenNLP | | | | | | | | | | | | + | | | | | | | | | | | | | | | | | | | | | | |
| Tomcat7 | | | | | | | | | | | | | | + | | | | | | | | | | | | | | | | | | | | |
| NLP Tool | | | | | | | | | | | | | | | + | | | | | | + | | | | | | | | | | | | | |

**Table 19** (continued)

| Filtering Types | Collaborative | | | | | | | | Content-Based | | | | | | | Hybrid | | | | | | | Knowledge-Based | | | | | Demographic | | | Reciprocal | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Tools Used** | **FW** | **DB** | **KB** | **CL** | **SW** | **DC** | **DEV** | **DA** | **RT** | **FW** | **DB** | **CL** | **DC** | **DEV** | **DA** | **FW** | **DB** | **CL** | **DC** | **DEV** | **DA** | **RT** | **DB** | **KB** | **DC** | **DEV** | **DA** | **FW** | **DC** | **DA** | **FW** | **DB** | **DC** | **DA** |
| LiJar | | | | | | | | | | | | | | | | + | | | | | | | | | | | | | | | | | | |
| NoSQL | | | | | | | | | | | | | | | | | + | | | | | | | | | | | | | | | | | |
| R Tool | | | | | | | | | | | | | | | | | | | | + | | | | | | | | | | | | | | |
| XGBoost Lib | | | | | | | | | | | | | | | | | | + | | | | | | | | | | | | | | | | |
| Career Builder | | | | | | | | | | | | | | | | | | | + | | | | | | | | | | + | | | | | |
| Lagou | | | | | | | | | | | | | | | | | | | + | | | | | | | | | | | | | | | |
| CasMos | | | | | | | | | | | | | | | | | | | + | | | | | | | | | | | | | | | |
| FitRel Reg. Tree | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | | | | | | |
| RMSPROP Opt | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | | | | | | |
| Tensor Flow | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | | | | | | |
| SparQL | | | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | | | | |
| OWL | | | | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | | | |
| SWRL | | | | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | | | |
| HR-XML | | | | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | | | |
| Eclipse IDE | | | | | | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | |
| Protégé | | | | | | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | |
| SoNITS | | | | | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | | |
| GitHub Jobs API | | | | | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | | |
| AXELOS | | | | | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | | |
| ELISE | | | | | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | | |
| JADE | | | | | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | | |
| VADB | | | | | | | | | | | | | | | | | | | | | | | | | + | | | | | | | | | |
| jQuery Mobile | | | | | | | | | | | | | | | | | | | | | | | | | | | | + | | | + | | | |

**Table 19** (continued)

| Filtering Types | Collaborative | | | | | | | | Content-Based | | | | | | | Hybrid | | | | | | | Knowledge-Based | | | | | Demographic | | | Reciprocal | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Tools Used** | **FW** | **DB** | **KB** | **CL** | **SW** | **DC** | **DEV** | **DA** | **RT** | **FW** | **DB** | **CL** | **DC** | **DEV** | **DA** | **FW** | **DB** | **CL** | **DC** | **DEV** | **DA** | **RT** | **DB** | **KB** | **DC** | **DEV** | **DA** | **FW** | **DC** | **DA** | **FW** | **DB** | **DC** | **DA** |
| Holland Assessment Scale | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | + | | | | |
| zhaopin.com | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | + | |
| Lucene IK Anal | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | + |

(1) *FW* Frameworks, (2) *DB* Database tools, (3) *KB* Knowledge Base/Markup Language tools, (4) *CL* Components & Libraries tools, (5) *SW* Software tools, (6) *DC* Data Collection & Platforms tools, (7) *DEV* Development tools, (8) *DA* Data Analysis & Classification tools, and (9) *RT* Ranking tools

opportunities for future advancements in JRSs [38]. The evolving landscape of JRS aims to enhance the recruitment processes of HR departments, selecting candidates who can significantly contribute to the improvement of company services, processes, and productivity. However, as JRS solutions become more sophisticated, there is a growing concern about the potential reinforcement of biases and prejudices, particularly against marginalized groups (e.g., criminal groups, disabled groups), in the recruitment process [142]. This arises from the comparison of skills and profiles of marginalized job seekers with those of former employees, or the attempt to identify candidates based on criteria that may inadvertently favor certain groups. This ethical issue underscores the need for JRS developers to prioritize the creation of more fair and equitable solutions.

Smarter JRS solutions should include measurable mathematical criteria optimized for normative concepts such as justice. In addressing this concern, the emphasis should be placed on developing JRSs enriched with advanced AI technologies that can provide fair and unbiased recommendations. Developers of advanced JRS solutions must strive to implement anti-bias measures, reduce existing inequalities, and detect differences among individuals, especially within the context of marginalized groups [142]. Moreover, transparent AI systems, such as those that incorporate Explainable AI (XAI) [49], can play a crucial role in ensuring fairness by enabling both employers and candidates to comprehend the reasoning behind specific decisions. This transparency helps increase trust in the system and reduces concerns about hidden bias or unfairness.

Data privacy is another significant consideration in the development of JRSs. The collection, processing, and storage of personal and professional data from candidates can raise privacy concerns. Compliance with privacy regulations (e.g., GDPR) should be a critical factor in the design of JRSs. The use of personal data without candidates' consent may be considered unethical and may increase concerns over trust. It is crucial for JRS developers to create systems that protect candidates'data privacy, ensuring user confidence in the system.

In conclusion, the future of JRS solutions will be shaped not only by their technological innovations but also by how these innovations follow the values of justice, equality, inclusivity, and data privacy. By addressing ethical concerns and minimizing bias, the JRS field can move towards a more equitable, transparent and trustworthy future for both employers and job seekers.

RQ6: What are the available standards applied in the JRSs?

HR Certification Institute [68] is an important accreditation body that has been setting HR standards for 50 years to ensure excellence in HR process management in small or large companies. It was established to set standards and provide certifications to individuals working in the field of HR.

European Skills, Competences, and Occupations (ESCO [50]) is a European multilingual dictionary portal that provides information about skills, competencies, and occupations. Some key supports provided by ESCO are listed below:

- *Multilingual Portal*. ESCO is available in multiple languages, offering information in 28 languages, including official EU languages and additional languages such as Icelandic, Norwegian, Ukrainian, and Arabic.
- *Comprehensive Information***.** The portal encompasses a vast amount of information, including 3,008 professions, 13,890 skills related to these professions, and definitions. This provided information facilitates a common understanding of language regarding professions and skills.
- *Electronic System Compatibility***.** ESCO is designed to be compatible with electronic systems, allowing for the inclusion of various services. These services can provide matching job seekers with job opportunities based on their skills, facilitating reskilling initiatives, and providing training opportunities for skill improvement.
- *Support for Job Mobility***.** The overarching goal of ESCO is to promote a common understanding of language between professions and skills. This supports job mobility and aims to create a more efficient labor market across Europe
- *APIs for Integration***.** ESCO provides Application Program Interfaces (APIs) that offer easy access to its classification lists. These APIs are designed for integration with other companies'software and can be accessed through web services.
- *Integration into HR Processes***.** The availability of ESCO's classification lists through APIs enables the early detection of existing and emerging professions along with their relevant competencies and skills. This information can be seamlessly integrated into companies'HR processes, contributing to the development of effective JRS applications.
- *Bias in AI Models***.** One of the key challenges in the application of JRSs is the potential bias in AI models used to match job seekers with opportunities. Biases in historical data or algorithmic decisions can lead to unfair or unequal outcomes in recruitment processes, which can hinder diversity and inclusion efforts.
- *Data Privacy Issues*. The use of AI and digital systems for JRSs raises significant concerns around data privacy. Ensuring that sensitive personal data is securely handled and that organizations comply with data protection regulations is critical for the ethical application of JRSs.
- *Integration Challenges*. Integrating JRSs with existing HR systems poses another significant challenge. Many organizations already use legacy HR systems, and ensuring smooth integration without disruption or the need for extensive system overhauls can be a complex process.

In summary, ESCO plays a crucial role in harmonizing language related to skills and professions across Europe, fostering job mobility, and supporting the development of JRS solutions by providing easily integrable data through APIs.

Additionally, there are other important classification resources such as DISCO [45], ISCO (ISCO [70]) and ISCED [69], etc., which determine the concepts and relationships of numerous occupations, skills, competencies, job descriptions, etc. These classification resources have been developed in the form of a taxonomy and represent the basis of a comprehensive terminology based on international standards (Martinez-Gil et al., 2016). These resources can actively contribute to the development of the knowledge bases

of JRS solutions by serving as a foundation for understanding and categorizing job-related information. Furthermore, the adoption of standardized classifications ensures that JRS systems work with high-quality and reliable data, leading to more accurate recommendations.

In conclusion, although existing frameworks such as DISCO, ESCO and ISCED provide valuable classifications, with the use of advanced description logics of ontologies of Semantic Web technology, JRSs can further increase their adequacy and efficiency in representing and reasoning of HR elements. This approach will enable both more complex reasoning in HR processes and interoperability in JRS solutions. HR elements supported by these frameworks are represented as concepts and can be referred as classes that are the basic components of ontologies. Additionally, semantic relationships can be created between classes in ontologies and detailed explanations can be determined. In summary, Due to ontology components, JRS solutions can offer more significant deductions.

#### Limitations

Although we believe we have provided a broad overview of work in the field of JRS, this review had some limitations:

First, due to time and resource constraints, this review focused on research and review articles published between January 1, 2010, and April 30, 2023. A total of 76 relevant studies were analyzed. However, studies on JRSs proposed in 2024 and beyond were excluded.

Second, while JRS solutions supporting filtering mechanisms such as collaborative filtering, content-based filtering, hybrid filtering, and knowledge-based filtering were prevalent in the literature, limited research was found on JRS solutions using demographic-based, reciprocal-based, popularity-based, or non-personalized filtering. Consequently, the findings may primarily focus on JRS solutions that utilize commonly applied filtering mechanisms.

Third, the scope of the review was narrowed to focus only on JRS solutions that support recruitment processes. The literature predominantly addressed solutions targeting candidate-job matching. However, future research could explore JRS solutions supporting other HR processes, such as employee satisfaction assessments, career planning recommendations, or performance management.

Fourth, this study also reviewed and compared other literature review papers on RSs and JRSs. This approach made the paper extensive and detailed. Nevertheless, we aimed to identify gaps and limitations in the field, believing that the findings will be valuable for future research.

Fifth, restricting the review to studies written in English may have excluded valuable insights from research published in other languages. Additionally, focusing on specific academic databases and disciplines may have overlooked relevant contributions from less prominent sources or interdisciplinary fields.

Finally, the inclusion of studies published between 2010 and 2023 may have limited the incorporation of more recent innovative approaches. Significant advancements in JRS solutions may have been achieved after 2024, driven by AI's innovative contributions. The application of Large Language Models (LLMs) and state-of-the-art machine/deep

learning algorithms could lead to more sophisticated and effective solutions in both RSs and JRSs. These advancements could foster the development of more precise and user-centered JRSs in HRM, addressing the growing turnover rates in various industries. Future studies could utilize this paper's findings to compare pre-2024 results with innovative approaches developed after 2024.

### Contributions

Our study provides a comprehensive evaluation of JRSs by examining 57 research papers published between January 1, 2010, and April 30, 2023. This review not only addresses gaps identified in the existing literature but also broadens the understanding of both JRSs and RSs.

Firstly, our development of a comprehensive taxonomy to classify JRSs based on filtering techniques used in RSs represents a significant contribution to the field. Compared to the general trends and methodologies provided by Lu et al. [88], our detailed classification of filtering techniques used in JRSs offers a clearer understanding of the impact of existing methods on JRS applications, including underutilized approaches such as popularity-based and social network-based filtering techniques. This structured framework fills an important gap in the literature.

Secondly, our in-depth analysis of evaluation metrics provides new insights. While various studies [35, 62, 63, 131, 147], have examined different evaluation metrics for RSs, our paper enriches this aspect by classifying these metrics into Predictive Accuracy Metrics, Classification-Based Accuracy Metrics, Ranking-Based Accuracy Metrics, and Non-Accuracy Based Metrics, and by addressing a broader range of RS performance metrics. This detailed classification enhances the understanding of performance evaluation for both RS and JRS solutions and assists researchers in the performance analysis process.

Thirdly, we have identified significant research gaps and trends in current studies. For instance, we highlighted the limited use of advanced filtering approaches and the relative scarcity of Reciprocal RS in JRS solutions. This finding aligns with concerns raised by Kwieciński et al. [81] regarding the need for more innovative approaches in JRSs. By identifying these gaps, we emphasize the need for further research and development in these areas.

Finally, our historical analysis of publication trends over the past 13 years reveals fluctuations in research interest and identifies years of significant developments. This temporal perspective contextualizes the current state of JRS research and predicts future trends, offering valuable insights into the evolving landscape of JRS solutions.

In conclusion, our study makes unique contributions by developing a comprehensive taxonomy, providing a detailed analysis of evaluation metrics, identifying research gaps and trends, analyzing historical publication trends, and offering practical insights for HR processes. These contributions advance theoretical understanding and provide practical guidance for future research and development efforts.

### Discussions

Although Stone et al. [139] interpreted technology as a challenge for the HR field, the researchers also emphasized that it offers new opportunities for HR. For example, it is

argued that technology often reduces the administrative burden in HR, increases efficiency, and enables the field to contribute to the strategic direction of organizations. Therefore, it is possible to say that HRM is continuously evolving due to rapidly changing workforce dynamics and increasing competition. Especially, the factors such as workforce diversity, digitalization, growing complexity in organizational processes, remote work, and the need to balance employee satisfaction with organizational efficiency pose significant challenges for HR practitioners. Additionally, identifying the right talent through complex, personalized recruitment processes, maintaining fair recruitment policies, training employees based on their skills and organizational needs, developing strategies to enhance employee engagement, and ensuring effective performance management are key challenges driving the ongoing evolution of HR practices. Accordingly, the use of effective technologies and data-driven decision-making processes has become indispensable to managing these increasingly complex HR processes and challenging factors. The rapid development of AI and ML technologies offers significant opportunities for JRSs [116, 149]. In particular, advances in deep learning and NLP technologies have improved the personalization and effectiveness of JRS. Furthermore, big data analytics and real-time decision-making capabilities help JRS adapt to dynamic labor market conditions. Recent advancements in LLMs have the potential to improve both RSs [48, 86, 130, 154, 165] leading to more sophisticated and effective solutions. These innovations could contribute to the development of more precise, user-centered JRSs in HRM, addressing rising turnover rates across industries. Future research should focus on optimizing the use of these technologies and developing new models for their application in organizational contexts.

## Conclusion

Recently, various factors such as the emergence of new professional fields, increased professional transitions based on individual career plans, growing complexity in Human Resources (HR) processes, and time pressure in recruitment have heightened interest in effective and high-performance Job Recommender Systems (JRSs). To expedite recruitment processes, correct matching, and reduce biases, JRS field has gained popularity across diverse industries, capturing the attention of both Recommender Systems (RSs) researchers and HR process practitioners. The aim of this research is to identify current trends in JRS solutions and provide developers and researchers with recent and relevant information. The findings of this study encompass several noteworthy implications, as summarized below:

- This SLR explored the similarities and comparisons between filtering techniques and development methodologies used in both RSs and JRSs. To achieve this, 19 literature survey studies were reviewed, with 8 focusing on RSs survey studies and 11 focusing on JRSs survey studies. However, some of the 11 JRS survey studies had certain limitations. Some studies did not review recently published articles, while others lacked information on potential future improvements to JRS solutions. Some studies did not explore evaluation metrics and methodologies, and some studies were not systematically structured. Additionally, some studies did not provide detailed information on JRS solutions such as a comprehensive hierarchical classification of the JRS field,

clear statistical information about JRS solutions, and the tools used in the development and evaluation of JRSs, among other aspects.

- In this study, a comprehensive taxonomy was developed after analyzing 19 literature review studies to categorize the filtering techniques and methodologies employed in both RSs and JRSs. This taxonomy was created to examine the impact of existing RS filtering techniques and development methodologies on JRS filtering techniques and methodologies. While JRS solutions utilizing collaborative filtering, content-based filtering, hybrid-based filtering, and knowledge-based filtering are well-represented in the literature, those applying demographic-based filtering, reciprocal-based filtering, popularity-based filtering, and non-personalized filtering remain underrepresented in current research. Future studies in the JRS field could be further advanced through the exploration of these rarely used alternative filtering methods by HR professionals and technology developers.
- Additionally, in this study, evaluation accuracy metrics commonly used in both RS and JRS have been determined and compared based on the RSs filtering types. The outcomes of this study can provide guidance to technology developers in the HR field to determine which filtering mechanisms align with specific performance metrics and use them in future JRS solutions.
- As next, a search query from January 1, 2010, to April 30, 2023, was formulated by investigating the following keywords:"(Recommender < OR > Recommendation < OR > Recommending < OR > Matching < OR > Decision Support < OR > AI) [AND] (Job < OR > Skill < OR > Job Opportunities < OR > Recruitment))". In addition to these keywords, we did not use other related terms such as "Content-Based Filtering" and only included those in English within the scope of the research. A total of 527 articles were retrieved through the search query, and after applying inclusion and exclusion criteria, a total of 57 research articles related to JRS filed were identified and examined. Each article was classified according to the developed taxonomy and assessed based on the publication year, their main focus, applied techniques, used data, employed tools, performance and accuracy metrics, and future development plans were analyzed and compared.
- Our findings are based on the 57 articles selected from both academic journals and conference publications. The number of conference papers is 46, while the number of academic journal articles is 11. This indicates that academic publications on JRSs are quite limited and suggests that research in the JRS field may be facing certain constraints. For instance, challenges in interdisciplinary studies such as HR and engineering, the insufficient or unstructured of HR data sets, investments made in Web 2.0 solutions, and barriers such as the inability to integrate AI-supported autonomous features, are possible significant factors influencing these limitations.
- Analyzing the publication rates in the literature revealed that 2011 had the least interest in JRS research studies, while 2017 showed the highest interest. Over the last 13 years, it has been observed that the number of studies on JRS solutions is quite limited, and the distribution of these studies over the years is characterized by fluctuations and uncertainty. Nevertheless, given the advancements and trends in AI, along with the increasing complexity and intensity of HRM processes, we believe

that interest in JRS solutions will experience significant growth in the future, gaining considerable attention from HR professionals and technology developers.

- In 57 research articles, various aspects of JRS applications were discussed, including the development of a JRS application in 24 articles. Recommendations for suitable professions based on social/demographic information and skills of individuals were explored in 6 articles. The development of innovative job-candidate matching algorithms was discussed in 5 articles. Identifying and recommending appropriate team members were topics in 4 articles. Recommendations for suitable training modules based on skill deficiencies of individuals were addressed in 4 articles. The development of reciprocal-based JRS solutions, considering mutual approval of both parties (job seeker and HR staff), was discussed in 3 articles. The creation of a career path planning system was the focus of 2 articles. Employee satisfaction assessments were explored in 2 articles. A comparison of different JRS applications involving various techniques was discussed in 2 articles. Standardization and homogenization of professional knowledge and subject expertise using semantic techniques were addressed in 2 articles. Determining the busiest job field based on analyzing job posting data was discussed in 1 article. The correlation between job tenure estimation and turnover intention was explored in 1 article, and the effectiveness of combining different filtering techniques was discussed in 1 article. In future JRS solutions, HR professionals and technology developers may focus on developing advanced job-candidate matching algorithms, exploring reciprocal-based solutions, integrating AI and semantic technologies for personalized recommendations, investigating career path planning, expanding research on demographic factors, and analyzing job market trends to predict skill demand.
- The methods applied in the reviewed 57 studies are categorized according to the filtering methodologies used. The findings are various similarity methods, such as TOPSIS, Loglikelihood, and Cosine Similarity, were utilized in 16 research studies. Additionally, Word/Sentence Vector Models, specifically Word2 Vec/Skip-Gram, were employed in 9 studies. The application of JRS studies encompassed both supervised and unsupervised ML models. Supervised learning, observed in 9 studies, involved classification algorithms like XGBoost, LSTM, and BERT. On the other hand, 4 studies focused on unsupervised learning techniques, utilizing methods like K-means and Markov Model. Regression algorithms, including SVM, Naive Bayes, and Random Forest, were featured in 12 studies. Knowledge-based approaches, incorporating Ontology and Semantic Web, were applied in 6 articles. Additionally, TF-IDF, Data Mining, Matrix Factorization Model, Multiple Relational Graph, and A/B Test were employed in varying frequencies across 5 to 3 studies. Individual models like RP3Beta, Maximum Entropy, Generalized Linear Mixing (GLMix), Interaction-based relationships, and Text Mining Technology each found mention in 1 article. Furthermore, techniques such as O*NET, ESCO, ISCO vector models, LFIT, and CART algorithms were employed in 3 studies, while 2 articles explored the use of ranking algorithms. In future JRS solutions, HR professionals and technology developers may focus on diverse similarity methods, with TOPSIS, Loglikelihood, and Cosine Similarity being popular. More emphasis on unsupervised learning (e.g., K-means, Markov), knowledge-based approaches

(Ontology, Semantic Web), RSs using effective LLM models (e.g., GPT, BERT, T5, LLaMA, LNet, RoBERTa, etc.) could improve JRS applications.

– In the majority of cases, JRS solutions in existing literature primarily facilitate one-way matching. Specifically, these solutions typically involve either identifying suitable candidate resumes and recommending them to HR personnel or presenting suitable job opportunities to job seeker(s). Our investigation indicates a notable scarcity of Reciprocal-based JRS solutions in the literature. Over the period from January 1, 2010, to April 30, 2023, only 3 RRS-based JRS solutions were identified. This emphasizes the considerable demand for effective and high-performance RRS-based JRS solutions that can actively involve both parties in the recruitment process, facilitating online matching and reconciliation.

As a result, methods and applications in JRS solutions are a very important research area for both researchers and real-world developers, especially in the effective and bias-free development of the recruitment process in the field of HRM. The findings of this research can guide all future research and development in the field of JRS for developers, practitioners and academics working in this field. In particular, studies conducted/to be conducted after 2024 to examine the impact of AI innovations on JRS solutions can investigate the use of Large Language Models and state-of-the-art Machine/Deep Learning algorithms to develop more sophisticated and effective systems in the field of HRM. In addition, future researches can follow the latest developments after 2024 to analyze the extent to which user-based JRS applications contribute to the increasing turnover trends of individuals and advances in HRM solutions. Finally, this study provides guidance to understand the current challenges in JRS solutions and to identify standards that can enhance the functionality of future JRS solutions.

**Abbreviations**

| | |
|---|---|
| JRS | Job recommender system |
| AI | Artificial intelligence |
| HR | Human resources |
| HRM | Human resources management |
| ML | Machine learning |
| NLP | Natural language processing |
| NER | Named entity recognition |
| MAE | Mean absolute error |
| MSE | Mean squared error |
| RMSE | Root mean squared error |
| NMAE | Normalized mean absolute error |
| MAUE | Mean absolute user error |
| R2 | R Squared |
| EV | Explained variance |
| PCC | Pearson correlation coefficient |
| ROC | Receiver operating characteristics |
| Log-Loss | Logistic loss |
| AUC | Area under curve |
| P(k) | Precision@k |
| SPE | Specificity |
| MAP | Mean average precision |
| AP | Average precision |
| HR | Hit Ratio |
| MRR | Mean reciprocal rank |
| NDPM | Normalized distance-based performance measure |
| MLE | Maximum likelihood estimation |
| ARHR | Average reciprocal hit ratio |
| SVM | Support vector machine |
| CG | Cumulative gain |

| | |
|---|---|
| DCG | Discounted cumulative gain |
| KNN | K-Nearest neighbor |
| iDCG | The ideal DCG |
| KG | Knowledge graph |
| NDCG | Normalized discounted cumulative gain |
| GNN | Graph neural network |
| ANN | Artificial neural networks |
| MYA | A chatbot developed by start-up MYA Systems |
| SLR | Systematic literature review |
| WoS | Web of science |
| MLE | Maximum likelihood Estimation |
| PMI | Pointwise mutual information |
| GPA | Graph based approach |
| DSSM | Deep semantic structured models |
| DSS | Decision support system |
| CGA | Continuous genetic algorithm |
| GA | Genetic algorithm |
| Fuzzy-AHP | Fuzzy analytical hierarchy process |
| TOPSIS | Technique for order of preference by similarity to ideal solution |
| LFIT | Learning from interpretation transition |
| ILP | Inductive logic programming |
| T-GBRT | Gradient boosting regression tree and time factors |
| MBL | Multi behavior learning module |
| EPL | Explicit preference learning |
| FoDRA | Four dimensions recommendation algorithm |
| DISCO | European dictionary of skills and competences |
| ISCO | The International standard classification of occupations |
| ISCED | International standard classification of education |
| ESCO | European skills competencesand occupations |
| MCTS | Monte-Carlo tree search |
| ITFACP | Adaptive context partition |
| DNN | Deep neural network |
| BWM | Best–Worst Method |
| SNRs | Social network relationships |
| HRCI | HR certification institute |
| GBDTs | Gradient boosted decision trees |
| TÜBİTAK | Turkish scientific and technological research council |
| RRS | Reciprocal recommendation system |

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